Higher Scores, Not A Better Model: A Large Multiple-Choice Lift That Is Mostly Inducible, Off-Axis, and Does Not Transfer to Generation

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Abstract

Contrastive-correctness parameter-efficient fine-tuning (the LoFiT/ITI family) is widely used to sharpen a model's multiple-choice judgment cheaply. On a frontier 31B instruction model (Gemma 4 31B IT) it buys a large cold multiple-choice lift: ARC-Challenge +35.0, HellaSwag +17.0, Winogrande +19.0 (scoring-face, no chain-of-thought), plus TruthfulQA-mc1 +37.25, which we cite separately throughout as an answer-prior selection lift and exclude from the mechanism. We ask what that lift *is*, and answer with a quantified causal attribution rather than a new method.

Much of the lift is *inducible without the contrastive objective*, and how much depends on capacity and on the task. A *same-parameterization* control — the identical 48 head-pairs and 12,336 trainable parameters, same corpus and steps, ordinary cross-entropy with no term against the incorrect option, three seeds — recovers 36.2% of the ARC lift (sd 5.0 over three training seeds; item-sampling uncertainty not included), while a non-contrastive LoRA with 1,800×–58,000× more parameters recovers ~62% (paired-bootstrap fraction-CI [51%, 73%], flat across rank r8–256; a larger r512 flank at ~117,000× does not improve on it and can fall below base). The two bars quantify disjoint uncertainties — training runs and item sampling — and we do not test the difference between the two fractions. Inducibility is also strongly task-dependent: the same-parameterization control recovers 79.5% of Winogrande's lift and 69.5% of HellaSwag's but 0.9% of TruthfulQA-mc1's, i.e. nothing (on each task's headline metric; the four *lifts* are robust to that choice, the recovered *fractions* are not — under raw accuracy HellaSwag's 69.5% reads 8.9% and ARC's 36.2% reads 26.9% — so we cite every fraction as an acc_norm statement and rest no mechanistic claim on them or on their cross-task ordering) — raising the correct option's probability suffices where the task scores continuation plausibility, but not where a plausible distractor must be beaten. That places the lift on a matched-control spectrum — random ≈0% → same-parameterization ~37% → higher-capacity SFT ~62% → contrastive 100% — with an on-axis reference (W_{know}) at 6.4%, *statistically indistinguishable from zero* (exact McNemar $p = 0.136$), counted with the floor rather than as a rung. To our knowledge no prior work runs matched controls and reports these *fractions*; that attribution is the load-bearing novelty.

The contrastive adapter's write is *causally off-axis of degree* — off the measured linear correctness axis at each intervened head, the strongest such 1-D discriminant we can build, not the whole correctness manifold — and the identifying evidence is the negative arms: a push *parallel* to the correctness direction recovers 0% of the lift, and a norm-matched *random* perpendicular push recovers ~0% across three seeds — so the learned perpendicular direction is privileged, not any perturbation of the right size. (A perpendicular push recovers ~100%, but the offset is already almost entirely perpendicular — $\cos(\theta_{\text{mc}}, v_{\text{inf}}) \approx -0.003$ — so that arm is near-tautological and we do not lean on it.) The lift is also *undeployable as a generator*: it does not transfer (gsm8k -24.2 ± 7.3 pp across three independently trained seeds — the sign is invariant, the magnitude is not — a mechanistic multi-step-reasoning loss; MMLU -4.2 pp spillover; out-of-task, GPQA cold-MC ~0 while the base reasons GPQA out at CoT 0.70 — single-run and directional, with 31/198 traces unparsed; a re-run that persists token counts shows the unparsed traces are ones that ran out of budget rather than answers, and bounds the accuracy in $[0.60, 0.91]$). A per-item arbiter bounds it: the base already solves 124/124 of the fixed ARC items under chain-of-thought, against 41/47 on the items the adapter fails to fix ($p = 1.4 \times 10^{-4}$) — the arbiter discriminates rather than restating a ceiling. Re-run on the *entire* ARC-Challenge test split, outside the n=400 draw the other diagnostics share, it replicates with far more power: the adapter fixes 444 items and the base already answers 442/444 = 0.9955 of them under chain-of-thought, against 134/150 = 0.8933 on the items neither arm fixes (Fisher OR 26.4, $p = 1.4 \times 10^{-8}$).

A *read-only* linear selector recovers the lift (ARC selector 0.943 across the full n=400 evaluation set, not only base-wrong items) without touching generation, improving on the adapter read-only at the same measured cost — the read leg of a read≠write dissociation (the selector capability itself is prior work, NoVo/CORAL, used here as evidence, not claimed as new). We likewise build on, rather than lead on, the established fact that cold-MC scoring under a chat template is a lossy readout of what the model knows (Confidence-Manifold, Inside-Out, KAPPA). The effect, its off-axis geometry and the causal decomposition all replicate cross-family (Qwen 14B: same-sign lift on all four tasks, three clearly and Winogrande marginal at n=400; whitened $\cos(\theta_{\text{mc}}, v_{\text{inf}})$ −0.002, and the lift again carried by the perpendicular component, not by the parallel one). Nor is the effect an artifact of saturated four-option benchmarks: on MMLU-Pro — ten options, base cold 0.2610 — the same adapter lifts scoring +12.20 pp and again fails to transfer. So what tracks whether the exploit fires is not benchmark difficulty: both benchmarks are hard and unsaturated, and they differ in whether the adapter's training distribution covers them — only the covered one lifts, a protocol-bound inference: benchmark identity carries format, domain and option count with it (Section 4.7). Practically, this is a caution for the contrastive-PEFT-for-multiple-choice program, drawn from one adapter of that family rather than from its published recipes: a cold-MC lift of this size can be an off-axis, non-transferring readout edit, beaten on accuracy *and* on measured latency by the most trivial baseline there is, simply asking the base to emit the answer letter. That last comparison puts the base in a lettered-MCQ format, a *different readout* from the cold-MC option-continuation scoring the adapter operates on, so it is not a like-for-like scoring comparison: it contrasts two readouts of the *same* base, which is the point rather than a confound. On measured cost (median per-item latency, batch 1) that baseline is 0.970 on base-wrong items against the adapter's 0.758 — 0.980 against 0.855 across all 400 items — at about half the adapter's measured latency as we implement the two (181 ms vs 354 ms; our scoring harness runs the ~4 option continuations as separate batch-1 forwards, and we did not time a variant that batches them), and it ties a self-consistency reasoning baseline that costs ~150× more. A benchmark gain of this kind should not be read as a capability gain. This is a mechanistic study of one observable, not an architecture.

1 Introduction

Contrastive-correctness parameter-efficient fine-tuning — the family that learns per-head additive offsets from contrastive pairs of correct and incorrect answers, of which LoFiT [65] and inference-time intervention (ITI) [33] are representative — is an attractive way to raise a model's multiple-choice score at near-zero parameter cost. Applied to a multiple-choice discrimination objective on a frontier 31B instruction model (Gemma 4 31B IT), it does exactly that: cold multiple-choice scoring on ARC-Challenge rises by +35.0 accuracy points and on TruthfulQA mc1 by +37.25, with gains on HellaSwag (+17.0) and Winogrande (+19.0).

A benchmark movement of this size invites the conclusion that the model has become better — that the adapter has *sharpened the model's judgment*. It has not. The lift does not transfer to generation (gsm8k −24.2 ± 7.3 pp over three independently trained seeds). On the items it fixes, the base model already reaches the right answer by reasoning (124/124 under chain-of-thought). A read-only linear selector on the same activations recovers the lift without touching the weights or the decoder. And the most trivial baseline imaginable beats it, and does so for *less* measured latency: asked to emit the answer letter in a single forward, no chain of thought, the base recovers 0.970 of the base-wrong items against the adapter's 0.758, at about half the latency of our scoring implementation — while tying a reasoning baseline that costs some 150× more. The score moved; measurable capability did not. The comparison behind that last contrast is between two readouts of the same base, not a like-for-like scoring test (Section 5.4).

This paper asks what that lift *is*, and answers with a quantified causal attribution rather than a new method: most of it is what non-contrastive supervised fine-tuning in the same multiple-choice format already buys, and it re-ranks a lossy readout rather than adding capability. The contrastive adapter's own write lives in a learned direction that is off-axis to the correctness geometry measured at the 48 heads it intervenes on.

Much of the lift is inducible without the contrastive objective. If the lift were a specific property of *contrastive-correctness* training, an ordinary, non-contrastive supervised fine-tune should not reproduce it. It largely does — and how much it reproduces turns out to depend on two things, capacity and task, which is why we run the control at both ends of the capacity range.

At *identical* capacity — the same 48 head-pairs and the same 12,336 trainable parameters as the adapter, the same corpus and steps, ordinary cross-entropy on the correct answer with no term against the incorrect one, three seeds — the control recovers **36.2**% of the ARC lift (the 5.0 we quote alongside it is an sd over those seeds, not an item-sampling interval). With 1,800× to 58,000× more parameters, a non-contrastive Align-LoRA [30] recovers about **62%**, flat across a rank sweep (r8/32/64/256: 59/63/60/65; the slope's confidence interval includes zero, so within that grid the residual is not a gap more rank closes). A best-effort r512 flank at ~117,000× does not extend the plateau — it recovers 0.22 at lr1e-4 and falls below base at lr2e-4 — so more capacity past r256 buys nothing here.

Read correctly, these matched controls *are* an attribution method, and the fractions are the result, not a deflation. They span a series — a norm-matched random direction ≈0% → same-parameterization control ~37% → higher-capacity generic SFT ~62% (paired-bootstrap fraction-CI [51%, 73%]) → the contrastive adapter 100% — that places much of the lift on fine-tuning *in general* rather than on the contrastive objective, without letting us name one number as "the" generic fraction: it moves with capacity. The task dependence is sharper still. The same three same-parameterization seeds recover 79.5% of the Winogrande lift and 69.5% of HellaSwag's but 0.9% — nothing — of TruthfulQA-mc1's, even though TruthfulQA is in the control's own training corpus. These are each task's headline metric, and the caveat travels with the fractions themselves, not only with their ordering: the four lifts are robust to the choice of metric, the recovered *fractions* are not — under raw accuracy HellaSwag's 69.5% reads 8.9% and the same-parameterization control's 36.2% on ARC reads 26.9% — so every fraction in this paper is an acc_norm statement, and no mechanistic claim rests on one or on their cross-task ordering (Section 6). Read as a mechanism, raising the correct option's log-probability is enough where the task scores the plausibility of a continuation, and not enough where a plausible distractor has to be beaten. The strongest on-axis reference we can build (a Fisher-LDA direction W_{know}) sits at 6.4%, but its interval includes zero (exact McNemar $p = 0.136$), so we count it with the floor rather than as a distinct rung. We report these as the best recovery observed under the generic-SFT family we tested rather than as a bound on generic fine-tuning: within each rank/LR/epoch grid the fraction is flat (the r512 flank recovers *less*, 0.22, or breaks below base, so it does not raise the ~62%), yet across capacity the fraction plainly moves, from ~37% to ~62%. What a different architecture, placement or search budget would recover, we do not know.

The clean test for the higher-capacity arm is ARC; TruthfulQA is excluded there because its matched corpus leaks into the evaluation (held-out, that control collapses from 92% to 30% recovery — the signature of memorizing train-on-eval — while the contrastive adapter still generalizes held-out at 0.83).

Neighboring work establishes the recognition–production gap qualitatively (Section 2); to our knowledge no prior work runs a *matched* control and reports the recovered *fraction*, which is the load-bearing novelty.

The recovery direction is off-axis to correctness — off-axis of degree, and established causally. Where the recovered lift lives relative to the model's own correctness geometry is not a matter of inference from weights but of a direct causal test. Pushing an offset *perpendicular* to the functional correctness direction recovers ∼**100%** of the lift; pushing it *parallel* recovers **0%**; and a norm-matched random direction inside the same perpendicular complement recovers ∼**0%** across three seeds. Of these three arms, the perpendicular one is the least informative: because the offset's cosine with the correctness direction is already ≈ -0.003, the perpendicular component *is* essentially the whole offset, so its ∼100% is close to a restatement (Section 4.4). The weight falls on the other two — an on-axis push at the same scale does nothing, and an equal-norm random direction in the same subspace does nothing either.

Two facts follow at once. The lift is genuinely off the measured linear correctness axes — an on-axis push does nothing, and we mean by this the strongest 1-D linear correctness direction we can build *at each intervened head*, not the whole correctness manifold (Section 3.3) — yet the *learned* perpendicular direction is privileged, since an equal-norm random perpendicular direction does not reproduce it. This closes the standard objection that an off-axis direction is unidentifiable because *any* norm-matched perturbation would work: it is closed on the harness side (the same machinery that yields 100% for the perpendicular push yields only 6.4% for the on-axis reference, so the on-axis failure is real, not an injection artifact) and on the write side (the learned perpendicular push vs the shuffle control).

Comparing the net activation effect of each method, the geometry is a *method-dependent spectrum*, not a single fact: both the contrastive adapter and generic SFT are mostly off-axis (cosines 0.088 and 0.199 with the correctness direction), but generic SFT is 2.3× more on-axis.

We apply a caution to ourselves here — a zero-cost weight-space preview reported "off-axis" (ratio ∼1.05) and the clean activation experiment refuted it (ratio 3.93); we report only the clean experiment.

Cold-MC scoring under a chat template under-expresses what the model knows — established setup we build on. That cold multiple-choice scoring under a chat template is a lossy readout of a model's knowledge is, by now, documented across model families with causal validation: a correctness probe reaches 0.80–0.97 AUC across eleven models, and on the subset where output-based methods are measured against it they reach only 0.44–0.64 [14], the internal-vs-external knowledge gap averages 40% [22], and the residual stream carries distinct knowledge and prediction subspaces [44]. We take this as setup, not as our contribution, and confirm it holds in our channel with a per-item arbiter.

When we take the 124 ARC items that *only* the adapter fixes in cold scoring and ask the base model to answer them by generating a chain of thought, the base solves 124/124 (1.000) — indistinguishable from its CoT accuracy on items it already scores correctly cold (200/202 = 0.990). What makes that informative rather than circular is the other side of the same run: on the items the adapter *fails* to fix, the base solves only 41/47 (0.872, $p = 1.4 \times 10^{-4}$). The arbiter is therefore tracking which items the adapter moves, not restating that ARC is easy under chain-of-thought. The lift is not new capability; it is a re-ranking of cold scores toward answers the base can reason its way to but that no-reasoning scoring misranks.

This is why the lift does not transfer to generation (gsm8k loss in every condition), and the no-transfer is not an artifact of ARC being easy: on GPQA, where the base must genuinely reason and does (CoT accuracy 0.7071, well above chance — a floor, since 31/198 traces did not parse and are counted wrong), the cold-MC lift is itself ∼0 (base 0.4747 → adapter 0.4697, CI crossing zero), so a scoring shortcut has no generative headroom to add even out-of-task.

(How the readout misranks on ARC — confident burial of the gold under fluent surface forms — and how far the present-but-swamped signal survives, we characterize in Section 5.1; here it is the established backdrop against which the *attribution* is the result.)

A read-only selector recovers the lift without the pathologies — the read leg of read≠write, on established setup. Because the correctness signal is present in the activations and merely under-expressed in the cold output scores, it can be read off directly, with no weight edit at all. That a read-only probe or readout can *select* MC answers and recover an accuracy lift is itself established — NoVo selects answers by voting over truth-correlated attention-head norms (+19 points on TruthfulQA-MC1) [28], and CORAL re-scores per-option probabilities with a frozen correctness probe (+14% on ARC-Challenge/HellaSwag) [40] — so we treat the read-only selector as setup, not as a new method.

Our use of it is as the *read leg* of a read≠write dissociation: a read-only linear selector on the same per-head activations the adapter writes to recovers the ARC lift (selector accuracy **0.943**, three PCA/fit seeds; fixes 181 base-wrong, breaks 5 base-right, exact McNemar $p = 3.7 \times 10^{-47}$) *without touching generation*, so it carries none of the adapter's deployment costs (no generation loss, no retrieval spillover) while scoring higher than the *adapter* at the same measured latency. Unlike NoVo/CORAL — which present such a selector as a calibration/accuracy *fix* — we use it as causal evidence that the lift the adapter writes off-axis is read-recoverable on-axis without the write's damage.

We are precise: 0.943 exceeding the adapter's 0.853 is *not* "the probe beats the adapter as capability" (different information sets — 12288 activation dimensions vs collapsed output logits); it is a read-side statement that the cold-MC channel discards recoverable signal. Its feature set is held out by item but not by feature (the 48 cells are selected on the training split), so cross-benchmark generalization is the natural strengthening.

Two failure modes, reported separately. The two headline tasks do not fail the readout the same way, and we do not blend them. ARC, HellaSwag, and Winogrande carry the question-conditional readout story (the ARC lift retains 78% under a PMI / decision-conditional control; the choices-only lift is only +0.0375, CI crossing zero).

TruthfulQA is a second, *answer-prior / surface-form Goodhart* mode [21, 24, 39]: the choices-only lift is +0.2225 and the adapter selects the correct option 0.5725 of the time with no question at all. The TruthfulQA gain does generalize held-out (Δ_{UNSEEN} mc1 +37.8 ≈ Δ_{SEEN} +36.9, so it is not memorization — and a generic Align control collapses held-out to +7.9, confirming the contrastive lift is genuine), but we read it strictly as *answer-prior selection on the mc1 metric*, never as a truthfulness gain and never as evidence for the off-axis mechanism, which is anchored on ARC.

This is a mechanistic study of an observable effect, not an architecture. We make no claim about routing as a contribution and no claim about a redesigned model. The object of study is one measurable phenomenon: what a frontier model's cold-MC scoring misses, and what contrastive-correctness PEFT actually does when it recovers part of it. Where we describe a routed system at all, it is strictly a mitigation vehicle that sends generation-face traffic to the unmodified base; it is not advanced as a result.

Scope, discipline, and what is not yet validated. We characterize this on Gemma 4 31B IT and corroborate the discrimination signature cross-family and cross-scale where artifacts exist. We hold ourselves to a verify-before-claim discipline: every number traces to its source run or is marked, and we freeze no prose on un-executed evaluations.

Concretely, the headline scoring/gen magnitudes are pre-registered with kill-rules (in the sense made precise in Section 6). The anti-floor 5-shot rule has executed and the lift survived. The gen-face canonical gsm8k sweep (R6) has now run: −25 pp, flat across the generation budget, so the loss is mechanistic, not a clipped chain of thought; two further independently trained seeds put the loss at −24.2 ± 7.3 pp, so the sign replicates while the magnitude is seed-dependent and we cite it with that spread. The contrastive-specific residual is bounded by the controls we ran, not by a proof: the Align-LoRA fairness flank does not raise the ~62% within its grid (r512 recovers 0.22 / breaks below base), while at matched capacity the generic fraction is lower (~37%) — so the residual is grid-relative, and we say so. The earlier V7-mc end-of-turn figure (60→4%) is **retracted**: it and the analogous D.1 numbers came from a tokenizer bug that scored termination on the wrong turn-stop token, and the one clean re-measure shows no collapse (Section 5.5). The D.1 open-gen degradation is real but modest (social 96→76%, creative 68→12%), adapter-dependent, and off the V7-mc headline. A temperature/null-space account of the offset is reported refuted (Section 5.6), now confirmed by causal patching, not merely direct-logit attribution. We mark each in place rather than leaning on it.

Contributions. The object of this paper is a dissociation — a large multiple-choice score gain that is not a capability gain — and the contribution is the causal attribution that explains it, together with the deployment pathologies the recovery introduces.

- **Inducibility-as-attribution.** Two non-contrastive controls bracket the objective's contribution. A *same-parameterization* control (identical 48 head-pairs and 12,336 parameters, CE-plain, three seeds) recovers **36.2**% of the ARC lift (sd 5.0 over the three seeds, not an item-sampling interval); a *higher-capacity* Align-LoRA (1,800×–58,000× more parameters) recovers ~62% — paired-bootstrap fraction-CI **[51%, 73%]**, flat across rank r8–256, and a larger r512 flank (~117,000×) does not raise it. Recovery is also strongly task-dependent (Winogrande 79.5% / HellaSwag 69.5% / ARC 36.2% / TruthfulQA **0.9**% at matched capacity). Read as an *attribution spectrum* — random ≈0% → same-parameterization ~37% → higher-capacity SFT ~62% → contrastive 100%, with the on-axis reference (W_{know}) at 6.4% indistinguishable from zero ($p = 0.136$) and therefore counted with the floor — much of the lift is what non-contrastive SFT buys back, and how much depends on capacity and task rather than being one constant. Every control we ran drops the contrastive term while keeping supervision on correct-option continuations in multiple-choice format, so the class the attribution names is that one, not fine-tuning in general. To our knowledge no prior work runs matched non-contrastive controls to *attribute* an MC lift this way and report the fractions. The interval is wide, and we read the claim at its resolution: most, not a specific percentage.
- **A causal off-axis-of-degree geometry, non-identifiability closed both sides.** A parallel push recovers 0% of the lift and a norm-matched random perpendicular push ~0% (three seeds) — so the off-axis direction is learned and privileged, not any perturbation of the right size. The perpendicular push recovers ~100%, but since the offset is already nearly all-perpendicular that arm is close to tautological (Section 4.4); the parallel and shuffle arms carry the identification. The net-effect comparison is a method-dependent spectrum (contrastive cosine 0.088 vs generic SFT 0.199), and the lift has no layer locus (it scales with offset magnitude, correlation 0.985); a training-time on-axis penalty does not rescue transfer (see Section 6), adding *necessity* to this sufficiency.
- **No transfer, on an established lossy-readout setup.** Cold-MC scoring under a chat template under-expresses knowledge the model holds — established [14, 22, 44] — and we confirm it (the fixed items are CoT-accessible, 124/124 on ARC, against 41/47 on the items the adapter does *not* fix, $p = 1.4 \times 10^{-4}$) and show the no-transfer holds *out-of-task* (GPQA cold-MC ~0 at base CoT 0.70, single-run and a floor at 31/198 unparsed) and on generation (gsm8k −25 pp). That generation cost is coupled to *how* the offset is injected, not to the lift (a generic SFT recovers the lift but preserves gsm8k 0.96→0.95); a previously-claimed end-of-turn collapse is **retracted** as an eos-token artifact.
- **The read leg of read≠write (built on established setup, not a new method).** A read-only linear selector on the same per-head activations recovers the ARC lift (accuracy 0.943; McNemar $p = 3.7 \times 10^{-47}$) without touching generation — improving on the *adapter* read-only at the same measured cost, though both a reasoning test-time-compute baseline and a single-forward letter emission of the base outscore it, the latter at half its measured latency — which is why we hold the selector to an evidentiary role

(Section 5.4). That a read-only probe-selector recovers an MC lift is established [28, 40]; our contribution is its *use* as the read leg of a read≠write dissociation auditing an off-axis write neither studies.

- **Two clean negatives, briefly.** The offset is *not* a temperature/null-space mechanism (refuted causally) and *not* a token-frequency edit (frequency control closed); and the lift is task-heterogeneous — ARC carries the question-conditional mechanism while *TruthfulQA is a second, answer-prior failure mode* (choices-only +0.2225), generalizing held-out but reported separately, never as the off-axis mechanism.

Scope at a glance. So the reader carries the boundaries once rather than re-encountering them piecewise: everything is single-model (Gemma 4 31B IT), and the re-train replication is partial rather than absent — the fine-grained causal arms (arbiter, inducibility control, recovery spectrum) are computed on the canonical adapter, while the headline lift, the net-effect geometry and the gsm8k loss are replicated across three independently trained adapters (Table 12). That mechanistic chain — arbiter, calibration, PMI-residual, probe, selector, geometry — is convergent analysis developed *after* the effect was discovered, on one fixed 400-item slice, not an independent confirmatory replication (the arbiter, the probe and the selector have since been re-run outside that slice, Section 5.9). The *effect* and its off-axis *geometry* are further replicated on Qwen 14B; the mechanism claims are anchored on ARC (TruthfulQA is a separate answer-prior mode; HellaSwag/Winogrande corroborate); every claim is protocol-bound to cold-MC scoring under a chat template (without the template there is no lift to study, Section 4.1); n=400 denotes the first 400 items of each lm-eval split, with a full-split run confirming every headline delta (Section 4.1); the headline anchors are low-CoT-headroom benchmarks by design (Section 6); and the routed system is a mitigation vehicle, not a contribution.

Notation. Full definitions in Section 3.6. **V7-mc** — the contrastive-correctness adapter under study (48 attention heads, additive per-head offsets). $\boldsymbol{\theta}_{\mathbf{mc}}$ — its learned offset (aggregate write). $\boldsymbol{v}_{\mathbf{inf}}$ — the per-head functional correctness direction (mass-mean estimate); $\boldsymbol{W}_{\mathbf{know}}$ — the whitened Fisher-LDA on-axis reference. $\mathbf{off}_{\parallel}$ / $\mathbf{off}_{\perp}$ — the offset's components parallel / perpendicular to v_{inf}. **Cold-MC scoring** (= *scoring-face*) — log-likelihood ranking of options under the chat template, no generation; **generation-face** — evaluation of produced text. **Recovery fraction** — (arm − base)/(adapter − base) on the task's headline metric (Section 3.6).

2 Related work

2.1 The recognition–production gap

A robust body of work establishes that what a model can *recognize* (rank correctly among options, or encode in its hidden states) outruns what it can *produce* (generate correctly), and that this gap is real rather than an evaluation artifact. Probing work shows models often encode the correct answer in their representations while generating an incorrect one [41]; the original inference-time-intervention work reports a large gap between a truthfulness probe's accuracy and the model's generation accuracy [33]; and recent evaluation work argues that answer-matching on free generation is a stricter, more faithful test than multiple-choice scoring [12]. Two recent results establish the specific, causal-and-geometric form of the gap that we take as setup: a correctness probe reaches 0.80–0.97 AUC across eleven models, and on the subset where output-based readouts are measured against it those readouts (P(True), token entropy, semantic entropy) reach only 0.44–0.64, with the direction validated causally by activation steering — it shifts hallucination rates by 9.1 pp on Qwen2-7B, while norm-matched random and orthogonal directions move them by 0.8 and 0.3 points and neither reaches significance [14]; and a formal internal-vs-external knowledge measure finds an average 40% hidden-knowledge gap across open-weight models [22]. Two scope conditions travel with the first of these and we state them because our own setup inherits them: its probes are teacher-forced rather than read off cold multiple-choice scoring, and its authors report the internal advantage as regime-specific — large on adversarial misconceptions, closing on standard QA — so what it establishes for us is that correctness is linearly decodable where an output-level readout is not, not a general ranking of internal over external estimators. We take this lossy-readout fact as *established setup and claim no priority on it* — it is the backdrop, not our contribution. Our per-item arbiter (the base solves under chain-of-thought 124/124 of the items the adapter fixes in cold scoring) confirms the gap in our channel. It also shows that between *faces* the asymmetry runs the other way — generation succeeds precisely where the cold discriminative readout fails — which is the direction the generative-AI paradox reports: generative capability is not contingent upon, and can therefore exceed, the ability to understand the same kind of output, with models outperforming humans at generation while falling short of them on measures of understanding [60]. Which capacity outruns which is thus a question about the pair of channels being compared, not a fact about models in general. Our contribution is the *converse* direction and a *quantified causal attribution*: we show that a cheap fine-tune recovers most of the cold-scoring deficit along a learned, off-axis direction, and that the recovery does not transfer back to generation — a property of what the adapter does, not only of the gap's existence. Closer to our read-only selector specifically, the gap has already been *operationalized* as an answer-selection method: an activation-derived readout selects the multiple-choice answer and recovers an accuracy lift with no weight edit — by attention-head-norm voting [28] or a frozen correctness probe re-scoring per-option probabilities [40]. We claim no priority on that capability either; we use it only as the read leg of a read≠write dissociation (Section 5.4), where the novelty is the contrast with an off-axis *write* these works do not study. A complementary line *does* study a causal write, but in the generation face rather than our cold-scoring channel: instruction-tuned models often fix their answer before reasoning, and a linear probe on the pre-CoT last-token activation decodes that answer at ~0.9 AUC while steering along the probe direction flips the produced answer in over 50% of cases, exceeding orthogonal baselines [18]. This sharpens rather than blurs our claim: in that decision/generation face an answer direction is genuinely writable, whereas in our cold-MC scoring channel the correctness direction is decodable (0.955, Section 5.4) yet the on-axis write fails (W_{know} 6.4%, Section 4.4) and the off-axis write that does lift never transfers back to generation (Section 5.1) — so our read≠write dissociation is *channel-specific*, not a claim that answer directions are never writable. Its two steering failure modes when a wrong answer is induced — non-entailment and confabulation, i.e. reasoning from a false belief — also parallel the reasoning-chain breakage our contrastive injection produces on gsm8k (Section 5.5).

2.2 Multiple-choice scoring of instruction-tuned models is an unreliable readout

That cold multiple-choice scoring is a poor proxy for an instruction-tuned model's behavior is established, and we claim no priority on it. First-token / log-likelihood option scores are severely misaligned with the model's actual text answer — mismatch rates exceeding 60%, worst on models heavily fine-tuned on conversational or safety data [54] — and the text answer is the more robust quantity when the two disagree [53]. Cold MC scoring is further sensitive to option ordering and symbol identity [45, 68], is poorly calibrated without correction [67], and instruction tuning polarizes the answer distribution [19]. One recent result is especially close to our framing: the chat template itself inflates a model's confidence in "its own" answer in a way removable by a parameter-free reframing [48]. A second is close in mechanism but sits on the opposite population, and we flag the mismatch rather than borrow the conclusion: on *intermediate pretraining checkpoints* — small, *not* instruction-tuned models — emitting the option symbol is a skill separable from knowing the answer, and fine-tuning on multiple-choice format first disentangles the evaluation of knowledge from the emerging ability to follow the format [11]. Our model is instruction-tuned, i.e. the population that work treats as already past the difficulty, so it does not license the claim that our base cannot emit the symbol

— and indeed ours can, which is why the one-forward letter-emission baseline is so strong (Section 5.4). What transfers is the weaker and still useful point: symbol emission and knowledge are separable enough that a protocol can score one while intending the other. Selection bias is moreover introduced by supervised fine-tuning itself, tracing to weak multiple-choice symbol binding — the A/B/C/D symbol–content association — which strengthening during SFT both reduces and turns into an accuracy gain [64]; this bears directly on our matched-SFT control and on the trivial letter-emission baseline we find outscores the adapter (Section 5.4). We build directly on this literature — the protocol (cold MC under a chat template) is a lossy readout, which we treat as *established setup rather than as our thesis* — and our contribution is past the diagnosis: we quantify how much of the deficit a generic fine-tune recovers (~62% on ARC, a matched-control attribution), locate that recovery geometrically (off-axis of degree, established causally), and show it does not transfer. Critically, and unlike a "the lift is a formatting artifact" reading, the recovered lift is *real* signal: it survives a same-scaffold PMI decomposition (ARC retains 78%), so the readout is genuinely leaving knowledge on the table, not merely mis-normalizing it.

2.3 The mirror in compression: the benchmark illusion

The closest single neighbor in the same channel (log-likelihood scoring vs greedy generation) comes from model compression. Pruned models can pass multiple-choice evaluation while failing to answer the same question in open generation; the correct answer is usually not erased but *demoted* in rank, reappearing under beam search, sampling, or one in-context example [59]. That is our phenomenon's mirror: there, MC scoring *overstates* usable capability (the answer is demoted but MC-recoverable); here, cold MC scoring *understates* it (the answer is CoT-accessible but cold-misranked) and a fine-tune *promotes* it in the MC channel without helping generation. The shared lesson — multiple-choice scores are not generation capability — is established; the constructive, opposite-signed instance, and its off-axis mechanism, are ours.

2.4 The knowledge–prediction gap and KAPPA (our foil, not our competitor)

Most directly adjacent is work on the knowledge–prediction gap in multiple-choice questions: models fail MCQs whose answers they encode internally, and KAPPA identifies distinct knowledge and prediction subspaces in the residual stream and aligns them with a closed-form, inference-time intervention [44]. We position KAPPA as a foil rather than a competitor, and explicitly *not* as a head-to-head baseline we beat. KAPPA operates where the correct option is present and decodable in the residual (options-present MCQ); our effect lives in cold options-absent text-continuation scoring, where we find no evidence that the precondition holds — a faithful k-class knowledge probe in our channel decodes the gold option at 0.317 against chance 0.25, while the same probe for the greedy option decodes at 0.542. We state this as a *limit on power rather than a demonstration of absence*, because the design cannot separate the two: the gold figure rests on a 120-item dev split, whose Wilson 95% interval $[0.240, 0.404]$ contains chance *and* contains improvements larger than 10 pp, so an equivalence test at that margin does not pass; and the point itself is a maximum over a 40-cell (layer $\times$ regularization) grid, which biases it upward exactly where it is being read as a null. What we can say is that KAPPA's mechanism is not *detectably* instantiated in our channel at this resolution; the relationship is a channel mismatch, and we make no claim of out-performing it. One apparent tension deserves a sentence here, because a reader who reaches Section 5.4 will otherwise trip on it: our read-only selector decodes the gold at **0.943**, far above this 0.317. The two probes read *different channels*. The 0.317 is a k-class probe on the *prompt-final residual* — a single vector with the options in context, which is precisely the channel KAPPA operates on — whereas the 0.943 selector reads the *per-head activations of each option continuation* (four separate forwards, arg-max over the per-option score). KAPPA's precondition fails in our channel, and the selector does not restore it; it succeeds by reading somewhere else. We further reconcile KAPPA's reported free-form *generalization* with our no-transfer result (Section 5.5), since the two superficially conflict: KAPPA's own free-form evaluation carries its gain almost entirely on a single answer-prior-shaped task (TruthfulQA +2.5) while it *regresses* on reasoning generation (GSM8K 91.6→90.7) and is flat on BBQ-Age (+0.2) — so KAPPA, like our lift, does not transfer to reasoning generation, which corroborates rather than contradicts our finding.

2.5 Answer priors and surface-form competition

Our task-heterogeneity finding — that the TruthfulQA half of the lift is largely an answer-prior / surface-form effect — rests on a well-developed line. Surface-form competition shows that highest-probability scoring is biased by how answers are phrased, motivating PMI-style domain-conditional normalization [29]; choices-only probing shows models answer MCQs above chance with the question removed [4, 5]; and choice-sensitivity admits parameter-free fixes [13]. We use the same PMI/decision-conditional and choices-only controls to separate the two tasks: ARC is question-conditional and carries the mechanism, whereas TruthfulQA is largely answer-prior, which is why we report it held-out as evidence the lift is real and never as evidence for the off-axis mechanism.

2.6 The capability tax of interventions, and the specificity question

Adjacent work shows that adapting a model along one axis taxes others: knowledge injection under PEFT erodes reasoning, with the reasoning-critical information distributed across the singular spectrum rather than localized [8], and supervised fine-tuning degrades pretrained capabilities because its targets diverge from the model's autoregressive distribution [37]. Inference-time steering can be applied selectively, choosing when and how much to intervene [26]; and steering vectors that cleanly control a target property in multiple-choice or toy settings degrade in free-form generation, inducing degenerate repetition and factual hallucination at the strengths control requires [9] — a control–quality trade-off that mirrors the creative degeneration our contrastive write produces (Section 5.5). These motivate our deployment-pathology results, but they also raise the sharpest reviewer objection to an off-axis claim: that *any* norm-matched perturbation works, so an off-axis direction is not identifiable [52], and even random directions can move behavior [32]. We answer this directly, and with the same pair of controls that work runs on its own steering result — the learned direction against a norm-matched random one and against an orthogonal one, neither of which reaches significance there [14]: a norm-matched random offset recovers ≈0% of our lift (so the direction is learned, not arbitrary), the strongest on-axis reference recovers 6.4%, and the trained offset has an optimal magnitude (an inverted-U strength sweep), which a generic perturbation would not. We also note that reported specificity failures of activation directions are cross-model; within-model — our setting — directions remain specific [51], so that result does not subsume our control. A related limit on the *write* side sharpens what an off-axis control claim must contend with: a dual-stance evaluation of a sycophancy-reduction direction finds that although sycophantic and factual agreement occupy geometrically *distinct* subspaces, a single steering direction projects equally onto both and cannot differentially target either — a case where representations readable from activations are not separately writable through them [10]. That is the writability gap our read≠write dissociation (Sections 4.4 and 5.4) makes explicit from the opposite side: there a writable direction fails to *separate* two readable targets, whereas here the on-axis correctness direction is readable yet fails to *drive* the lift (W_{know} 6.4%) while an off-axis one does. A distinct mechanistic hypothesis for any logit-level intervention is temperature regulation — confidence-regulation / entropy neurons write into the unembedding null space to rescale the logit temperature without re-ordering the argmax [49] — which we test and rule out for our offset in Section 5.6.

2.7 PEFT, representation editing, and the fine-tuning baseline

Our adapter is from the localized-fine-tuning / representation-intervention family: LoFiT learns additive per-head offset vectors on a sparse set of attention heads [65], a descendant of inference-time intervention [33], and related to representation fine-tuning [62]; concept- and feature-level steering benchmarks find that fine-tuning and prompting often outperform sparse-feature methods [63]. A recent principled account establishes a first-order equivalence between activation-space interventions and weight-space updates and finds the two play distinct, complementary functional roles [1]; this is why we compare methods by their *net effect on activations* rather than by the weight-space offset, which we show is a misleading proxy (Section 4.4). The decisive baseline our inducibility result requires is ordinary supervised fine-tuning: prior domain-adaptation work establishes SFT as a strong, cost-effective default for multiple-choice accuracy while degrading open-ended generation [6]. Our matched Align-LoRA control instantiates exactly this baseline, and our finding — that it recovers ~62% of the contrastive lift — places most of the effect on fine-tuning in general rather than on the contrastive objective.

3 Method

3.1 Model and contrastive-correctness adapter

All results are on Gemma 4 31B IT, run in bf16 — the Hugging Face checkpoint `google/gemma-4-31B-it`, a `Gemma4ForConditionalGeneration` with 60 decoder layers, d_{model} 5376, 32 query heads and head dimension 256, evaluated under `transformers` 5.8.0 and `lm-eval` 0.4.12 [50].

Which snapshot, and why it is two. Our harness resolves the model by `revision=main` and lm-eval leaves `model_sha` empty, so no result file carries an explicit pin. The snapshot is nonetheless recoverable for most runs, because the harness records the model path verbatim and on our machines that path is a Hugging Face cache directory containing the snapshot hash; we extract it from every artifact and release the resulting map (`scripts/model_provenance.py`, `runs/provenance/`). The exception is the rented-GPU campaigns, which read a pre-staged copy under a flat path (`/workspace/models/`) that carries no hash — the canonical June trio among them. Those we date rather than resolve, which is weaker evidence and we mark it as such. Upstream published a revised checkpoint on 2026-07-20, so `main` moved mid-project and our runs straddle it. The June and early-July campaigns — the arbiter, the inducibility control, the recovery spectrum, the on-axis reference, and, by date rather than by hash, the headline trio (run 2026-06-10, four weeks before the newer template existed) — ran on `3548789868c5356dbf307c98e6f609007b82b3eb`; the late-July campaigns — the full-split scoring replication of Section 5.9, the anti-axis arms and the dense scale sweep — ran on `842da3794eaa0b77d5f08bae87a17459d91ff475`, which is `main` at the time of writing. We diffed the two snapshots file by file: `config.json` and `tokenizer.json` are byte-identical, so the architecture and the vocabulary did not change, while `chat_template.jinja` and `tokenizer_config.json` differ — the newer template is dated 2026-07-09 and its own header describes fixes to tool-calling loops and turn closures. Because every protocol here is template-bound (Section 4.1), we flag this rather than bury it. It cuts in the reassuring direction: the headline scoring lift is observed under *both* templates — the n=400 campaign under the older one and the n=1172 full-split replication (+36.26 pp) under the newer — so the central effect survives the change rather than depending on it. The remaining arms were each run under one snapshot only, and we do not claim more than that. We accordingly treat snapshot-level reproducibility as a limitation of these runs, and the released harness now stamps the resolved snapshot and a hash of the applied chat template into every result file.

The cross-family replication (Section 6) uses `Qwen/Qwen2.5-14B-Instruct` at revision `cf98f3b3bbb457ad`

`9e2bb7baf9a0125b6b88caa8` (`Qwen2ForCausalLM`, 48 layers, d_{model} 5120); that repository has been unmodified since September 2024, well before these runs, so `main` and the pinned revision coincide. The discrimination adapter (referred to as V7-mc) is a contrastive-correctness PEFT in the LoFiT/ITI family: it learns additive per-head offset vectors on a sparse set of attention heads from contrastive pairs of correct and incorrect answers under a multiple-choice discrimination objective. Concretely, each pair is (question, correct option, first-listed incorrect option), built from the TruthfulQA mc1 validation split (seed-0 permutation, 80% train side) and the native ARC-Challenge train split; training minimizes a margin loss on the summed continuation log-probability under the chat template, $-\log p(\text{correct}) + \gamma \cdot \log p(\text{wrong})$ with $\gamma = 0.5$, for 800 steps, with the deployed checkpoint selected by a validation-accuracy monitor. Two construction details a reader of the code will find, stated here rather than left to be discovered. First, because the HF TruthfulQA mc1 release lists the gold answer at index 0, "first-listed incorrect option" means every TQA pair contrasts the dataset's first two options — a fixed positional split rather than a sample over distractors; this touches no mechanism claim (those are anchored on ARC, where option order carries no gold information) and is consistent with — plausibly contributes to — TruthfulQA's answer-prior character (Section 5.8). Second, the checkpoint-selection monitor was, due to dataset ordering in the validation slice, computed on TruthfulQA items only: checkpoint selection never saw ARC signal, so the ARC lift is not checkpoint-selected. The causal analyses use the 48 heads carrying the adapter's offset (selection criterion in Section 3.6). Those 48 (layer, head) targets are unique (no aliasing) and the offset modulates a sparse, *non-contiguous* set of 12 layers — [30, 31, 32, 33, 34, 36, 37, 38, 39, 40, 49, 55] (deterministic across runs; note the gap at layer 35) — so references below to a "layer ~30–55 span" denote this discrete set's bracket, not a contiguous band.

3.2 Evaluation protocol: scoring-face vs generation-face

We distinguish two protocols and report absolutes only with their protocol attached. **Scoring-face** is cold multiple-choice scoring: log-likelihood ranking of options with the chat template applied and no generation (lm-eval canonical, `--apply-chat-template`, n=400, acc_norm), with identical flags across arms (one documented exception, the random-offset control, in Section 3.5). The n=400 sample is the first 400 items of each task's evaluation split in native order (lm-eval `--limit 400`), not a random draw — identical across arms by construction, and not doing selection work: the R4 full-split run confirms every headline delta within the n=400 confidence interval (Section 4.1, Section 6). This is deliberate — it yields apples-to-apples deltas against the adapter — and it is why our base absolutes (e.g. ARC ≈0.50) are scoring-face floors and must not be compared to a third-party-reported ~91% chain-of-thought [57] figure (the protocol caveat in the Table 1 caption). The same scoring-face protocol — identical log-likelihood option ranking, chat template, and lm-eval canonical flags — is also applied out-of-task to GPQA [46], where the cold-MC lift is itself ~0 (base 0.4747 → adapter 0.4697, −0.5 pp); this cold-MC GPQA measurement is distinct from the generation-face GPQA evaluation described below. **Generation-face** evaluates the produced text: gsm8k [16] under `generate_until` with exact-match; a per-item arbiter that runs base generative chain-of-thought (thinking-on, greedy, k=1) aligned to the cold-MC `doc_ids`, on the items the adapter fixes; and GPQA under thinking-on greedy decoding with the generation budget raised to 8192 tokens (at 2048 the base truncated 96% of generations, producing a spurious 0.15). The per-item arbiter is run across multiple anchors — ARC, GPQA, HellaSwag, and Winogrande — to test whether the fixed items are already CoT-reachable. One caveat is load-bearing here: HellaSwag and Winogrande are not native letter-MCQ tasks, so to run the generative arbiter on them we reformulate each item as a lettered MCQ (HellaSwag: the four continuations → A/B/C/D; Winogrande: the two options → A/B). That reformulation puts the arbiter in a *different* format from the cold-MC scorer (which scores option continuations without letters), so on HellaSwag/Winogrande the arbiter anchors *knowledge* in a letter-MCQ format rather than replicating the cold-MC channel — the same format gap that ARC and GPQA already carry (where the arbiter is nonetheless informative). We therefore read the arbiter as a knowledge-accessibility probe, not as a channel-faithful readout.

3.3 Functional correctness direction and on-axis references

Per attention head we estimate a functional correctness direction v_{inf} (the mass-mean difference between correct and incorrect inference-form activations). The estimation corpus is the inference-family contrastive pairs (ARC-Challenge native train split plus HellaSwag), so on ARC — where every on-axis recovery fraction is anchored — the items that estimate the direction and the test items that measure recovery are disjoint by construction, the same train/test boundary the adapter itself respects (Section 6). To make the on-axis causal test as strong as possible, we also construct W_{know}, a whitened Fisher-LDA correctness direction — the strongest on-axis reference we can build locally, and genuinely distinct from the mass-mean estimate (cos with v_{inf} 0.379). "Strongest" is a construction criterion, not an outcome criterion: the whitened Fisher-LDA direction is the optimal linear correctness discriminant under the class covariance, which the raw mass-mean ignores — it is not chosen for recovering the least lift, and Section 4.4 reports both references' recovery side by side. Because optimality under the class covariance is a population statement and the covariance here is estimated, we also check the two references *as readers* rather than arguing from construction alone: on the base-wrong cold-MC population both separate gold from distractors at or above the level of a probe fitted on that same space (W_{know} 0.9613, v_{inf} 0.9646, fitted probe 0.9545), and paired over the same items the gap between them is -0.0034 with a CI95 of $[-0.0135, +0.0067]$ — so "strongest" is not contradicted by outcome, and neither is it established by it (Section 4.4).

Scope of "off-axis". Both references are *one direction per head*, not a single global axis: the geometry is a family of per-head linear correctness axes, one at each intervened head. Everything we call off-axis is therefore off the measured linear correctness axis at each intervened head, which is the strongest such direction we can build there but not the whole geometry of correctness: correctness information has been reported to occupy a low-dimensional subspace of two to eight directions rather than one, with a causal rank bounded at five under distributed alignment search [14]. Our negative arms show that the learned offset is not carried by the best 1-D discriminant; they do not show that it lies outside every correctness-bearing subspace. Testing k-dimensional subspaces with norm-matched parallel and residual re-injection is the natural follow-up and we do not claim its result here. The on-axis causal control substitutes an offset built along v_{inf} (and along W_{know}), unit direction scaled to the per-head norm of the trained offset across all 48 heads, so that *direction* is the only variable, gated by a sign-safe three-zone control gate with a null-task check (the gate, and the Gate-A reproduction check it presupposes, are defined in Section 3.6).

3.4 Net-effect geometry and the inducibility control

To compare methods fairly in activation space (rather than via the weight-space offset, which we show is a misleading proxy), we measure each method's *net effect on activations* and decompose it against v_{inf}, reporting the off-axis/on-axis split as a ratio relative to a random direction.

The inducibility question — what fraction of the contrastive lift a *non-contrastive* objective recovers — is asked with two controls, because the answer depends on how much capacity the control is given.

(i) Same-parameterization control (the objective, isolated). The identical LoFiT apparatus as the adapter: the same 48 head-pairs from the same selection file, the same 12,336 trainable parameters ($\alpha_h \in \mathbb{R}$, $\theta_h \in \mathbb{R}^{256}$ per head), the same corpus, optimizer steps, batch size, learning rate, sequence length, train/validation split and best-checkpoint rule. The *only* change is the loss: ordinary cross-entropy on the correct continuation, $-\log p(\text{correct})$, with no term pushing down the incorrect option. Three seeds vary the batch-visit order with the data split held fixed, so the three runs see the same examples. Because the intervention space is held exactly fixed, this control attributes the difference to the *objective* and nothing else. That equality is recorded inside the released checkpoints rather than inferred: each offset file carries the training `config`, whose `loss_type` reads `direct` for the adapter and `plain_ce` for the three control seeds, with the same head-selection file, the same 48 (layer, head) pairs in the same order, and the same `top_k`, step count, learning rate and corpus; the 12,336 is counted on the tensors themselves ($\theta \in \mathbb{R}^{48\times256}$ plus $\alpha \in \mathbb{R}^{48}$), not inferred from file size, and the released training log states it independently. One scope note: the adapter's `config` predates the field that records the training seed, so the equality is verified over the fields both sides record — which are all of those that fix the parameterization — and not over that one.

One asymmetry in that control deserves stating, since it touches the arm the attribution rests on. The *objective* is genuinely non-contrastive — the gradient carries no term against the incorrect option — but the *checkpoint-selection rule*, which we held identical to the adapter's precisely to keep the apparatus fixed, is not: validation tracks whether the correct option outscores the incorrect one, so the distractor enters the choice of which step to keep even though it never enters the loss. So "only the loss changed" is exact for the training signal and not for model selection. Two things bound the consequence. The rule only bit in one of the three seeds — in seeds 1 and 2 the selected checkpoint *is* the final one, and only in seed 0 does the best-by-validation step (600) differ from the last (800). And in that seed the rule selects the *worse* ARC checkpoint, which is why the fixed-final variant, which consults no validation signal at all, reports a *higher* recovered fraction (39.7% ± 3.1) than the best-checkpoint figure we quote (36.2% ± 5.0). The contamination therefore deflates this control rather than inflating it, and the residual we attribute to the contrastive objective is, if anything, conservative. We quote the best-checkpoint number as the headline and carry the ~36–40% range throughout.

(ii) Higher-capacity control (ordinary SFT at scale). A non-contrastive Align-LoRA: ordinary supervised fine-tuning on a data- and step-matched corpus, swept over LoRA rank (r8/32/64/256) and scored through the same three-zone control gate. This arm is *not* capacity-matched — LoRA on the text decoder's q/k/v/o projections across 60 layers carries ~22M trainable parameters at r8 and ~1.44B at r512, i.e. 1,800× to 117,000× the adapter's 12,336 — and we report it as such. The recovery fraction we quote (~62%) is the r8–r256 plateau, i.e. 1,800×–58,000×; the r512 point is a separate best-effort flank and is always reported as one. The mismatch is conservative for the inducibility claim (a control with far more capacity that still recovers only part of the lift), but it is the reason the same-parameterization arm exists: only (i) licenses reading the residual as contrastive-specific *at equal capacity*.

Both recovery fractions, the matched corpus, and the gate are defined in Section 3.6.

3.5 Additional controls

We further run: a norm-matched random offset (three seeds) to test whether any perturbation of the right size reproduces the lift — this arm is the one exception to the identical-flags statement of Section 3.2, having been scored with a 2048-token context window against 4096 for the canonical trio and the other spectrum arms; task and `--limit` match, and no truncation is reachable at either setting because the longest prompt in this protocol is ≈228 tokens (Section 6), but we state it rather than let "identical flags" cover it; a same-scaffold PMI / decision-conditional decomposition and a choices-only (question-removed) control to separate question-conditional gains from answer priors [29]; layer-band patching (restricting the offset to layer windows) to test for a layer locus; and a within-format difficulty stratification on ARC — ARC-Easy vs ARC-Challenge under the identical cold-MC scoring protocol (acc_norm), stratified by base-wrong items — which holds format and distribution fixed and varies only difficulty, isolating the contribution of difficulty from that of distribution distance.

3.6 Formal definitions

This subsection makes the operational terms used throughout precise — each grounded in the analysis code. The maps from every headline number to the run and the artifact that produced it, so no two item populations are silently conflated, are in Section B.

Capability. Throughout, *capability* is used in a readout-independent sense: accuracy the base attains under *any* readout we measured, at comparable or lower cost. A reader who reserves the term for a fixed readout will read our claim as being about the cold-MC channel specifically — and on that reading the adapter does improve it, which is the dissociation this paper is about rather than an objection to it.

The trained offset θ_{mc}. θ_{mc} denotes the adapter's learned per-head additive offset — the aggregate write direction across the 48 heads (Section 3.1). The geometric and causal decompositions of Section 4.4 ($\mathrm{off}_{\parallel}/\mathrm{off}_{\perp}$, whitened cosines) operate on it.

Recovery fraction. For each anchor task the recovery fraction of an arm is $\mathrm{frac} = (\mathrm{arm}-\mathrm{base})/(\mathrm{v7mc}-\mathrm{base})$, the fraction of the contrastive lift the arm reproduces (`scripts/analyze_r1_align_frac.py:6,99`; returns NaN when $|\mathrm{v7mc} - \mathrm{base}| < 10^{-9}$). The per-task headline metric is fixed in code (`TASK_METRIC`, `:36--41`): **acc_norm** for ARC-Challenge [15] and HellaSwag [66], plain **acc** for Winogrande [47] and TruthfulQA-mc1 [35] (the latter two define no length-normalized variant). Every "fraction of the lift" in this paper is this quantity at these metrics.

Gate-A (canonical reproduction). Before any causal arm is trusted, `scripts/check_canonical_repro.py` re-runs the canonical R1 baselines and aborts unless they reproduce: in *reproduce* mode $|\text{measured} - \text{expected}| \leq 0.05$ for both the base (ARC-Challenge acc_norm 0.505) and the adapter (0.855) (`:69,87--91`); in *null* mode the zero-offset wrapper ($\alpha=\theta=0$) must be behaviorally identical to the base, $|\text{null} - \text{base}| \leq 0.02$ (`:16`), certifying the lift is not a wrapper artifact. A FAIL exits non-zero and blocks the dependent batch. The Gate-A base values are the exact reference the on-axis arms are compared against (e.g. $\mathrm{off}_{\parallel}$ recovering to 0.505 = "0%", Section 4.4).

Three-zone control gate. The inducibility fraction is read through a sign-safe three-zone gate (`scripts/check_canonical_repro.py:70--150`) that keeps numerator and denominator harness-internal (the native HFLM for the control's own effect, the V7 wrapper for the contrastive lift) so no cross-harness bias enters the ratio. Two sign-safety checks precede any fraction: a **setup** gate requires the contrastive lift to be real and positive (v7mc − base > min_effect, min_effect 0.01) and a **convergence** gate requires the control's own effect to be non-trivial and positive (arm − base ≥ 0.01), so a no-op or harmful control cannot be scored. The surviving fraction is binned at frac_lo = 1/3 and frac_hi = 2/3 into three zones (`:71,73,139--150`); the ~62% ARC recovery lands in the middle zone, which is what we report as "zone C" against these thresholds. The thresholds bin the *point* estimate, so the zone assignment does not depend on which interval accompanies it (paired bootstrap [51%, 73%], delta-method [45%, 85%]; Section 5.2). A separate null-task check (MMLU [27]) guards that a causal arm does not damage an unrelated task (null MMLU +0.07 pp at PASS, Table 7).

Whitened cosine. Geometric alignments (e.g. $\cos(\theta_{\mathrm{mc}}, v_{\mathrm{inf}}) = -0.003$, $\cos(v_{\mathrm{inf}}, v_{\mathrm{ret}}) = +0.283$) are Mahalanobis cosines under the per-head pooled class-activation covariance, in the 256-dim per-head activation space. The canonical headline values come from an eigendecomposition whitening (`scripts/probe_oq1_functional_angle.py::_whitening_from_cov`, reused verbatim by `characterize_theta_mc.py`): $W = V \cdot \mathrm{diag}(\lambda^{-1/2}) \cdot V^{\top}$ from the eigh of $\mathrm{cov\_common} = \frac{1}{2}(\mathrm{cov\_inf} + \mathrm{cov\_ret})$, eigenvalues floor-clamped at 10^{-6} — a floor that never binds in the actual computation (zero clamped eigenvalues; median condition number ~894). Each per-head covariance is estimated from $N = 2400$ activation samples (1200 correct + 1200 incorrect) against $d = 256$ dimensions, an N/d ratio of ~9.4. A related additive-ridge formulation, $M = (\text{cov} + 10^{-2} \cdot \text{mean}(\text{diag cov}) \cdot I)^{-1}$, appears in a separate probe-direction sanity script and (as a shrinkage on the covariance) in the $\text{off}_{\|}/\text{off}_{\perp}$ construction; it is not the source of the headline cosines. Robustness: sweeping that ridge factor over two orders of magnitude ($\{10^{-3} \ldots 10^{-1}\}$) moves $\cos(\theta_{\text{mc}}, v_{\text{inf}})$ only within $[-0.0032, -0.0021]$ (always inside the null band, sign stable — stronger regularization pushes it *toward* zero) and $\cos(v_{\text{inf}}, v_{\text{ret}})$ within $[+0.284, +0.325]$ (sign and magnitude stable), so neither headline geometric fact is an artifact of the regularization choice (`runs/oq1_functional_axis/ridge_sensitivity.json`).

The 48 heads. The causal analyses use the 48 (layer, head) cells carrying the adapter's offset. These are the **top-K = 48** heads ranked by per-head probe accuracy (`best_acc` = the max of the Fisher- and Ridge-CV probes), a pure ranking slice with **no accuracy-threshold gate** (`scripts/probe_v7_lofit_head_selection.py:1029--1032`); the 0.60 accuracy threshold in that script only annotates the report, it does not select heads ($K = 48$ follows the LoFiT recommendation).

Align-LoRA corpus and budget. The inducibility control is a non-contrastive LoRA fine-tune under a plain causal-LM cross-entropy loss (`scripts/train_align_lora.py`), on a corpus **matched to the V7-mc training data**: the matched builder (`scripts/build_align_control_corpus.py`) imports only the TruthfulQA and ARC splits used to train V7-mc, byte-identically (TQA = the first 80% of the validation split under a seed-0 permutation; ARC = the native ARC-Challenge train split), and the control is swept over LoRA rank $r \in \{8, 32, 64, 256\}$ at 2 epochs, effective batch 8 (batch 1 × grad-accum 8), max sequence 1024. **"Budget-matched" means the same training examples and the same optimizer steps as the contrastive adapter — not the same FLOPs** (the two objectives differ in per-step cost). ARC is the clean anchor; TruthfulQA is excluded from the fraction because its matched corpus overlaps its evaluation (Section 5.2).

4 Results

4.1 The headline lift and its non-transfer

The adapter lifts cold multiple-choice scoring substantially (Table 1: ARC +35.0, TruthfulQA mc1 +37.25; transfer benchmarks HellaSwag +17.0, Winogrande +19.0) and loses on generation in every condition (Table 2: gsm8k loss, canonical magnitude **−25** pp per R6 — flat across a generation-budget sweep, so a mechanistic multi-step-reasoning breakage, not truncation). The generation loss that anchors the trade is gsm8k, which is verified and canonical (the R6probe/L3 per-run forensic on the v7mc arm: 295/400 correct, 79 clean-wrong with a normal-terminating wrong `####` numeral, 26 non-terminating reasoning loops, 0 empties) and R6-confirmed; lambada (−32) was an *illustrative* custom-token-norm scorer and the lm-eval confirmation (R9, `lambada_openai` template-off) is **not yet run**, so we do not count it among the verified headline losses (Table 2). The scoring lift and the generation loss are the two halves the rest of the paper reconciles.

The lift is not an artifact of the zero-shot floor, but the anti-floor control resolves into *two* distinct quantities and we report both rather than let one stand in for the other. A paired 5-shot control (pre-registered eval R2, same n=400) raises the base itself — ARC 0.505→0.635, HellaSwag 0.540→0.610, Winogrande 0.6575→0.765 — closing **37% of the ARC headline, 41% of HellaSwag's, and 57% of Winogrande's**. Against that better-prompted base, the **anti-floor** quantity (a@0 − b@5: the zero-shot adapter against a base given every prompting advantage, which is what the kill-rule asks) is ARC **+22.0** pp, HellaSwag **+10.0** pp, Winogrande **+8.25** pp. The **matched-shot** quantity (a@5 − b@5: whether the adapter still wins when few-shot is granted to both arms) is ARC **+22.25** pp (acc_norm, paired McNemar over base-wrong items: 99 fixed vs 10 broken, $p \approx 10^{-19}$), HellaSwag **+17.5** pp, Winogrande **+13.75** pp. On ARC the two nearly coincide (+22.0 vs +22.25) because the adapter is already saturated there (v7mc@5 ≈ v7mc@0); on Winogrande they diverge sharply (+8.25 vs +13.75), and the anti-floor figure is the one we treat as the bound. Per-item on ARC (198 base-wrong at 0-shot), the adapter fixes 151 (76.3%) against few-shot's 69 (34.8%) and covers 64/69 = **92.8**% of the few-shot-fixed items (87 only-adapter, 5 only-few-shot); saturation is task-specific — v7mc@5 ≈ v7mc@0 on ARC (0.8575 vs 0.855) and TruthfulQA (0.81 vs 0.810), while few-shot still stacks on the adapter for HellaSwag (0.785 > 0.710) and Winogrande (0.9025 > 0.8475). The headline remains the zero-shot +35.0, and the anti-floor bound on it is **+22.0**.

The lift is also protocol-bound in a second, sharper sense: with the chat template removed entirely (R1b, same n=400 harness), *both* arms collapse to near-chance on ARC (base acc_norm 0.25, adapter 0.285) — there is no lift to exploit outside the chat-template readout. This is direct evidence that the object of study is the readout protocol itself, and it scopes every claim in this paper to cold-MC scoring under a chat template.

The TruthfulQA +37.25 is reported separately and carries **no** anti-floor control (the lm-eval `truthfulqa_mc1` task fixes num_fewshot=0, so both arms ran 0-shot in R2); only ARC/HellaSwag/Winogrande have a real 5-shot control. We further scope the +37.25 as a **selection lift on the mc1 metric** (paired-significant, Holm-corrected p_{corr} 6.26×10^{-38}), **not** a truthfulness gain and never as evidence for the off-axis mechanism (it is largely answer-prior, Section 5.8).

The held-out-plus-mc2 control has now **run** (full set n=817): the lift **generalizes held-out** (Δ_{UNSEEN} mc1 +37.8 ≈ Δ_{SEEN} +36.9; mc2 Δ_{UNSEEN} +26.1), so it is real beyond the mc1 selection metric and is **not** memorization (the matched train covers ∼79.9% of the eval, yet the held-out gain is undiminished), whereas a generic Align control collapses held-out (mc1 +7.9), marking the contrastive lift as the genuine, generalizing one. We therefore report it as a real, held-out, generalizing **answer-prior** lift on mc1/mc2 (Section 5.8) — never as truthfulness and never as the off-axis mechanism.

4.2 The fixed items are CoT-reachable — an upper bound on knowledge-accessibility

The non-transfer is not a failure to teach the model anything; it is that the model already knew. Of the 124 ARC items the adapter fixes in cold scoring (and the on-axis offset does not), the base solves 124/124 (1.000) under generative chain-of-thought — the rate of the items it already scores correctly cold (base-right control, 200/202 = 0.990) — so the fixed items are knowledge the base reaches by reasoning but that cold scoring misranks (Table 3). A negative control decides whether that number discriminates or merely restates a ceiling, and it is available from the same run: on the adapter's **residual** — the 47 items that are base-wrong and that the adapter does *not* fix — base CoT accuracy falls to **41/47 = 0.872** (CI $[0.748, 0.940]$), against 151/151 on the base-wrong items it *does* fix (Fisher $p = 1.4 \times 10^{-4}$) and 392/400 = 0.980 over all of ARC ($p = 0.0015$). The arbiter therefore carries information about which items the adapter moves. We attach the magnitude caveat rather than oversell it: ARC sits near its CoT ceiling in every population, so the contrast is statistically clear but absolutely modest, and the residual figure landed between the two thresholds we pre-registered for this control (≤0.85 "discriminates" / ≈0.97 "saturated") — we read it as discriminating on the strength of the contrast, not of the point estimate. That caveat is a statement about ARC, and we have since run the same partition where it does not apply: on MMLU-Pro, which leaves the base 44.50 pp between cold and CoT, the fixed cell reads 0.8462 against 0.5919 on the residual — +25.42 pp, Newcombe95 [+18.41, +31.49], twice the ARC contrast with the control cell 41 pp from the ceiling instead of 13 (Table 3, Section 4.7). We read the 124/124 as an *upper bound* on knowledge-accessibility, not a proof of perfect recall, and the MMLU-Pro figure is what that upper bound costs when the ceiling is removed: 0.8462, not 1.000, so roughly a sixth of the items the adapter fixes there are *not* reachable by the base under greedy CoT. It is a greedy k=1 arbiter, and for HellaSwag/Winogrande it runs in a letter-MCQ format distinct from the cold-MC channel (Section 3.2). Its robustness is corroborated independently: base self-consistency [55] CoT at k=3 also reaches 124/124 on this set (Section 6), so the result is not an artifact of the single greedy chain. This holds out-of-task too: on GPQA the base solves the items under chain-of-thought at 0.7071 (arbiter), well above chance, so there is no generative headroom a scoring shortcut could supply, and we therefore did not build a verifier-guided arm (a design choice following from the no-headroom result, not an independent test). And measured directly, the cold-MC lift on GPQA is itself ~0 (base 0.4747 → adapter 0.4697, −0.5 pp, CI crossing zero): the no-transfer is confirmed out-of-task (GPQA is out-of-task and distribution-distant — it is also the B.2 spillover task at −6.6 pp — so this is a no-transfer-out-of-task result, not a no-transfer-to-hard one; Section 5.1 separates difficulty from distribution with within-format and second-anchor controls). The adapter is not inert there — it fixes ~21 of 104 base-wrong items at a 0.202 fix-rate and breaks ~22 base-right ones for a net of about −1 — but that exchange is churn within noise (no McNemar test), not compensatory damage. Note the converse implication we do not hide: GPQA leaves ~23 pp of cold-scoring headroom (cold 0.4747 vs CoT 0.7071) that the adapter fails to recover — the adapter is not a general repair of the lossy readout but an in-distribution exploit of it, which is exactly what the off-axis-of-degree geometry predicts.

4.3 Much of the lift is inducible without the contrastive objective, and how much depends on capacity and on the task

Two non-contrastive controls bound how much of the lift the *objective* is responsible for, and they answer different questions because they sit at opposite ends of the capacity range (Table 5).

At identical capacity. A *same-parameterization* control — the identical 48 head-pairs and identical 12,336 trainable parameters as the adapter, the same corpus, steps, split and checkpoint rule, with ordinary cross-entropy on the correct answer and *no* term against the incorrect one, three seeds — recovers **36.2**% ± **5.0** of the ARC lift (best-checkpoint; 39.7% ± 3.1 at the fixed final step, so ~36–40% is the honest range). This is the control that isolates the objective, because the intervention space is held exactly fixed.

At much higher capacity. A non-contrastive Align-LoRA — data- and step-matched but with 1,800× to 58,000× more trainable parameters (Section 3.6) — recovers ~62%, flat across a rank sweep (59/63/60/65; slope-CI [−1.4, +3.4] includes zero) and not raised by a best-effort r512 flank (frac 0.22 at lr1e-4, converged but overfitting; below-base at lr2e-4).

So the fraction a generic objective recovers is *not* a single constant: it moves from ~37% at the adapter's own capacity to ~62% with three-to-five orders of magnitude more parameters. We therefore report inducibility as a capacity-dependent quantity rather than as one number, and we do not claim a ceiling for generic fine-tuning: across parameterizations, more capacity coincides with more recovery, while *within* each rank grid we tested the fraction is flat (the r8–r256 slope-CI includes zero; note (ii) of Table 6).

It is also strongly task-dependent — the same three same-parameterization seeds recover **79.5**% ± **1.4** of the Winogrande lift and **69.5**% ± **6.4** of HellaSwag's, but **0.9**% ± **3.3** of TruthfulQA-mc1's, i.e. nothing, even though TruthfulQA is in the control's own training corpus (Table 5). The pattern is legible: where the task is scored by the plausibility of a continuation (Winogrande, HellaSwag) simply raising the log-probability of the correct option induces most of the lift, whereas where the task requires *beating a plausible distractor* — which is precisely what TruthfulQA-mc1 is built to test — the contrastive term is doing work that ordinary cross-entropy does not. ARC sits between the two. "Most of the lift is generic" is therefore true of Winogrande and HellaSwag, and of ARC only at high capacity; it is false at matched capacity on ARC and false throughout on TruthfulQA. TruthfulQA is in any case excluded from the higher-capacity control as train-on-eval leakage (held-out, the highest-rank control collapses 92%→30%).

This ordering is a metric statement, and we say so here rather than only in Section 6. The inducibility fractions are computed on the same per-task metric as the headline lifts, which for HellaSwag is acc_norm. Recomputed on raw accuracy the ordering does not survive: HellaSwag's 69.5% becomes 8.9% for this control and 92.6% becomes 19.1% for the higher-capacity arm, while ARC's same-parameterization 36.2% becomes 26.9% in Gemma and 27.2% becomes 59.7% in Qwen. The four *lifts* are robust to the choice; the cross-task *ordering* of inducibility is not, and no mechanistic claim in this paper rests on it.

4.4 The recovery is off-axis to correctness — of degree, on a measured spectrum

We now locate the recovery relative to the model's own correctness geometry. Per attention head, we estimate a functional correctness direction v_{inf} and compare it both to the adapter's trained offset and, more carefully, to the *net effect each method has on activations*. The trained offset alone is a misleading proxy: its whitened cosine with v_{inf} (Section 3.6) is -0.003 (within the random-direction null), it places only 6.4% of its mass in the top principal components (a quantity unrelated to — and only coincidentally equal to — the W_{know} recovery figure below), and its norm is 4.3× the class gap — a near-orthogonal weight-space picture. But the offset is a weight-space proxy, and a zero-cost preview of the net-effect comparison from weights alone reported "off-axis" (ratio ∼1.05); the clean experiment — comparing the net effect on activations — refutes that simplicity.

The clean comparison yields a **spectrum**, not a single off-axis fact. A norm-matched **random** direction (three seeds) recovers ≈0% of the lift (ARC acc_norm 0.50–0.52 ≈ base 0.505): the learned direction is specific, not any perturbation of the right size. The strongest **on-axis** reference we can construct locally — a whitened Fisher-LDA direction W_{know}, genuinely distinct from the mass-mean (cos with v_{inf} 0.379) yet orthogonal to the offset (cos with θ_{mc} 0.0014) — recovers only **6.4**% of the ARC lift (null 0.500 / adapter 0.853 / W_{know} 0.522; fraction 0.064, far below the pre-registered 0.30 kill-rule and statistically indistinguishable from zero — exact McNemar $p = 0.136$, with the paired-bootstrap recovery CI $[-0.009, +0.136]$ and the Wilson accuracy CI $[0.474, 0.571]$ both including chance, which *strengthens* the on-axis-fails reading), and essentially nothing on TruthfulQA; a raw mass-mean v_{inf} recovers 11% (+4.0 vs +35.0; $z = -6.63$). The two on-axis references are not equally null, and the paired test is what separates them: over the same 400 items v_{inf}'s small recovery is *nominally* detectable (exact McNemar b=12/c=28 $p = 0.017$, paired-bootstrap frac-CI95 $[+0.027, +0.197]$, excluding 0), while W_{know}'s is not — though this p is uncorrected and would not survive the Bonferroni over the ∼25 tests of the campaign (Section 5.9), and each arm rests on a single injection run. The weaker reference recovers a little; the stronger one does not clear its own null — which is the ordering the off-axis reading predicts, and the opposite of what a weak-reference objection would produce. We therefore read that ordering as suggestive and rest nothing on it.

A non-rejection is not by itself an absence, so we bound what this design could have seen. Freezing the backward cell at its observed rate (10/400) and enumerating the exact two-sided McNemar power at n=400, $\alpha = 0.05$, the minimum detectable effect at power 0.80 is **12.6**% of the ARC lift (+4.45 pp of accuracy, i.e. an arm accuracy of ≈0.5445); conditioning instead on the 29 observed discordant pairs gives 9.2%, and the MDE moves only between 10.2% and 16.2% when that backward rate is halved or doubled. The design therefore carries power **1.0000** against both reference effects this paper calls real elsewhere — the 36.2% the same-parameterization control recovers and the 50% mark — so the on-axis null does exclude an on-axis effect of any magnitude we report as real. It does *not* license reading W_{know} as exactly zero: the design is blind below ∼9–13% of the lift and the 6.4% point estimate sits inside that blind band. What is declarable is an **upper bound**, not a zero — and the MDE (12.6%) nearly coincides with the upper end of the paired-bootstrap recovery interval already reported (+13.6%), which is the design face and the estimation face of the same precision, not a coincidence.

The ordering of the two on-axis references deserves a sentence, because a reader may suspect the smaller number was chosen for convenience: the better-constructed discriminator (W_{know}, the covariance-optimal linear separator, Section 3.3) recovers no *more* of the lift than the plain mass-mean (6.4% vs 11%) — improving the on-axis estimate does not improve recovery, which is itself evidence against a knowledge-deficit account. Both figures sit far below the 0.30 kill-rule, and the conclusion is unchanged whichever is taken as the on-axis ceiling.

The on-axis references are not bad readers, so the on-axis null is not an estimation artifact. A null arm invites the objection that the direction, not the axis, is what failed: W_{know} inverts a covariance estimated at N=2400 against d=256, and whitening amplifies exactly the worst-estimated eigendirections, so "the on-axis direction fails causally" and "the estimated on-axis direction is a poor reader" would look alike from the causal arm alone. They are separable, because a direction can be scored as a reader without touching a GPU: the weight vector of the linear functional *is* the injected blob, so we score gold against distractors with it on the same 12,288-dimensional activations and the same base-wrong population (n=198) where the fitted probe reaches 0.9545. W_{know} reads **0.9613** (CI95 $[0.9428, 0.9781]$) and v_{inf} 0.9646 ($[0.9461, 0.9815]$) — at or above the best linear reader *fitted* on that space, whose own interval is $[0.9343, 0.9731]$. The reference distribution re-randomises the *weights*, not the labels: a random unit direction per head at the same per-head norms (1,000 draws) reads 0.5018 on average, band $[0.2391, 0.7694]$. So the on-axis directions are excellent discriminants that, injected norm-matched, still buy 6.4% and 11% of the lift. The null is causal.

The same measurement bounds the word "strongest" (Section 3.3): paired over the same items, $W_{\text{know}} - v_{\text{inf}} = -0.0034$ (CI95 $[-0.0135, +0.0067]$, $P(W_{\text{know}}$ worse$) = 0.682$) — nominally behind, interval including zero, so as readers the two are indistinguishable and neither ordering is declarable. And the trained offset read the same way is a *poor* correctness reader: θ_{mc} scores 0.7525 ($[0.7020, 0.7997]$), with 3.5% of norm-matched random directions doing at least as well. The direction the adapter writes is not the direction that reads correctness — the read≠write dissociation, measured from the read side.

This off-axis picture does not hinge on the on-axis references being weak estimators of the true correctness axis: the offset is off-axis **even to the probe's own decision direction** — the empirically optimal correctness reader — the in-space probe of Section 5.3, whose base-wrong AUC is 0.955 (distinct from the Section 5.4 selector's accuracy, 0.943 at n=400 / 0.914 on base-wrong) — at pooled cosine $|0.008|$, inside the null band (the per-head $|0.066|$ sits at the edge of the ~0.061 per-head null, so this upgrade rests on the pooled statistic).

The two *methods*, measured by net activation effect, are both **mostly off-axis** — the contrastive adapter at cosine 0.088 (ratio 1.74 vs a random direction) and generic SFT at cosine 0.199 (ratio 3.93) — but generic SFT is **2.3× more on-axis** than the contrastive adapter. The recovery is thus off the correctness axis *to a degree that depends on the method*: random ≈0% (cos≈0.05, ratio 1.0) → W_{know} 6.4% (the on-axis reference) → generic SFT ~62% (cos 0.199, ratio 3.93) → contrastive 100% (cos 0.088, ratio 1.74). Note the two orderings are deliberately **not** monotone in each other: recovery fraction and on-axis-ness are different axes of the spectrum (the most off-axis direction, random, recovers nothing; the more on-axis method, generic SFT, recovers *less* than the contrastive one) — the spectrum is a two-axis object, not a single scale.

This refines the earlier "purely off-axis" framing into the central figure of the paper, and it is consistent with the earlier finding that the two functional correctness faces are themselves co-located (whitened cosine $v_{\text{inf}} \cdot v_{\text{ret}}$ +0.283, BCa95 $[+0.276, +0.291]$, null ±0.007, n=316 heads): the recovery is off the v_{inf} face by a method-dependent amount. It is off the v_{ret} face as well, and this we now measure rather than assume. Over the 48 heads the recipe intervenes on, $\cos_{\text{w}}(\theta_{\text{mc}}, v_{\text{ret}}) = +\mathbf{0.0076}$ (BCa95 $[-0.0033, +0.0187]$), inside an *isotropic* null band of ±0.0178 on the mean of n=48 — iid unit directions in $\mathbb{R}^{256}$, which states what chance direction gives at this n and deliberately preserves no law of the data, so it is not a permutation control. The component of v_{ret} perpendicular to v_{inf}, which is where an alignment invisible to the v_{inf} cosine would have to hide, is equally null (+0.0090, BCa95 $[-0.0027, +0.0209]$).

A floor is only informative if the estimator can see a ceiling in the same population, so we report the internal positive control: restricted to those same 48 heads and computed with the same whitening, $\cos_{\text{w}}(v_{\text{inf}}, v_{\text{ret}})$ reads $+\mathbf{0.2586}$ (BCa95 $[+0.2449, +0.2712]$) — the n=316 co-location above, seen through the 48-head window. The instrument registers co-location where there is co-location; against θ_{mc} it registers none, with a CI ceiling 14× below. The two distances are also statistically the same distance: paired within head, $\cos_w(\theta_{mc}, v_{ret}) - \cos_w(\theta_{mc}, v_{inf}) = +0.0109$ (sd 0.0558, paired t 1.35, $p = 0.183$; Wilcoxon $p = 0.260$), so the offset is not nearer one face than the other. This closes a deflationary account the earlier framing left open — that the lift is the adapter pushing on the *retrieval* face, which would explain the knowledge-benchmark lift, the MMLU spillover and the non-transfer at once. That account needs a cosine of the order of +0.2586; it gets +0.0076.

This co-location appears to sit in tension with work reporting that recall and reasoning occupy *causally separable* circuits [20]: if the correctness faces share a substrate, separately ablating recall and reasoning should not be possible. The tension dissolves at the level of description, because the two results measure different objects. That work measures task-circuit separability (in Qwen and LLaMA): ablating one functional circuit leaves the other intact, "separable but interacting." We measure the *direction of a trained offset* relative to the functional correctness direction, in Gemma 4 31B IT. A substrate can host circuits that partition sufficiently for separate ablation *and* carry functional directions that are co-located, with our +0.283 measurement living in that work's "interacting" seam rather than denying its separability. Our novel layer is neither: the direction the adapter actually moves is off-axis to both functional faces — measured against each, with the positive control above — which is why a partition-based account of the spillover, and the hope of re-selecting heads to avoid it, does not hold.

A direct causal decomposition: the off-axis push is privileged, not any perturbation of the right size. The spectrum above is read from each method's net activation effect; a second, more direct experiment decomposes the adapter's *own* offset against the correctness direction and asks which component carries the lift. We split the trained offset, per head, into a component parallel to the functional correctness direction ($\text{off}_{\parallel}$) and a component perpendicular to it ($\text{off}_{\perp}$), and inject each through the identical recovery harness used for every arm above, at its natural norm from the decomposition ($\text{off}_{\parallel}$ is consequently ∼16× smaller in norm than the W_{know} arm — the equal-norm control inside the perpendicular complement is the shuffle arm below, norm-matched per head to $\text{off}_{\perp}$).

The perpendicular component recovers essentially all of the lift (ARC acc_norm 0.8525, ∼100%; HellaSwag, Winogrande, TruthfulQA 99–104%), while the parallel component recovers none (ARC acc_norm 0.505 = the Gate-A base exactly, 0%; raw acc 0.5025). Given $\cos(\theta_{mc}, v_{inf}) \approx -0.003$, $\text{off}_{\perp} \approx$ the full offset by construction; the informative arms are $\text{off}_{\parallel}$ (an on-axis push at the offset's own scale does nothing) and the shuffle (equal-norm, same subspace, learnedness as the only variable).

Because the *same* harness that yields 100% for the perpendicular push yields only 6.4% for the on-axis W_{know} reference, the on-axis failure is a real property of the geometry, not an insensitivity of the injection method — the harness is direction-sensitive. This closes the non-identifiability objection [52] on the *harness* side.

We close it on the *write* side as well: a random direction sampled inside the *same* perpendicular complement, norm-matched per head to $\text{off}_{\perp}$ (sanity check: $|\cos|$ to $v_{inf} \approx 5 \times 10^{-17}$, $|\cos|$ to the learned $\text{off}_{\perp} \approx 0.047 \approx 1/\sqrt{256}$), recovers ∼0% of the lift across three seeds (ARC fraction −0.04 / −0.01 / −0.01; the lone non-zero, Winogrande ∼12%, is a small lift in noise) against $\text{off}_{\perp}$'s ∼100%. Same harness, same norm, same subspace — the only variable is *learnedness* — so it is the learned structure *inside* the off-axis complement that lifts, not orthogonality-plus-norm. This mirrors the control pair that work runs on its own steering result, where the learned direction moves behavior and neither the norm-matched random nor the orthogonal direction does [14]: here both orthogonal controls (the on-axis reference and the shuffled perpendicular) have an effect on the lift indistinguishable from zero while the learned off-axis push does not; and reported specificity failures of activation directions are *cross-model*, whereas within-model — our setting — directions remain specific [51], so that result does not subsume this control.

(These two "off-axis" readings are not in tension: $\text{off}_{\parallel}$ 0% is a decomposition of the trained offset in *weight space*, whereas the cosine 0.088 is the *net effect on activations* — different objects on different metrics, both saying the on-axis component does not carry the lift.)

A norm 4.3× the class gap also invites the question whether a linear on/off-axis decomposition is still meaningful at the offset's operating point. The dose–response sweep (Section 5.5) is the empirical check: the response to scaling the offset over $s \in [0, 1]$ is smooth and monotone — the scoring lift saturates, then the generation damage onsets — with no discontinuity or chaotic regime anywhere on the axis, so the injection operates on a graded response surface where the decomposition is interpretable. We nonetheless scope the geometric claims to the trained offset at its trained scale rather than asserting magnitude-invariance.

A training-time constructive control closes the causal argument from the other side (necessity, complementing this sufficiency): an on-axis training penalty toward v_{inf}, swept over λ, does *not* rescue transfer at matched lift (fork (b), Section 6) — so the +35 is the off-axis mass by necessity, not only by construction. Together, the off$_\perp$/random spectrum (sufficiency) and fork (b) (necessity) make the off-axis reading a two-sided causal claim.

The response is not proportional to projection onto the learned direction. A natural model of everything above is that a push buys lift in proportion to how much of it lands along θ_{mc}. That model is testable because it is *antisymmetric in scale*, and it fails. Crossing sign against axis on ARC (acc_norm, n=1172, paired) gives: θ_{mc} at +1, +36.26 pp CI [+33.28, +39.25] — this run's own reading of the headline lift, on the full test split rather than the canonical n=400 draw; and for a separately trained anti-axis, −0.5 → −1.96, −1 → −3.41 CI [−6.23, −0.60], −2 → −4.35, +1 → **−28.07** CI [−31.23, −25.00]. The proportional prediction for that last cell is +4.68: it misses by ∼33 pp and gets the sign wrong. At −2 the observed −4.35 against a predicted −9.36 saturates sublinearly.

The decisive fact, though, is not any single cell — it is that *all four* scales lose accuracy. An antisymmetric model cannot move both directions to the same side of zero, so the dial is unilaterally negative and the response is not proportional to its projection onto θ_{mc}. The rank data say the same thing qualitatively: of the four sign-by-axis combinations, three disperse the correct option below base and only one concentrates it.

One category caveat, because the arm invites a misreading: the anti-axis is 98.3% perpendicular to θ_{mc} ($|\cos| = 0.129$), at comparable norm. It is a small-known-projection control, *not* an on-axis one — W_{know} remains the on-axis arm and the 6.4% rests on it. This corroborates the family's pattern without being a member of it. One run per cell.

The learned direction is reproducible, so it is not seed noise. An off-axis direction invites the reading that it is whatever idiosyncratic perturbation this training run happened to land on. It is not. Two independently trained runs of the contrastive objective over the same 48 heads agree at a median per-head cosine of **+0.9985** (5th percentile 0.9958), against an RMS floor of 0.0625 for random directions in d=256. The objective picks out an almost deterministic direction, roughly three orders of magnitude away from chance agreement. This also forecloses averaging as a remedy. Replicate-averaging can only cancel a component that *varies* across replicates, and the test separates the two cases cleanly. Averaging three norm-matched *random* offsets collapses their norm to a ratio of **0.5875**, right at the $1/\sqrt{3} = 0.5774$ that independence predicts — which is the instrument working, not the finding. Averaging three independently *trained* offsets leaves a ratio of **0.9209**: almost nothing cancels, because there is almost nothing that differs. Averaging is a remedy for noise, and this direction is not noise.

The contrastive and generic net effects are neither the same nor independent. Section 5.2 establishes by *outcome* that a generic fine-tune recovers much of the lift; the geometric counterpart asks how much direction the two methods actually share, which no earlier measurement had computed — both net effects had been compared to v_{inf}, never to each other. They share **13.5–18.8**% of their energy (the range spans five ranks of the estimator; the spread across ranks exceeds the tolerance we fixed beforehand, so we quote the range and never a point) against 0.62% for a matched null. The overlap concentrates where the adapter pushes hardest: cosine +0.4825 on the ten heads carrying most energy against +0.0773 on the ten carrying least.

The tempting inference from that — that the shared part *is* the generic, on-axis component, and the contrastive residual is the off-axis remainder — does not hold: subtracting the generic direction leaves the residual's on-axis budget essentially unchanged at every rank tested. What the two methods share is not the part that points along the correctness axis. We report the concentration finding because it survived the control it invited rather than because it is clean: the obvious artifact is attenuation by measurement noise on weakly-driven heads, so we estimated per-head reliability by split-half within a single forward pass and found it at 0.9998, with reliability and push magnitude correlated at −0.204 (CI [−0.558, +0.140]) — the artifact route is not open. All of this is measured on one benchmark's prompts.

4.5 No single-layer locus: the lift scales with total offset magnitude

The recovery has no causal layer locus (Table 8). Across a layer-band patching sweep, the recovered fraction correlates with the *total magnitude* of the learned offset ($\Sigma|r|$) at **0.985**, far more than with the number of heads patched (0.832). (We read this as *no single-layer locus* rather than a strict magnitude-over-head-count causal claim: 0.985 and 0.832 are coupled correlations from the same band sweep, not a test that isolates magnitude from head-count; what the sweep licenses is that no privileged layer repairs the readout, not a proof that head-count is causally inert.) The natural temptation — to read "layers 36–40 carry 87% of the lift" — is a head-count artifact: that band simply contains 71% of the adapter's heads (which span L30–55). There is no privileged layer at which the lossy readout is repaired; the recovery is a property of the aggregate magnitude of an off-axis-of-degree push distributed across heads, consistent with the over-scaled offset (Section 4.4) and the coherent-sum logit signature (Section 5.5).

4.6 The causal spine replicates in a second model family

Everything above is one model. We repeated the causal decomposition on Qwen2.5-14B-Instruct — a different family, different tokenizer, different pre-training — with the same pipeline and 6,192 trainable parameters, in two runs. A gated reference run established the *effect* on this family: a same-sign cold-MC lift on all four tasks (ARC +18.0 / HellaSwag +7.8 / TruthfulQA-mc1 +15.8 / Winogrande +3.0), with the null-adapter arm exactly at base (Section 5.9). The decomposition run reproduces it at +16.25/+7.25/+15.00/+3.25, inside the ±3 pp band fixed before looking: ARC acc_norm moves 0.4975 → 0.6600, a lift of **+16.25** pp, smaller than Gemma's +35.83 but built by the same machinery. That run's null-adapter arm sits 1.5 pp above the reference and its adapter arm −0.25 pp — mixed signs, the signature of BF16 non-determinism rather than the one-directional ≈24 pp a broken chat template produces — so every fraction below divides by *this* run's own base and adapter, never across runs.

The decomposition transfers with it. Splitting the trained offset against the functional inference direction and re-injecting each part separately, the *off-axis* component carries the lift (off_perp = **+106.2**% recovery) while the on-axis component does not (off_par = −6.2%), and the strongest on-axis direction we can build recovers −18.5% — as in Gemma, the on-axis arm buys nothing. The learned direction is privileged rather than merely orthogonal: three norm-matched random directions inside the same complement recover −16.9/−16.9/−13.8% (mean −15.9 ± 1.8), so off_perp clears its own noise floor by ∼122 pp in the same units, against ∼101 pp in Gemma. Read against that floor, the on-axis arm does not beat same-norm noise in *either* family.

Non-transfer to generation replicates too, and harder: gsm8k falls 0.945 → 0.455, **−49.0** pp, twice

Table 1: Headline scoring-face MC lift (V7-mc, Gemma 4 31B IT, n=400, chat-template, lm-eval canonical). The cold-scoring lift the whole paper is about. Same flags both arms; never cited without the gen-face row (Table 2) and the protocol caveat (base ≈0.50 is loglik-MC scoring-face, *not* a third-party-reported ~91% gen-CoT figure — a blog, not Google's official technical report). TruthfulQA-mc1 is reported separately as an answer-prior selection lift (mechanism-excluded; it generalizes held-out — Table 9), never as a truthfulness gain and never as evidence for the off-axis mechanism.

Benchmark	Base	V7-mc	Lift	Source artifact
ARC-Challenge (acc_norm)	0.505	0.855	**+35.0**	`msap-a1-trio-b2-planner-2026-06-10`
TruthfulQA-mc1	0.4375	0.810	**+37.25**	same
HellaSwag (acc_norm)	0.540	0.710	**+17.0**	same
Winogrande	0.6575	0.8475	**+19.0**	same

Pre-registered with kill-rules; the anti-floor kill-rule **R2 has executed (**n=400**) and the lift survived**. The paired 5-shot control, its two named quantities (**anti-floor** a@0 − b@5: ARC **+22.0** pp / HSwag **+10.0** pp / Wino **+8.25** pp; **matched-shot** a@5 − b@5: ARC **+22.25** pp, McNemar b=99/c=10, $p \approx 10^{-19}$ / HSwag **+17.5** pp / Wino **+13.75** pp; few-shot closes 37/41/57% of the respective headlines), the per-item coverage, and the task-specific saturation are reported in Section 4.1. **TQA carries no anti-floor control** (lm-eval `truthfulqa_mc1` fixes num_fewshot=0; both arms 0-shot) → cite the +37.25 separately, with no "irreducible-to-few-shot" claim. The artifact-backed replication is verified.

Paired significance (n=400**).** All four lifts are **paired-significant** by an exact McNemar test computed over byte-identical base/adapter items (prompts byte-identical 400/400 per task, 0 truncation): ARC b=11/c=151 $p = 1.32 \times 10^{-32}$, TQA b=6/c=155 $p = 1.57 \times 10^{-38}$, Winogrande b=23/c=99 $p = 1.99 \times 10^{-12}$, HellaSwag b=39/c=107 $p = 1.63 \times 10^{-8}$. They survive **Holm–Bonferroni** correction across the four anchors (m=4): p_{corr} ARC 3.95×10^{-32}, TQA 6.26×10^{-38}, Wino 3.98×10^{-12}, HSwag 1.63×10^{-8} — every anchor at or below 1.63×10^{-8}, **none dropped**. *Scope caveat:* this certifies the effect **exists and is paired-significant**, not that it is on-axis — the on-axis ceiling is W_{know} 6.4% (Section 4.4). *Metric note:* the headline mixes acc_norm (ARC/HSwag) with acc (TQA/Wino, which define no acc_norm); on HSwag acc_norm ≤ acc (+17.0 vs +22.25), so the length-normalized headline is the conservative figure, not a length artifact.

Gemma's −24.2. Crucially the turn-termination rate is preserved (1.000 → 0.985), with stop ids resolved from the model's own generation config — so the damage is multi-step reasoning, not the EOS collapse an earlier counting bug once suggested (Table 10). Generation also shortens, 191 → 133 tokens on average.

Two honest limits on how far to read this. First, **this is replication in two families, not universality**: two points do not establish a law, and the arms differ in magnitude throughout. Second, two cells are not readable here and we do not report them as agreement: Winogrande's lift is only 3.25 pp, too small a denominator for a recovery fraction to mean anything, and the shuffle floor on HellaSwag comes out *positive* (+31%) rather than near zero. The matched non-contrastive control orders the tasks HellaSwag 114.9% > ARC 27.2% > TruthfulQA −58.3%, the same ordering Gemma gives *under acc_norm* — a caveat that matters, because under raw accuracy the ordering does not survive in either family (Section 6).

4.7 The lift is not an artifact of saturated benchmarks: MMLU-Pro

Every anchor above sits near the base's chain-of-thought ceiling, which invites the obvious objection: perhaps the exploit only exists on easy four-option benchmarks the model has effectively already solved. We tested it on MMLU-Pro [56] — ten options, chance 0.100, and a base that scores **0.2610** cold, barely 2.6× chance. It is not saturated by any reading. Four arms, 1,000 items, paired on every `doc_id`.

The lift appears anyway. The adapter scores 0.3830 against the base's 0.2610: **+12.20** pp, CI [+9.10, +15.30], with disjoint Manski bounds (the cold channel has no unparsed items, so the bounds are the point estimates). Because the metric ordering burned us once before (Section 6), we recomputed under both defensible definitions: raw argmax reproduces the scored prediction 1,000/1,000 in every arm, and the lift is +12.20 under plain acc and **+15.00** CI [+11.70, +18.20] under acc_norm. The two non-contrastive arms

Table 2: Gen-face: V7-mc loses on generation (Gemma 4 31B IT). The other half of the trade. The verified anchor is **gsm8k**, the direct "ARC-in-CoT" test (`generate_until`, exact_match on the generation): V7-mc loses in every condition. **Direction (loss) is robust; the canonical magnitude is** −25 pp (R6 kill-rule PASS: base 0.9775 vs v7mc 0.7275, apples-to-apples template-ON 5-shot, flat across cap {512, 2048} → mechanistic multi-step-reasoning breakage, not truncation). **Across three independently trained adapters the loss is** −24.2 ± 7.3 pp (range −16.5 to −31; Table 12): the *sign* replicates in 3/3 seeds, the magnitude varies by about a factor of two, so the −25 here is this run's value and near the middle of the observed range rather than a constant of the method. The retrieval-side **lambada** [43] loss (−32) is reported here only as an *illustrative* signal: it comes from a custom token-norm scorer, and the lm-eval confirmation (R9, `lambada_openai` template-off) is not yet run, so the inference/retrieval trade-off in this paper rests on gsm8k alone, not on lambada.

Benchmark	Base	V7-mc	Δ	Source artifact
gsm8k (A1 run, strict-match)	0.9775	0.7275	**−25.0**	A1 run, lm-eval canonical; confirmed R6
gsm8k (day3, mu_only)	~0.955	0.905	−5	`runs/phase4_routed/day3/`
gsm8k (day3, romulo)	~0.955	0.785	−17	same
gsm8k (v7.5-baked, custom eval, n=200)	0.955	0.17	**−78**	`runs/v7_5_lofit/eval_gsm8k_v75.json`
gsm8k canonical magnitude	0.9775	0.7275	**−25.0**	R6 (n=400 paired, apples-to-apples template-ON 5-shot, cap sweep {512, 2048}=0.7275; template-OFF worse: base 0.730/v7mc 0.035, −69 pp; R6probe agreed −24 pp)
lambada (illustrative)	0.64	0.32	−32 (ppl 11.6→2523)	`runs/v7_lofit_gemma4_31b_chat/eval/` — **custom token-norm scorer**, audit-verified 2026-06-10

separate as the inducibility spectrum predicts: the higher-capacity LoRA moves +5.10 CI [+2.50, +7.80], while the *same-parameterization* plain-CE arm — the one that recovers 36.2% on ARC — is null under both definitions here (−0.60 CI [−3.10, +2.00] and +0.50 CI [−2.10, +3.10]), the same collapse it shows on TruthfulQA-mc1 and further evidence that inducibility is task-dependent rather than a constant of the objective.

And it still does not transfer. Under chain-of-thought the paired difference is −8.00 pp, CI [−10.90, −5.10]: the interval excludes zero on the wrong side of zero, which is not transfer. We do not declare even that much — the three-way Manski bounds *overlap* ([0.604, 0.845] against the base's [0.684, 0.905]), so the honest statement is that the scoring lift buys nothing generative here, not that it costs something.[1]

And the per-item arbiter runs here, where it can fail. On ARC the arbiter is a contrast near a ceiling, and we say so. This benchmark is the falsifying case: the base is at 0.2610 cold and 0.7060 under CoT, so "the base already reached these items by reasoning" is a claim with 44.50 pp of room to be wrong. Partitioning the same paired traces by what the adapter's cold scoring did to each item, and reading base CoT accuracy inside each cell:

[1]These bounds were recomputed after we found that the harness's `unparsed` field does not count truncated traces; the real denominator was ~5× larger (base 41 against 221). The conclusion did not change and now holds a fortiori, since widening the unknown set can only widen both brackets.

Table 3: The fixed items are CoT-accessible; no transfer even on a hard task (the per-item arbiter). The core result. The adapter "fixes" items the base already answers by reasoning; the gain is re-ranking, not capability — and this holds where the base must genuinely reason (GPQA).

Quantity	Value	Source artifact
ARC items adapter fixes in cold MC but on-axis offset does not	124 (mc_only; mc∩vinf=27 / mc_only=124)	`surface_stratified_arc_challenge.json`
Of those 124, base solves under generative CoT	**124/124** = **1.000** (fixed-parser re-run: n_unparsed 5→0; the earlier "123/124" was a parser artifact — doc 128's numeric option label "4" the letter-parser could not emit, now parses correct; ≈ base-right control 0.990)	`arbiter_arc_cot_v2.json`
Negative control — adapter's residual (base-wrong items it does *not* fix)	**41/47** = **0.872** CI[0.748, 0.940] vs **151/151** on the ones it fixes (Fisher $\mathbf{p = 1.4 \times 10^{-4}}$) and 392/400 = 0.980 over all of ARC ($p = 0.0015$) → **the arbiter discriminates**; robustness in the premise-file scaffold 66/73 = 0.904 vs 326/327 ($p = 3.6 \times 10^{-5}$). *Same run, same protocol* (re-partition, no new generation). Point estimate fell between the pre-registered ≤0.85 / ≈0.97 thresholds; reading rests on the contrast, and the absolute magnitude is modest (ARC near its CoT ceiling throughout)	`arbiter_negative_control_b2.json` (`analyze_b2_arbiter_negative_control.py`)
Base-right control (already-correct-cold items)	**200/202** = **0.990** (fixed-parser run; the superseded pre-fix run gave 196/202 = 0.970)	`arbiter_arc_cot_v2.json`
Base GPQA accuracy under CoT (no-transfer is out-of-task, not difficulty-bound)	**0.70** (139/198; thinking-ON greedy, max_new=8192; thinking_rate 1.000) — an earlier execution of the *same* protocol; the per-item arbiter run scores 140/198 = 0.7071, differing by one item. The two share `build_prompt`, `parse_answer` and `build_messages`, so this is the same apparatus re-run, not independent corroboration	`msap-phase1-20260624 b1_gpqa_base_cot_8k.json`
GPQA cold-MC scoring lift (adapter, out-of-task)	**−0.5** pp (base 0.4747 → adapter 0.4697; CI crosses 0); arbiter base GPQA CoT 0.7071; fix-rate 0.202 over 104 base-wrong	`msap-phase1-20260624`
The same contrast where it can fail — MMLU-Pro (base cold 0.2610, base CoT 0.7060: 44.50 pp of headroom, against ARC's ~13)	Base CoT on the items the adapter fixes, **165/195** = **0.8462** (Wilson [0.7889, 0.8901]), against **322/544** = **0.5919** (Wilson [0.5501, 0.6324]) on the base-wrong items it does *not*: **+25.42** pp, Newcombe95 [**+18.41**, **+31.49**], Fisher $\mathbf{p = 2.43 \times 10^{-11}}$, OR 3.79. Bounding the 41 unparsed base traces both ways gives [+19.73, +27.99] pp	`runs/paired_gen_dl/arbiter_mmlu_pro_cot.json`
Same partition, remaining two cells	Items both arms score right, 169/188 = 0.8989 (Wilson [0.8476, 0.9343]); items the adapter breaks, 50/73 = 0.6849 (Wilson [0.5714, 0.7800]). The fixed cell sits with the already-right cell, not with the residual	*ibid.*
Reading	fixed items CoT-accessible; cold-MC misranks; no scoring headroom out-of-task (GPQA lift ~0); and the fixed-vs-residual contrast survives at twice the size on a benchmark with 44.50 pp of headroom	—

MMLU-Pro note — what the null is, and what differs from the ARC row. The reference distribution permutes the label "the adapter fixed this item" *within* the base-cold-wrong stratum (n=739), which preserves the conditional law P(base CoT correct | base cold wrong) and both cell sizes; it does not preserve the marginal base-CoT rate over all 1,000 items, which is not the quantity at issue. Realised exactly as the Fisher exact test on the 2×2. The scaffold differs from the ARC row and we say so: ARC partitions an lm-eval scoring run, MMLU-Pro partitions the cold scoring of the same paired-generation run that produced the traces — same items, same `doc_id`, same prompt. Unparsed traces are bounded, not adjudicated: all 41

Table 4: How the readout misranks, and that the lift is not interpretable re-scoring. The *how* of the readout failure (confident burial under fluency) and the channel result (the lift does not reduce to re-scoring the output scalars). All ARC-Challenge, Gemma 4 31B IT, base-wrong items unless noted.

Quantity	Value	Source artifact
Base buries gold confidently — chosen-vs-gold z-gap (adapter-independent $S_{\text{cot\_coldwrong}}$, n=192)	$\mathbf{1.45}\sigma$; gold-rank median **2**; frac rank≥2 **0.59**	`cold_mc_calibration_arc_challenge.json`
Consistency check (S_{flip}, n=151, adapter-conditioned)	z-gap $\mathbf{1.37}\sigma$; gold-rank median **2**; frac rank≥2 0.54	same
Gold-vs-distractor naive AUC (base-wrong, *below* chance)	**0.394** CI[0.356, 0.430], n=204	`residual_discrimination_arc_challenge_base.json`
Base raw pick = gold ($S_{\text{cot\_coldwrong}}$ vs base-right control)	**0.19** vs **0.81**	cold_mc_calibration
Entropy "flatness" is a softmax-temperature artifact	acc_norm answer-entropy 0.920 ($S_{\text{cot\_coldwrong}}$; length-norm compresses spread ~50×); report rank/z-gap, not entropy	cold_mc_calibration
Adapter re-ranking is **promotion-dominated** (S_{flip})	gold-mass +0.35 > wrong-mass removed +0.21; Δentropy −0.19	cold_mc_calibration `re_routing`
Present-but-swamped: PMI-residual AUC (same scalar channel, different populations)	base-wrong **0.574** CI[0.520, 0.627] n=204 *(intermediate, CI_{lo}<0.55)* / adapter residual **0.667** CI[0.584, 0.744] n=73 *(claim rests here)*	`residual_discrimination_arc_challenge_{base,mc}.json`
Surface ceiling (supervised, held-out, base-wrong AUC) vs adapter	logistic **0.458** / GBM **0.462** / on-test oracle stack **0.502** (19% recovery) / PMI **0.574** vs adapter **0.830**	`offaxis_vs_stack_` + `supervised_surface_ceiling_arc_challenge.json`
PMI applied to the adapter / adapter re-rank explained by surface	PMI **hurts** 0.818→0.758; R^2(re-rank|surface) **0.024**	`offaxis_vs_stack_arc_challenge.json`

Reading. The cold readout buries the gold under fluent surface forms (z-gap 1.45σ, naive AUC 0.394 *below* chance), the correctness signal is *present but swamped* (PMI-residual clears chance on the adapter arm 0.667 n=73; base 0.574 n=204 intermediate — **lossy, not empty**), and no interpretable re-scoring of the output scalars recovers the lift (ceiling 0.574 ≪ adapter 0.830; PMI *hurts*; R^2 0.024). The 0.830-vs-ceiling gap is **information-set-confounded** (activations vs collapsed scalars) → a **channel** claim, not a "magic direction" (the higher-capacity non-contrastive SFT arm recovers ~62% of the ARC lift the same way) nor "empty channel". **The in-space activation-probe has now run and resolves this gate:** a logistic probe on the *same* per-head `o_proj`-input activations (held out by item, same cold-MC option-continuation distribution, preprocessing fit inside each fold, within-item permutation null) recovers gold-vs-distractor on ARC base-wrong at **AUC 0.955** CI[0.934, 0.973] (null 0.494), against the **0.502** collapsed-output surface-cap; a surface-form control on the option string caps at **0.442** (≈ chance) → the 0.955 reads correctness the output scalars discard, **not** a length/format artifact. This **widens and cleans the channel gap (**0.502→0.955**)** and supersedes the preliminary wrong-space ~0.65. **Three guards hold the read down to a channel claim** (match Section 5.3): (1) the probe reads 12288 activation dims while the adapter only nudges collapsed output logits — different information sets, so 0.955 is **not** "probe beats adapter" and **not** a representational edit; (2) a linear probe decoding correctness ~0.95 is near-universal in capable LMs → it shows the info *exists* in the activations, **not** "pure miscalibration / signal intact upstream" — we keep **lossy** and drop "pure miscalibration"; (3) the signal is **diffuse** — a random-48 head set decodes it at 0.949 CI[0.924, 0.971] (null 0.492) ≈ the adapter's 48 heads → a **read-test** that the cold-MC channel is lossy *model-wide*, **decoupled** from the **write-tests** that establish the adapter's specificity (W_{know} 6.4%, random norm-matched *offset* ~0% recovery of the lift — a different operation; read ≠ write). TruthfulQA is **anchored out** of the channel claim (probe 0.974 but surface-form control 0.639 + gold-first artifact 1.000 ⇒ answer-prior, Section 5.8); the channel claim is anchored on **ARC**.

Table 5: Inducibility is capacity- and task-dependent. Two non-contrastive controls. The *same-parameterization* control holds the intervention space exactly fixed (identical 48 head-pairs, identical 12,336 parameters) and varies only the objective, so it isolates the objective; the *higher-capacity* Align-LoRA is data- and step-matched but carries 1,800×–58,000× more parameters across the r8–r256 plateau (117,000× at the r512 flank). The fraction a generic objective recovers moves with both capacity and task → the effect is **partly, not wholly, special to the contrastive objective**. ARC is the clean test for the higher-capacity arm; TQA is excluded there as train-on-eval leakage.

Quantity	Value	Source artifact
*At identical capacity — same-parameterization control (*12,336 *params, CE-plain, 3 seeds)*		
ARC lift recovered	fraction **36.2**% ± **5.0** (best-ckpt; 39.7% ± 3.1 final-ckpt) → honest range ~36–40%	`runs/e1_plain_ce/`
Task heterogeneity (same three seeds)	Winogrande **79.5**% ± **1.4**; HellaSwag **69.5**% ± **6.4**; ARC **36.2**% ± **5.0**; TruthfulQA-mc1 **0.9**% ± **3.3** (*nothing*, despite TQA being in the control's own training corpus)	same
At 1,800×–58,000× *capacity — higher-capacity Align-LoRA*		
ARC lift recovered (rank sweep r8/32/64/256)	fraction **59**/**63**/**60**/**65** = ~**62**%, flat	`msap-phase1-20260624` (rank sweep)
Within-sweep capacity slope	slope-CI [−**1.4**, +**3.4**] (includes 0); paired-bootstrap frac-CI [**51**%, **73**%], delta-method ~[45%, 85%] (zone C, binned on the point)	same + `l1_paired_bootstrap.json`
Best-effort capacity flank (r512, causal fairness, n=400)	lr1e-4 frac **0.22** (converged, acc>base; overfit ep2→ep4, robust ep2=ep4); lr2e-4 **below base** (broken, excluded like r256hi) → **does not raise** ~62% **within this grid**	`msap-align-fairness-20260702`
TQA excluded — leakage (proof)	frac 41/66/72/92 monotone↑; held-out 20%: r256 **92**%→**30**% (CI95 [0.79, 1.06] on n=400 → [0.05, 0.55] on the n=82 stratum, paired bootstrap; the intervals do not overlap); the rank sweep flattens, r8=r256=0.524 on the full n=164 held-out slice; contrastive generalizes held-out (0.83)	`runs/align_fairness/tqa_heldout_stratum_ci.json`
Residual not explained by the tested controls (ARC)	~**60**–**64**% at matched capacity; ~**38**% against the higher-capacity arm	derived

Two r256 series, and which one the interval uses. The rank-sweep row above reports r256 at fraction 0.65 (accuracy 0.7325), taken from the de-risking rank sweep; the paired-bootstrap interval [51%, 73%] is computed on the per-item canonical arm, where r256 reads accuracy 0.7225 and fraction **0.6214**. These are different runs of the same configuration, not a discrepancy in one: the sweep establishes flatness across ranks, the canonical arm supplies the point and the interval, and the pre-specified zone binning is applied to the canonical 0.6214. Per-item files for r8/r32/r64 are not released, so the *slope* across ranks rests on the parametric sweep rather than on a per-item re-derivation.

The objective is non-contrastive; the selection rule is not. The same-parameterization arm's loss carries no term against the incorrect option, but the checkpoint rule we inherited from the adapter validates on whether the correct option outscores the incorrect one. It bit in one seed of three (seeds 1–2 select the final step; seed 0 selects step 600 over 800), and there it picked the *worse* ARC checkpoint — which is why the fixed-final variant, using no validation signal at all, reads 39.7% ± 3.1 against the 36.2% ± 5.0 we quote. The contamination deflates this control, so the contrastive-specific residual is conservative. Provenance: `runs/e1_plain_ce/offsets_seed*.log.json`.

Note. We report these as the best observed recovery under the generic-SFT family we tested, not as a bound on generic fine-tuning in general. Two facts set the limits of the claim. (i) *Capacity moves the fraction*: the same non-contrastive objective recovers ~37% at the adapter's own 12,336 parameters and ~62% with three-to-five orders of magnitude more, so no single number is "the" generic fraction and we do not assert a ceiling. (ii) *Within each grid the fraction is flat*: the r8–r256 slope-CI includes zero and the r512 flank does not raise it (it recovers less at lr1e-4 and breaks below base at lr2e-4, both excluded by the same convergence-pathology rule that excluded r256hi), so the residual is not a gap that more rank closes *within the tested rank/LR/epoch grid*. What we cannot exclude is a different architecture, placement, optimizer or search budget doing better.

Interval validation and the r512 flank. Because delta-method (Wald) intervals on ratio estimators can be unstable when the denominator's sampling error matters, the fraction interval is cross-checked twice. An independent bootstrap-of-the-ratio (200k parametric draws from the published arm accuracies at n=400, independent-binomial resampling) gives r256 frac CI [0.50, 0.80] against the delta-method [0.45, 0.85], and a bootstrap slope CI of [−2.5, +4.6] pp per log2-rank step containing zero, both matching the published reading (`runs/oq1_functional_axis/frac_ci_bootstrap.json`). That check was conservative by construction — resampling each arm independently discards the correlation induced by scoring the *same* 400 items in all three arms — and we flagged that a paired design would only tighten it. It does: resampling items with replacement and carrying base, generic and contrastive together (200k reps, seed 0, zero degenerate replicates) gives [**0.51**, **0.73**], 44% narrower than the delta-method interval and contained in it, with the measured item-level correlation (r=0.42

Table 6: Geometry spectrum: off-axis of degree, method-dependent. Apples-to-apples: the *net effect on activations* of each method, decomposed against the functional correctness direction v_{inf}. `ratio` = on-axis alignment of the net effect with v_{inf}, normalized to a random direction (1.0 = like random; higher = more on-axis than chance).

Direction	Recovery of the lift	cos with v_{inf} (ratio vs chance)	Source artifact
random norm-matched (3 seeds)	≈**0**% (acc_norm 0.50–0.52 ≈ base 0.505)	0.05 (1.0)	`msap-phase1-20260624 derisk/ offsets_ random_s*`
on-axis maximum (W_{know}, Fisher-LDA)	**6.4**% (ARC null 0.500 / mc 0.853 / wknow 0.522; **statistically indistinguishable from zero**: exact McNemar b=10/c=19 $p = 0.136$, paired-bootstrap frac-CI95 $[-0.009, +0.136]$ and Wilson acc-CI $[0.474, 0.571]$ both include chance)	— (cos with v_{inf} 0.379; cos with θ_{mc} 0.0014)	`runs/wknow_ causal/` + `msap-wknow-onaxis-20260623`
contrastive adapter (net effect)	100% (by construction)	**0.088 (1.74)**; across three seeds 0.094 ± 0.008 (1.85 ± 0.14); direct θ_{mc} offset cos 0.0014	`msap-phase1-20260624 derisk/ contrastive_ direction`; `runs/v7_lofit_ seed{1,2}/`
same-parameterization control (net effect, 3 seeds) — *same intervention space, objective changed*	~**37**% (ARC)	**0.136 ± 0.017** (2.68 ± 0.34)	`runs/e1_plain_ ce/neteffect_ seed*`
higher-capacity generic SFT (net effect)	~**62**%	**0.199 (3.93)**	`msap-phase1-20260624 derisk/align_ direction`

Verdict. All three methods are **mostly off-axis** (cos<0.2, ~78–85° from v_{inf}), but by different degrees, and the ordering is informative: the contrastive adapter is the *most* off-axis (1.85), the same-parameterization control is intermediate (2.68), and higher-capacity generic SFT is the least (3.93, 2.1× more on-axis than the contrastive adapter) → **off-axis of degree, not universal.**

The same-parameterization arm is the load-bearing comparison here, because it holds the intervention space exactly fixed (identical 48 head-pairs, identical 12,336 parameters) and changes *only* the objective: with the parameterization controlled, the contrastive objective lands **further off-axis** (1.85 vs 2.68) *and* recovers substantially more (100% vs ~37%). The off-axis character is therefore not an artifact of the LoFiT parameterization — it tracks the objective.

Two axes, not a ladder. Recovery does not increase monotonically with off-axis-ness: among the two *non*-contrastive arms the more on-axis one (higher-capacity SFT, 3.93) recovers *more* than the more off-axis one (same-parameterization, 2.68), because there capacity is the dominant variable. The spectrum is organized by two factors — objective and capacity — and should not be read as a single ordered scale.

The direct θ_{mc} offset (cos 0.0014, a weight-space proxy) was misleading; the clean net-activation-effect experiment is the canonical measurement. **Forward filter:** a zero-cost read-space preview said "off-axis" (ratio ~1.05); the clean experiment refuted it (3.93). Report only the clean experiment.

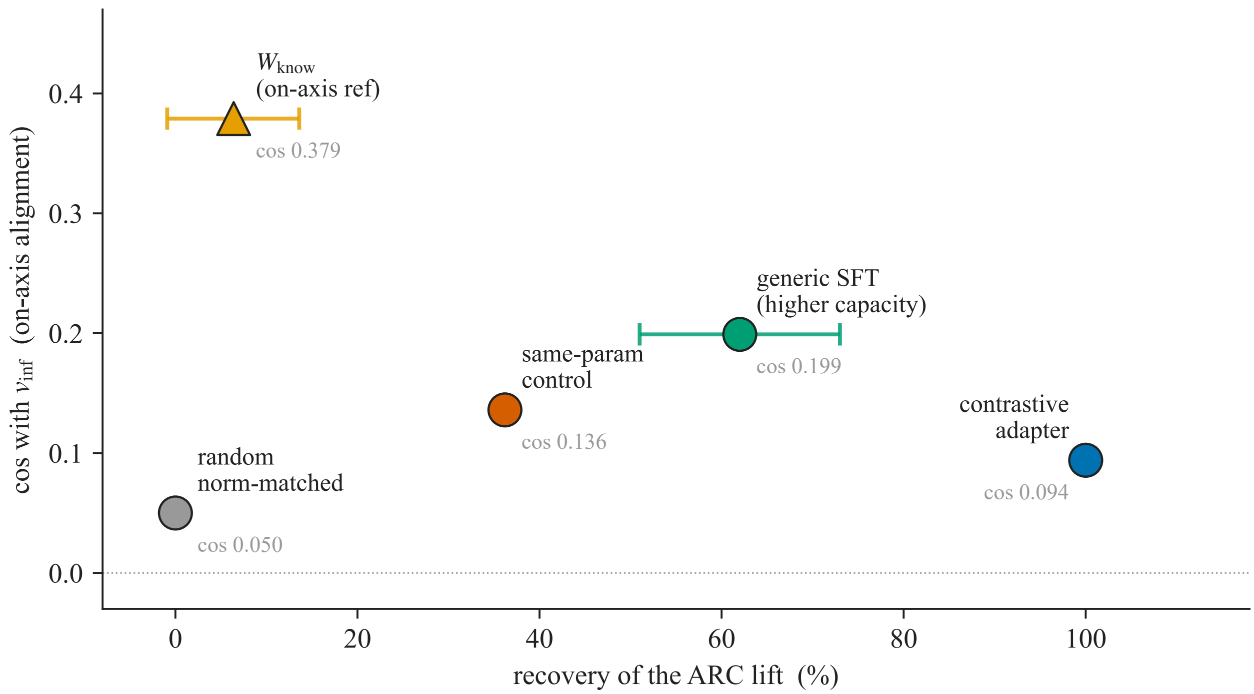

Figure 1: The matched-control spectrum on two independent axes. x: fraction of the ARC lift each direction recovers; y: cosine with the functional correctness direction v_{inf} (net effect on activations for the four methods; the reference direction itself for W_{know}, marked ▲). The two orderings are not monotone in each other: the most on-axis point (W_{know}, cos 0.379) recovers only 6.4%, while the top recoverer (the contrastive adapter, 100%) is the *least* on-axis (cos 0.094). **The two non-contrastive controls separate the two factors at work.** The *same-parameterization* control (identical 48 head-pairs and 12,336 parameters; only the objective differs) recovers ~37% at cos 0.136, while *higher-capacity* generic SFT (1,800×–58,000× more parameters) recovers ~62% at cos 0.199: with the parameterization held fixed the contrastive objective is both further off-axis and far more effective, whereas added capacity buys recovery while moving *toward* the axis. So how much a generic objective recovers depends on capacity, and where the recovery lives depends on the objective — two factors, not one ordered scale. Horizontal bars are the published 95% intervals on the recovery fraction where one exists — W_{know} [−0.9%, +13.6%] and generic SFT [51%, 73%] (paired bootstrap); the random floor (three seeds), the same-parameterization control (three seeds, sd 5.0 pp) and the contrastive anchor (100% by construction) carry none, and we do not synthesize one. **W_{know}'s interval crosses zero and overlaps the random floor:** its 6.4% is statistically indistinguishable from no recovery (exact McNemar $p = 0.136$), and it is counted with the floor rather than as a rung. Values in Table 6.

Cell (by the adapter's cold scoring)	n	base CoT	Wilson95
fixed (base wrong → adapter right)	195	**0.8462**	[0.7889, 0.8901]
kept right (right → right)	188	0.8989	[0.8476, 0.9343]
broken (right → wrong)	73	0.6849	[0.5714, 0.7800]
not fixed (wrong → wrong)	544	**0.5919**	[0.5501, 0.6324]

Within the base-cold-wrong stratum (n=739) the fixed cell beats the residual by **+25.42** pp (Newcombe95 [+18.41, +31.49], Fisher $p = 2.43 \times 10^{-11}$, OR 3.79); bounding the 41 unparsed base traces both ways leaves [+19.73, +27.99] pp, so neither the sign nor the order of magnitude turns on how they are scored. A capacity account predicts the fixed cell at or below the residual — these are items the base got wrong cold, on a benchmark it is far from saturating. Instead the fixed cell sits beside the *kept-right* cell (0.8462 against 0.8989): the items the adapter "fixes" behave almost like the ones the base already scored right in the first place. What the arbiter also shows, now that the ceiling is gone, is its own limit — 0.8462 is not 1.000, so on

Table 7: On-axis causal controls (no on-axis reference reproduces the lift). The earlier causal arms, consistent with Table 6's spectrum: pushing *along* correctness does not recover the lift. (These on-axis causal arms were measured on the vinf_causal run, whose within-run MC baselines — ARC +35.0, TQA +36.5 — differ slightly from the A1 canonical of Table 1, +37.25; fractions are computed within-run.)

Quantity	Value	Source artifact
ARC lift recovered by on-axis v_{inf} (mass-mean, norm-matched)	fraction **0.11** (MC +35.0 / VINF +4.0; $z = -6.63$, unpaired and conservative — the base stderr enters both lifts). **Paired over the same 400 items the small recovery is *nominally* detectable**: exact McNemar b=12/c=28 $p = 0.017$, paired-bootstrap frac-CI95 [+0.027, +0.197] excludes 0 — uncorrected, and it would not survive the Bonferroni over the ~25 tests of the campaign, so we read the ordering as suggestive. On that reading the weaker on-axis reference recovers *little*, not *nothing* — unlike W_{know} below	`runs/vinf_causal/vinf_paired_significance.json` + `msap-vinf-causal-20260623`
ARC lift recovered by W_{know} (strongest on-axis ref)	fraction **0.064** (null 0.500 / mc 0.853 / wknow 0.522; ≪ 0.30 kill-rule; **indistinguishable from zero**: exact McNemar b=10/c=19 $p = 0.136$, paired-bootstrap frac-CI95 [−0.009, +0.136] includes 0)	`runs/wknow_causal/`
TQA lift recovered by either on-axis ref	~**0** (VINF −2.75; $W_{\text{know}} \approx$ null)	—
θ_{mc} vs v_{inf}	$\cos_{\text{w}}$ −**0.003** (within null); 6.4% mass in top PCs; norm 4.3× class gap (the top-PC 6.4% is coincidentally equal to the W_{know} recovery 6.4%, not the same quantity)	`theta_mc_characterization.json`
MMLU damage, on-axis v_{inf} vs mc	VINF −5.83 ≥ MC −4.30 (gate PASS, null MMLU +0.07 pp)	—

`process-LoFiT-PRM` (an on-axis training program) is **refuted causally** at eval time (a PRM corpus is not warranted); a *training-time* on-axis constraint **has now run and does not rescue transfer** (no λ passes the lift-matched gate, so the +35 is the off-axis mass by *necessity*; see Section 6).

Table 8: No layer locus: the lift scales with offset magnitude, not layer.

Quantity	Value	Source artifact
corr(recovered lift, total offset magnitude $\Sigma\|r\|$)	**0.985**	`msap-phase1-20260624 patch_layers/*.json`
corr(recovered lift, head count)	0.832	same
"Layers 36–40 carry 87%"	head-count artifact (that band holds 71% of adapter heads); adapter spans L30–55 — **not a causal locus**	same

a benchmark like this one roughly a sixth of the fixed items are outside the base's greedy-CoT reach, which is why Section 4.2 reads the ARC 124/124 as an upper bound rather than as perfect recall.

What this widens, and what it bounds, are different things. It widens the *scope*: the phenomenon is not a property of saturated four-option benchmarks. It bounds the *magnitude*: +12.20 here against +35.83 on ARC is roughly a third. Task-to-task variation of that size is already on record for this objective, albeit as a different quantity — the fraction of the lift a generic control recovers, which runs Winogrande 79.5% / HellaSwag 69.5% / ARC 36.2% / TruthfulQA 0.9% (Section 4.3).

The result also sharpens what the GPQA null in Section 4.2 does and does not show. Both benchmarks are hard and unsaturated, and both leave the base substantial cold-to-CoT headroom — GPQA ~23 pp (0.4747 → 0.7071), MMLU-Pro ~44 pp (0.2610 → 0.7060) — yet the adapter recovers approximately

Table 9: Task heterogeneity: ARC ≠ TQA. ARC is question-conditional and carries the readout story; TQA is largely answer-prior / surface-form Goodhart. The held-out + mc2 control has run: TQA generalizes held-out → reported as a real, generalizing answer-prior lift, not as the off-axis mechanism.

Task	Diagnostic	Value	Source artifact
ARC	PMI-DC retention of lift	**78**% (naive +0.3275 → PMI +0.2550)	`pmi_dc_arc_challenge.json`
ARC	choices-only lift	+**0.0375** CI[−0.005, +0.08] (not clear) — question-conditional, not answer-prior	`answer_only_arc_challenge.json`
TQA	choices-only lift	+**0.2225** CI[+0.1675, +0.2725] — answer-prior	`answer_only_truthfulqa_mc1.json`
TQA	adapter picks correct option with NO question	**0.5725** (base 0.35, chance 0.229) — surface-form Goodhart [4, 29]	same
TQA	PMI-DC retention	68%	`pmi_dc_truthfulqa_mc1.json`
TQA	paired significance of the +37.25	exact McNemar b=6/c=155, p_{corr} $\mathbf{6.26 \times 10^{-38}}$ (Holm m=4)	paired doc_hash 400/400
TQA	held-out generalization ≠ memorization (full set n=817; train covers ~79.9%, 653 SEEN / 164 UNSEEN)	v7mc $\mathbf{\Delta_{\text{UNSEEN}}}$ **mc1 +37.8** ≈ Δ_{SEEN} +36.9; mc2 Δ_{UNSEEN} +26.1 (gate Δ_{UNSEEN} mc1 ≥ + 15 AND Δmc2 > + 10 passed); generic Align **collapses held-out** (mc1 +38.6→+**7.9**; mc2 +24.1→ + 7.6)	`analyze_tqa_contamination_l4.py`, full 817

Note. The lift is task-heterogeneous. TruthfulQA is an answer-prior effect — a generalizing selection lift on mc1/mc2 (the held-out + mc2 control generalizes: Δ_{UNSEEN} mc1 +37.8, so it is neither memorization nor mc1-only, while a generic Align control collapses held-out to +7.9) — but it is not a truthfulness gain and not evidence for the off-axis mechanism. ARC is question-conditional, off-axis-of-degree, and non-transferable.

nothing on the first and +12.20 pp on the second. Difficulty therefore does not govern whether the exploit fires. What the two benchmarks *do* differ in is whether the adapter's training distribution covers them, and only the covered one lifts — a protocol-bound inference, since we vary benchmark identity rather than manipulating distance directly, and benchmark identity carries format, domain and option count with it. That is the same conclusion Section 5.1 reaches with its within-format controls, arrived at here by varying the benchmark instead of the format, and it is what "an in-distribution exploit of a lossy readout" predicts.

Finally, the generative pathology reproduces in its now-familiar form: degeneration rather than degraded reasoning. Repetition loops (distinct-4 < 0.35) run **0/1,000** for the base against 146 for the adapter and 298 for plain CE — an ordering that replicates on a second benchmark (Section 5.5).

Two caveats bind this section. None of the four arms is replicated. And it is not the converse arm we flag in the limitations (Section 6): the adapter here was trained on the same corpus as everywhere else and *evaluated* on MMLU-Pro, so it answers whether the effect survives an unsaturated benchmark, not whether contrastive PEFT trained in-distribution on one would produce it.

5 Discussion: the mechanism of the recovery

The scoring-face gain documented above invites a natural reading: the contrastive-correctness adapter sharpens the model's internal direction for answer correctness, so that cold multiple-choice scoring becomes more accurate. This section shows the gain is better understood as a *quantified causal attribution* than as a sharpening. We take as established setup that cold multiple-choice scoring under a chat template under-expresses knowledge the model holds (Section 2; 14, 22, 44); the result is what the adapter does about

Table 10: Deployment pathologies (coupled to injection method, not to the lift). The recovery introduces **two certified costs** — no transfer to generative CoT (multi-step reasoning breakage) and retrieval/recall spillover — that a generic SFT recovering most of the lift does not incur, so they trace to how the contrastive offset is folded into weights, not to the lift itself. A third, previously-listed cost — an end-of-turn "collapse" — is **retracted** as a measurement artifact (last row). Rows 2–4 are **family** evidence from sibling adapters (the D.1 SocialIQA and B.2/KR adapters), not the V7-mc adapter under study; V7-mc's own directly-measured MMLU spillover is −**4.2** pp (row 5).

Pathology	Value	Source artifact
No-CoT-transfer (multi-step reasoning)	gsm8k loss every condition (Table 2); contrastive baked →0.17. **R6 canonical (kill-rule PASS, apples-to-apples, template ON, 5-shot,** n=400**):** base **0.9775** vs v7mc **0.7275** = −**25** pp, flat across cap {512, 2048}=0.7275/0.7275 → **not truncation** (extra budget never rescues; v7mc cap2048 runs ~2h to the ceiling without improving; template-OFF is worse — base 0.730 / v7mc 0.035, −69 pp — so it is not a template artifact). Forensic (R6probe @cap2048): 79 clean-wrong (#### wrong number — reasoning resolved but defective) + 26 degenerate non-terminating loops already present @cap512 (the loop precedes the budget, not truncation). **Align SFT preserves** 0.96→0.95 **(NS)**	R6 (`runs/mechanism_v8/R6_*`) + R6probe (`R6probe_*`) + Table 2
Retrieval spillover (D.1 SocialIQA adapter — family evidence)	MMLU −**19.59** pp (0.8306→0.6347); K=12 not recovered (−20.80, Pareto-worse)	D.1 SocialIQA spillover audit
Spillover (KR SciQ [58] K=24)	GPQA Diamond −**12.1** pp (sciq +2.1 pp overshoot)	KR SciQ K=24 overshoot audit
Spillover (B.2 mc_discrimination)	MMLU −6.3 pp / GPQA −6.6 pp	B.2 phase-3 findings
Spillover MMLU (V7-mc direct, eval-bulletproof R10)	−**4.2** pp (0.8226→0.7805, ~2k stratified, 0-shot) — the V7-mc adapter under study spillovers MMLU itself; same sign as B.2 (−6.3 pp), smaller	`runs/eval_bulletproof/R10_*`
Spillover mechanism	$\cos_w(v_{\text{inf}}, v_{\text{ret}})$ **+0.283** BCa95[+0.276, +0.291], null ±0.007, n=316 — orthogonality refuted; co-location on a shared substrate	functional-axis geometry analysis
~~End-of-turn collapse~~ **retracted — eos-token artifact** (was: D.1 social 56→4, D.1 factual 48→4, V7-mc factual 48→4/60→4)	`eval_adaptive_verbosity` scored termination on `tokenizer.eos_token_id` (`<eos>`) but Gemma 4 closes a turn with `<end_of_turn>` (106). Corrected re-measure (`_resolve_stop_ids`): base_factual 48→**100**%, v7mc_factual 4→**100**% — **no collapse**; v7mc factual answers are correct + terse + terminate ("Paris"/"Au"/"1945"); the "degenerate loop" read earlier was junk *after* `<end_of_turn>`. The analogous D.1 cells were re-measured: no →4% collapse, only a modest open-ended-generation degradation (social 96→76%, creative 68→12%), adapter-dependent and off the V7-mc headline.	buggy: `EOS_v7mc_smoke.json`; fix: `mechanism_v8/EOS_verbosity_fixed.json`

it. Most of the recovery is what *generic* fine-tuning buys (Section 5.2); it lives in a learned direction that is off-axis-of-degree to the model's correctness geometry and is causally privileged within that off-axis complement (Section 4.4); it has no layer locus (Section 4.5); and it does not transfer to generation (Section 5.5).

The off-axis geometry is not a curiosity but the *expected* signature of the attribution we lead with, and reading it that way organizes the rest of this section. The cold readout does not fail for lack of knowledge; it fails by **surface-form swamping**: on the items the adapter fixes the base buries the gold beneath more fluent distractors (Section 5.1). Plausibly, though we do not measure this, suppressing surface form need not align with signalling correctness. The two causal arms make the point in opposite directions: injecting correctness *on-axis* (the strongest on-axis reference we can build, W_{know}) recovers only **6.4**% of the lift, bounding an on-axis contribution at ≤12.6% — the design's minimum detectable effect — which is the size a knowledge-deficit account would have to fit inside; pushing *off-axis* (perpendicular to correctness, $\text{off}_\perp$) recovers ~**100%**, which is consistent with the surface-swamping account. The off-axis direction is therefore *consistent with* a natural form for a fix to take when the pathology is one of format rather than of knowledge — an account under which a single off-axis push organizes the readout result (Section 5.1), the inducibility attribution (Section 5.2), the on-axis null (Section 4.4), and the deployment pathologies (Section 5.5). It organizes them without being shown to be their mechanism: the offset is *not* the retrieval direction ($\cos_w(\theta_{mc}, v_{ret}) = +0.0076$, Section 4.4), the mean-direction format control is null (cosine −0.0099 with the aggregate write, at the null floor), and the suppression signature appears in the *tail* of the offset's logit image rather than in its mean (Section 5.6).

5.1 The fixed items are chain-of-thought-accessible, the no-transfer is out-of-task (not difficulty-bound), and the readout misranks by confident burial

The cleanest evidence that the readout — not the model's knowledge — is the bottleneck comes from a per-item arbiter. We isolate the **124** ARC items the adapter corrects in cold multiple-choice scoring but the on-axis offset does not (Section 4.4), and for each ask whether the *base* model already reaches the right answer through generative chain-of-thought. It does: the base solves **124/124** (1.000) of these items under generative CoT (the fixed-parser re-run resolves the earlier 123/124: the one "failure" was a parser artifact — doc 128's numeric option label "4", which the letter-parser could not emit by construction — now parsing correct, n_unparsed 5→0), indistinguishable from its CoT accuracy on the items it already scores correctly cold (base-right control, 200/202 = 0.990 — read from the same fixed-parser run, not from the superseded pre-fix one, where it was 196/202). The items the adapter "fixes" are knowledge the base already reaches by reasoning; cold scoring simply misranks them.

What the arbiter is and is not. It establishes that the corrected items are within the *base*'s generative reach — a statement about latent knowledge and about the cold-MC readout that misranks it. It is not a paired base-versus-adapter generation comparison on those items, so on its own it does not measure what the adapter does to generation. That is measured separately, and negatively, by the generation-face results (gsm8k −24.2 ± 7.3 pp across three independently trained seeds, MMLU −4.2 pp, GPQA cold-MC ∼0 against the base's CoT 0.70). Where we say the lift "does not transfer", the scope is the generation channels we measure, not every conceivable generative use.

The arbiter discriminates: a negative control on the adapter's residual. A per-item arbiter of this shape invites an obvious objection — if the base solved *everything* under chain-of-thought, "the fixed items were already CoT-accessible" would be true of all of ARC and would say nothing about the adapter. The control is the complementary population, and it needs no new generation: the v2 arbiter run already covers all 400 items under the identical protocol (greedy k=1, same prompt, same fixed parser, `n_unparsed=0`), so re-partitioning it is protocol-identical to the headline by construction. On the adapter's **residual** — the 47 base-wrong items it does not fix — the base solves **41/47 = 0.872** (CI [0.748, 0.940]), against **151/151** on the base-wrong items it does fix (Fisher $p = 1.4 \times 10^{-4}$) and **392/400 = 0.980** over all of ARC ($p = 0.0015$). A related partition in the independent premise-file scaffold — which defines the n=73 residual of the PMI residual-discrimination analysis below; the calibration analysis runs on the lm-eval scaffold — agrees: $66/73 = 0.904$ against $326/327 = 0.997$, $p = 3.6 \times 10^{-5}$. It is *related* rather than identical, and we say so: the lm-eval partition conditions on base-wrong (47 against 151), while the n=73 is every item the adapter gets wrong, including 8 it broke from base-right. Restricting it to base-wrong for a like-for-like read gives $58/65 = 0.892$, between the two. So the 124/124 is not trivially true of all of ARC. Two honest qualifications travel with it. First, the magnitude is modest in absolute terms — ARC is near its CoT ceiling everywhere, and the discrimination is 0.872 against 0.980, not against something far lower. Second, the residual figure fell *between* the two thresholds we pre-registered for this control (≤0.85 → discriminates, ≈0.97 → saturated); we therefore rest the reading on the contrast and its interval (CI upper 0.940 excludes the saturated value) rather than on the point estimate clearing a bar it did not clear. The adapter is a *shortcut* to that knowledge — it nudges the scoring logits toward answers the base could reason its way to, without reasoning.

This directly explains why the lift does not transfer to generation (Section 4): on exactly these items the base is already at ceiling under CoT, so there is no generative headroom for a scoring shortcut to add. The obvious objection is that ARC is simply easy — near its CoT ceiling — so the arbiter cannot tell "recovers knowable" from "fabricates plausible." We corroborate the no-transfer out-of-task.

On **GPQA**, where the base must genuinely reason, its CoT accuracy is **0.7071** (arbiter; 140/198, thinking-on, with the generation budget raised to 8192 tokens — at 2048 the base truncated 96% of its reasoning, producing a spurious 0.15). That figure was a point estimate over a set we could not audit: the artifact kept no token counts, and the parser ends in a "lone letter in the last 100 characters" fallback, so a trace that stopped mid-sentence could not be told from one that answered badly — a parser cannot measure its own fallback. Re-running the arbiter with the counts persisted separates them, and the answer is lopsided: **39** of the 40 unparsed items **ran out of budget**, and exactly one is malformed with a complete trace. 62/198 (31.3%) exhausted the 8192-token budget in all. Counting the unparsed as wrong was therefore the right call — those traces carry no answer — but it is only half of the bias: 17 equally truncated traces *did* receive a letter from the tail fallback, and 7 of those entered as correct. Bounding it both ways (every truncated item wrong, then every one right) gives $[0.6010, 0.9141]$, an interval 31 points wide whose width is the cost of the cap; a second run of the nominally identical protocol reads 0.6616 against 0.7071 under greedy decoding. What survives all of this is the claim the number is used for: at chance 0.25, even the floor of 0.6010 is far above it. Two things follow. First, 0.7071 is well above chance, so the base *does* solve these by reasoning; yet the cold-scoring shortcut still has no generative headroom to add, and indeed the measured cold-MC lift on GPQA is ~0 (base 0.4747 → adapter 0.4697, −0.5 pp, CI crossing zero) — the no-transfer result restated out-of-task. This is a no-transfer-*out-of-task* result, not a no-transfer-to-*hard* one: GPQA is out-of-task and distribution-distant (it is the B.2 spillover task at −6.6 pp). Second, because the base already reasons GPQA out, a verifier-guided arm (base generates, adapter scores) has nothing to harvest, so we did not build one (a design choice following from the result, not an independent test). The readout-is-the-bottleneck result therefore survives an out-of-task control that the ARC arbiter alone could not provide.

Two added controls now separate difficulty from distribution, where previously a single out-of-task benchmark could not. *Within-format difficulty:* the cold-MC fix-rate on base-wrong items moves only **−0.10** as ARC difficulty rises (ARC-Easy 0.861, n=360 → ARC-Challenge 0.763, n=198, both burial-matched by gold-rank), whereas it collapses **−0.56** going out-of-task to GPQA (0.202) — so *distribution*, not within-format difficulty, drives the collapse. *A second pair of anchors:* on HellaSwag and Winogrande, where the lift also holds (+17 / +19), the base's CoT accuracy is high (0.83 / 0.9425) — like ARC's 0.97 and well above GPQA's 0.70, though the arbiter is run in a lettered-MCQ format distinct from the cold-MC channel on every anchor (Section 3.2), so it is a knowledge-accessibility probe rather than a channel-faithful readout — and the recovery is recoverability-gated (the adapter fixes CoT-right items more than CoT-wrong ones: HellaSwag 0.64 vs 0.31, Winogrande 0.76 vs 0.42), i.e. the lift rescues knowledge the base can already reason out. The no-transfer is therefore **out-of-task, not difficulty-bound** — supported now, not merely suggestive, on all three anchors. *(Power caveat: the hard cells are small — HellaSwag n=32, Winogrande n=12 — and the Easy→Challenge difficulty range stays below GPQA's, both base-CoT near 0.97; the burial gradient is shallow throughout, corr(fix, z-gap) −0.17.)*

How the readout misranks on ARC: confident burial under fluency. The arbiter shows the model *has* the answer; a calibration analysis on ARC shows *how* cold scoring loses it, and it is not indifference between close options but misplaced confidence. We measure this on an **adapter-independent** subset — the n=192 ARC items the base scores wrong cold yet solves under CoT ($S_{\text{cot\_coldwrong}}$, defined without any reference to the adapter) — so the signature cannot be an artifact of which items the adapter happened to fix. On these items the base's chosen (wrong) option beats the gold by **1.45σ** within-item, the gold sits at **median rank 2 of 4** (only 41% have the gold at rank 1), and a gold-vs-distractor ranking on the base-wrong items scores **AUC 0.394, *below* chance**. (This confident-wrong signature is the multiple-choice overconfidence aligned-model calibration work documents, which separates an *answer-decision* uncertainty from a *format-preference* uncertainty and attributes alignment overconfidence to their conflation [25]; our claim goes past calibration — the per-item arbiter shows the buried answer is *reachable*, at 124/124 under CoT, and a learned offset *exploits* the gap rather than re-calibrating it post hoc.) The gold is not a flat near-miss; it is actively *buried* beneath more fluent surface forms — the base's raw pick is the gold only **19%** of the time here, against **81%** on the items it scores correctly cold.

The re-ranking the adapter performs is **promotion-dominated** (it adds 0.35 of probability mass to the gold while removing 0.21 from wrong options, and sharpens the distribution, Δentropy -0.19), consistent with surfacing a buried-but-present signal rather than manufacturing one.

Methodological note we are forced to make: the apparent *flatness* of the char-normalized answer entropy (0.920 on this subset) is a softmax-temperature artifact — length normalization compresses the spread ~50× without reordering, while the raw distribution is peaked. We therefore report **rank and within-item z-gap**, which are invariant to that normalization, and not entropy, whose flatness flips with the metric. (The selection-conditioned S_{flip} subset, the 151 items the adapter flips, shows the identical signature — z-gap 1.37σ, gold-rank median 2 — and serves only as a consistency check; its conditioning on the adapter's flips is why the adapter-independent subset is the headline.)

This burial-and-recoverability signature is established on ARC; we **could not detect** it out-of-task on GPQA, but as a power-starved non-finding rather than a contrast: on the burial-matched hard cell the CoT-right and CoT-wrong fix-rates are equal (0.186 = 0.186), but n=43 with 31 items left unparsed (k=1 greedy decoding) contaminates that cell, leaving it indistinguishable from no measurement. We therefore report the recoverability-gating as **could-not-detect / power-starved out-of-task**, not as evidence against it.

The signal is swamped, not absent — "broken" means *lossy*. It does not follow that the cold channel is empty. After removing length and fluency (a PMI / decision-conditional residual), a gold-vs-distractor discrimination scores above chance on both arms in the *same scalar channel*: **AUC 0.574** (CI $[0.520, 0.627]$) on the base-wrong items (n=204), and **0.667** (CI $[0.584, 0.744]$) on the adapter's own residual (n=73, the items it does not fix). The correctness signal is present in the cold channel but *out-ranked* by fluency, not erased: the same PMI-residual channel carries gold-vs-distractor signal, rather than the channel being empty and the adapter supplying missing knowledge.

Honesty boundary: these are two arm readings on **different populations** (n=204 vs n=73), not a within-set rise; and the base-arm CI lower bound, 0.520, does not on its own clear a 0.55 "present" threshold — we call the base channel *intermediate* and rest the present-but-swamped claim on the adapter arm (CI lower bound 0.584 > 0.55), and raising power on the base arm did not rescue it: pooling base-wrong with $S_{\text{cot\_coldwrong}}$ — and every defensible pool variant — ran, and none clears 0.55. The adapter-independent $S_{\text{cot\_coldwrong}}$ (n=192, the items the base *provably* knows via CoT yet cold-MC misranks) reaches only CI lower bound 0.512, and even the literal union base-wrong $\cup$ $S_{\text{cot\_coldwrong}}$ (n=228, which is *pro*-gate — its 24 extra items are base-right under premise scoring) reaches 0.535, so the base channel stays *intermediate* on the strongest honest pool.

5.2 Much of the recovery is generic to fine-tuning; how much depends on capacity and on the task

If the lift were a specific consequence of the *contrastive-correctness* objective, an ordinary supervised fine-tune should not reproduce it. It largely does — and running the control at two very different capacities shows that "how much" is not a single number.

At the adapter's own capacity. The same-parameterization control (Section 3.4: identical 48 head-pairs, identical 12,336 trainable parameters, identical corpus/steps/split/checkpoint rule, ordinary cross-entropy on the correct answer only, three seeds) recovers **36.2**% ± **5.0** of the ARC lift, or 39.7% ± 3.1 if the fixed final step is used instead of the best-validation checkpoint — a ~36–40% range. Because the intervention space is held exactly fixed here, this is the arm that attributes recovery to the objective as such.

At 1,800×**–**58,000× **capacity.** A non-contrastive Align-LoRA on the same data budget, run through the same causal recovery machinery (a sign-safe three-zone control gate), recovers about **62%** of the ARC lift, and that fraction is **flat across a rank sweep** (r8/32/64/256: 59/63/60/65), with the sweep slope's confidence interval spanning zero ($[-1.4, +3.4]$) and a paired-bootstrap fraction interval of [**51**%, **73**%]. That interval resamples the 400 items with replacement and carries the base, generic and contrastive arms together, so the positive inter-arm correlation the three arms actually have (item-level r=0.42 between base and generic) is retained instead of discarded. It is therefore an interval over *item sampling, conditional on the one r256 Align-LoRA we trained*: the generic arm here is a single training run, so no training-run variance enters, and the interval should not be read as covering how much a re-trained control would move. It is 44% narrower than, and contained in, the delta-method [45%, 85%] we report as the conservative reading (zone C is binned on the point estimate, 0.62, and is unchanged). One-sided, the fraction exceeds one half in 98.2% of resamples — one minus the one-sided bootstrap p-value, not a posterior probability. Under raw accuracy instead of length-normalized accuracy the point moves to 0.61 and that figure to 96.9%. Within that grid, then, the residual is not a gap more rank closes; but the comparison with the same-parameterization arm shows that capacity itself moves the fraction, from ~37% to ~62%. We therefore do not name a ceiling for generic fine-tuning, and we treat the contrastive-specific residual as *grid-relative*: ~60–64% of the ARC lift at equal capacity, ~38% against the higher-capacity arm.

One caveat we apply to ourselves, symmetric with the information-set discipline of Section 5.3: matching examples and optimizer steps does not match the effective training signal per step — the contrastive objective sees each example with an explicit correct/incorrect contrast where the plain objective sees only the correct completion — so these fractions attribute the lift to *fine-tuning on this data* rather than to any stronger claim of per-step informational equivalence. Indeed the task breakdown suggests that contrast is exactly what the harder cases need: the same three same-parameterization seeds recover 79.5% ± 1.4 of the Winogrande lift and 69.5% ± 6.4 of HellaSwag's — where scoring rewards the plausibility of a continuation — but 0.9% ± 3.3 of TruthfulQA-mc1's, where a plausible distractor must be beaten, despite TruthfulQA sitting in the control's own training corpus. ARC, at ~37%, falls between the two regimes.

We restrict this control to ARC because TruthfulQA's matched corpus leaks into its evaluation. Across the rank sweep the TQA recovered fraction rises monotonically (41/66/72/91), the signature of memorizing training examples that recur at test; held out on 20% of TQA, the highest-rank control collapses from 92% to 30% recovery and the rank sweep flattens (r8 = r256 = 0.524 over the full n=164 held-out slice), while the contrastive adapter still generalizes held-out (0.83). The collapse is resolvable at that n and we state the intervals rather than the points: the recovery fraction is 0.92 CI95 [0.79, 1.06] over the n=400 set where all four arms cover the same items, against 0.30 CI95 [0.05, 0.55] over the n=82 held-out stratum (paired bootstrap, 10,000 resamples), and the two intervals do not overlap. One caveat travels with the fraction: the arms do *not* cover the same items — base and the rank sweep were evaluated over the full 817, the contrastive arm under a 400-item limit — so any recovery fraction is only defined on their intersection, and it is computed there. The matched corpus (TQA train-80%, seed 0) overlaps the evaluation set; TQA is therefore excluded from the inducibility control, and the ~62% generic figure is an ARC result.

5.3 The lift does not reduce to an interpretable re-scoring of the output scalars

The inducibility result attributes most of the lift to generic fine-tuning, but it leaves a sharper question: whether the recovered signal is a simple, interpretable re-weighting of the scores the cold readout already produces (length, PMI, an answer prior, a calibration temperature). If so, the "lost" knowledge would be recoverable by post-hoc re-scoring alone, without touching the model's activations. It is not (Table 4). We build the strongest such re-scorer we can — a supervised, held-out, AUC-objective combination of the interpretable surface features (conditional log-likelihood, answer-only log-likelihood, character length, PMI) — and ask how well it separates gold from distractor on the base-wrong items. It caps at **AUC 0.574** (a held-out logistic fit at 0.458 and a non-linear GBM at 0.462, an on-test-tuned oracle stack at 0.502, PMI alone at 0.574), against the **adapter's 0.830** on the same items. PMI does not merely fail to match the adapter — applied *to* the adapter it *hurts* it (0.818 → 0.758), and the adapter's re-ranking is only **2.4%** explained by the surface features (R^2 0.024). The recovered signal therefore lives **upstream of the collapsed output scalars**: no re-scoring of those scalars recovers the lift, which answers the "your baseline is just under-fit" objection — the ceiling is supervised and held-out, not a hand-tuned stack.

We are deliberate about what this licenses. The comparison is *information-set-confounded* — the adapter mutates **activations**, whereas the surface stack re-weights **collapsed output scalars** — so the gap (0.830 vs 0.574) does **not** license "a magic off-axis direction" (a generic SFT recovers ~62% the same way, Section 5.2) nor "an empty channel" (the residual is 0.574 > chance, Section 5.1). The defensible claim is about the **channel**: the lossy readout discards signal that survives in the activations and is not reconstructible from its own outputs.

We can now sharpen the channel claim with a same-space probe. A logistic probe fit on the *same* per-head `o_proj`-input activations the adapter writes to — no fine-tuning, held out by item, on the same cold-MC option-continuation distribution, scaler→PCA→logistic with the preprocessing fit inside each fold and a within-item label-permutation null — recovers gold-vs-distractor on ARC base-wrong items at **AUC 0.955** (CI $[0.934, 0.973]$, permutation null 0.494), far above the **0.502** collapsed-output ceiling. A surface-form control — the same probe on length/format features of the option string — caps at **0.442** (≈ chance), so the 0.955 reads correctness the output scalars discard, not a length/format artifact. This **widens and cleans the channel gap** (0.502→0.955) relative to the scalar-stack version (0.574→0.830).

We are careful about what it does **not** license. First, the probe reads 12288 activation dimensions while the adapter is constrained to nudge output logits. The two are not on the same axis, so 0.955 is **not** "the probe beats the adapter," and we do not read it as a representational *edit*. Second, that a linear probe decodes correctness from activations at ~0.95 is a near-universal property of capable language models [3, 38]. It shows the information *exists* in the activations the readout collapses, **not** that the model's own deployed readout is "merely miscalibrated." We therefore make a **sharpened channel claim** — the cold-MC output-scalar channel is lossy relative to the activation channel — and we do **not** assert "pure miscalibration" or "signal intact upstream" as anything beyond that channel statement.

Third, the recoverable signal is **broadly distributed**: a random 48-head set decodes it as well (0.949 ≈ the adapter's 48 heads), so the probe speaks to the cold-MC channel being lossy *model-wide*, not to the adapter's subspace being privileged. This is a read-test about the channel, decoupled from the write-tests that establish the adapter's specificity (a random norm-matched *offset* recovers ~0% of the *lift*, Section 4.4, a different operation). On TruthfulQA the same probe reaches 0.974, but its surface-form control is non-trivial (0.639, with a gold-first position artifact at 1.000), consistent with TruthfulQA's answer-prior character (Section 5.8); the channel claim is therefore anchored on ARC.

5.4 A read-only selector deploys the recovered signal without the pathologies

The channel result has a constructive consequence — but we scope it as the *read leg of a read≠write dissociation*, not as a new method, because the bare capability is already published. That a read-only probe or readout can *select* multiple-choice answers and recover an accuracy lift, with no weight edit and no generation, is established by at least two independent mechanisms: NoVo selects answers by voting over truth-correlated attention-head norms (+19 points on TruthfulQA-MC1, gains on >90% of 20 datasets) [28], and CORAL re-scores per-option probabilities with a frozen weight-decay correctness probe (+14% accuracy on ARC-Challenge/HellaSwag/OpenBookQA/Math-MC, 7B models) [40]. Both present the selector as a calibration/accuracy *fix*; our use is narrower and different — the read-only selector is the *evidence* that the lift the adapter writes off-axis is recoverable read-only, on-axis, without the write's damage (a read≠write dissociation neither studies, lacking an off-axis adapter to audit). If the correctness signal the cold readout discards is *present and read-only recoverable* in the activations, then one can recover the MC lift without writing any offset at all — and therefore without the deployment pathologies the write incurs (Section 5.5).

We deploy the same-space probe of Section 5.3 as a **read-only selector**: for each cold-MC item it reads the per-head activations and picks the option the probe scores highest, with no offset injected and no change to generation. On ARC it reaches selector accuracy **0.943** (± 0.002 over three PCA/fit seeds; the outer grouped-CV split is deterministic and identical across the three, so this dispersion measures algorithmic sensitivity and *not* sampling uncertainty; a repeated-partition estimate is given below), against the base's 0.505 and the adapter's 0.853, and the gain is not a correct-for-wrong trade: relative to the base it fixes **181** of the 198 base-wrong items and breaks only **5** of the 202 base-right ones (exact McNemar $p = 3.7 \times 10^{-47}$; an accuracy-space permutation null sits at 0.255). Because it never touches the weights or the decoder, it carries none of the adapter's costs — no retrieval spillover, no generation loss (no reasoning breakage) — so it **outscores the adapter at the same measured latency**. Whether it also wins on deployment grounds — against the *reasoning* baseline the no-headroom arbiter (Section 5.1, base 124/124 under CoT) suggests, and against the trivial one-forward baseline — is settled below by the deployment triangle: it does **not**. A reasoning TTC ties the most accurate baseline, and a single-forward letter-emission of the base beats the selector on accuracy at *half* its measured latency — which is exactly why we hold the selector to its *evidentiary* role (read≠write) rather than a deployment win.

We hold the claim to its read side and guard against an over-reading: 0.943 exceeding the adapter's 0.853 is **not** "the probe beats the adapter as a capability" — the probe reads 12288 activation dimensions while the adapter is constrained to nudge collapsed output logits, so the two are not on the same axis (Section 5.3). The claim is that the cold-MC channel *discards* a signal a linear reader recovers read-only. One caveat bounds the contribution to Tier-1: the 48-cell feature set is held out by *item* but selected on the training split, not held out by *feature*, so the decisive strengthening is cross-benchmark (ARC-Easy / OpenBookQA), which we flag as the immediate follow-up.

How much the 0.943 moves. The ±0.002 above varies only the algorithmic seed, so we re-estimate the selector over $R = 20$ genuinely distinct outer partitions — item-grouped by construction, so no option of an item crosses the train/test boundary, with the pipeline (scaler, PCA-200, L_2 logistic, inner C selection) and the algorithmic seed held fixed, leaving the partition as the only varying factor. That gives $\mathbf{0.9455} \pm \mathbf{0.0040}$ (range 0.9400–0.9550; the published configuration re-run in the same environment reproduces its 0.9433 ± 0.0024 exactly, so the two dispersions are comparable). Partition dispersion is thus $\mathbf{1.70}\times$ the published one: larger, as the objection anticipates, but below both the pre-registered 2× trigger and 0.4 pp in absolute terms, and the deterministic split proves if anything mildly conservative rather than favorable (the mean moves +0.0022, within one partition SD). No claim here moves; the recovery fraction computed at the partition mean is 1.266 against the published 1.260. Treating the two sources as independent puts total dispersion near 0.005, and that — not the ±0.002 — is the honest figure for how far 0.943 travels. It remains partition/refit variance and *not* sampling uncertainty: repeated cross-validation re-partitions the *same* n=400 draw, so replication outside that slice stays open.

The deployment triangle (Figure 2). The read-only selector is neither the strongest nor the fastest good way to answer these items — both distinctions belong to the base model asked correctly. On the 198 base-wrong ARC items we benchmarked four deployment routes on one homogeneous cost unit: median (p50) end-to-end latency per item at batch 1 on an RTX PRO 6000 (bf16), reported alongside the tokens each route actually consumes (Table 11). This matters because "one forward" is not one unit of work: cold-MC scoring and the selector must score *every option continuation* separately (~4 per ARC item, ~24.7 scored continuation tokens), whereas letter emission is a single forward plus two generated tokens. Everything in this paragraph is a statement about the four implementations we timed. In particular our harness issues an item's ~4 option forwards *serially at batch 1*; an implementation that batched them within the item is a fifth route we did not build or measure, so the latency gap below should be read as a property of these implementations rather than of scoring-vs-emission as such.

Measured, the routes are: the contrastive adapter's cold-MC scoring recovers **0.758** at **354 ms**; the read-only selector **0.911** at **360 ms**; prompting the base to emit the answer letter in one greedy forward, no chain of thought, recovers **0.970** (192/198; 2 items unparsed and counted wrong, so a floor) at **181 ms**; and a reasoning test-time-compute baseline (base self-consistency CoT, k=3) also recovers **0.970**, at **26.7 s** and 853.9 generated tokens summed over its three traces. So letter emission is not merely a same-latency tie — among the four routes as implemented it is both the fastest and the most accurate: it runs at about half the measured latency of the adapter and the selector, beats the adapter by 21 points and the selector by 6, and matches the reasoning baseline at ~150× less latency. The adapter scores below every other route, and is slower than the one that scores highest.

One caveat scopes this exactly as it scopes the arbiter (Section 3.2): letter emission puts the base in a lettered-MCQ format, a *different readout* from the cold-MC option-continuation scoring the adapter operates on — not a like-for-like scoring comparison. That is the point rather than a confound: it compares two readouts of the *same base* — loglik continuation ranking vs letter emission — which move ARC accuracy from 0.505 to 0.980 over the full n=400 evaluation set (0→0.970 on the base-wrong subset the triangle uses), with the faster-as-implemented readout winning. This is the paper's thesis, that the cold-MC channel is lossy, in its sharpest form: the knowledge is present and reachable by a different readout of the base, and the adapter's off-axis write recovers *less* of it (0.758) than the most trivial alternative does (0.970) — at lower measured latency in our implementations.

The selector outscores only the *adapter*; a reasoning TTC and a one-forward letter-emission of the base both beat it — so the LoFiT/ITI-for-cold-MC objective, as instantiated here, optimizes a readout that even asking the base for the letter already beats. We therefore report the selector as *evidentiary* (the correctness signal is read-only recoverable, read≠write) and **not** as a deployment win (letter emission outscores it at half its measured latency, on the implementations we timed). *Population note (do not conflate):* the four figures here (letter-emission 0.970, SC-CoT 0.970, selector 0.911, adapter 0.758) are all on the **198 base-wrong** subset; the selector's all-items accuracy (**0.943**, three PCA/fit seeds, across all n=400 evaluation items) and the adapter's all-items accuracy (**0.853/0.855**; Table 1) are computed on a *different* population and are not directly comparable to the 0.911 / 0.758 here. The 0.943 and 0.911 are the same selector on two populations, not two systems (and the canonical lm-eval value on this subset is 0.914; the 0.911 is the same selector re-fit inside the cost harness, see the table note). "All items" here means the n=400 harness sample, **not** the full ARC-Challenge test split — that phrase is reserved for the R4 full-split run (Section 6, ARC acc_norm +36.2).

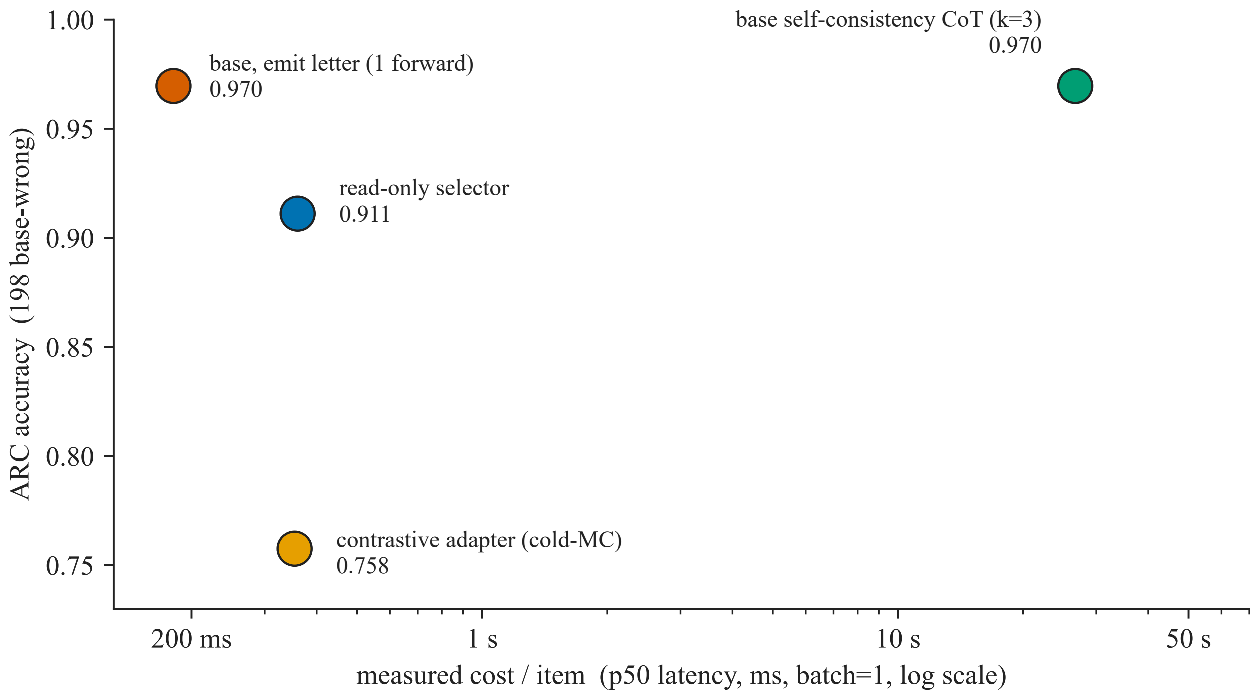

Figure 2: The deployment triangle on the 198 base-wrong ARC items, on measured cost. x: median (p50) per-item latency at batch 1, bf16, RTX PRO 6000 (log scale); y: ARC accuracy. One homogeneous unit for all four routes, replacing the earlier mixed forwards/tokens proxy — "one forward" is not one unit of work, since cold-MC scoring and the selector score every option continuation separately (~4 per item) while letter emission is one forward plus two tokens. Four routes: the contrastive adapter's cold-MC scoring (0.758, 354 ms) scores below every other route and is slower than the most accurate one; the read-only selector (0.911, 360 ms) improves on it read-only; **the base asked to emit the answer letter in one forward (0.970, 181 ms) is, among these four implementations, both the fastest and the most accurate — it matches the reasoning baseline at ~150× less latency and outscores the adapter and the selector at about half their measured latency**; base self-consistency CoT (0.970, 26.7 s, 853.9 generated tokens over three traces) ties letter emission at far higher cost. The LoFiT/ITI-for-cold-MC recipe, as we instantiate it — one adapter of that family, not LoFiT's or ITI's published recipe — optimizes a readout that the most trivial alternative — ask the base for the letter — already beats. **The comparison is evidentiary, not a like-for-like scoring test:** letter emission is a different (lettered-MCQ) readout than the cold-MC channel (Section 3.2), so the triangle contrasts two readouts of the same base — the lossy-readout thesis in its sharpest form — and the selector's role stays evidentiary (read≠write), not a deployment recommendation, since letter emission outscores it too. Cost is latency on this one accelerator at batch 1; we do not normalize FLOPs across readouts, and batched throughput (reported in Table 11) is not comparable between prefill-bound scoring and decode-bound generation. **The latency axis describes the implementations we timed:** the scoring routes issue an item's ~4 option forwards serially at batch 1, and we did not measure a variant that batches an item's options, so the ordering on x is not a claim about the routes in principle.

read ≠ write, geometrically. The selector and the off-axis write together make a four-way dissociation the paper rests on. A reader *reads* correctness from the activations (selector 0.943); an *on-axis* write fails to recover the lift (W_{know} 6.4%); an *off-axis* write succeeds (off$_{\perp}$ ~100%, shuffle ~0%, Section 4.4); and — closing the loop — the probe's decision direction is geometrically aligned with the correctness axis it reads, *not* with the axis the adapter writes. Per head, the probe direction's signed cosine with the functional correctness direction v_{inf} is +0.22 (median +0.20, above the ~0.06 null band), while its cosine with the adapter's write direction θ_{mc} is |0.066| (at the edge of the ~0.061 per-head null band, not clearly within); pooled, +0.209 vs |0.008| (clearly within the null — the read≠write conclusion rests on the pooled statistic). The read axis is v_{inf}; the write axis is off-axis to it. We mark this geometric leg **moderate, not a slam-dunk**: the alignment is

Table 11: Deployment routes on the 198 base-wrong ARC items, measured cost (the numeric view of Figure 2). Latency is median (p50) / 95th percentile per item at batch 1, bf16, single RTX PRO 6000; tokens are generated tokens for the generation routes and scored continuation tokens for the scoring routes. Batched throughput is listed for completeness but is *not* comparable across route families (prefill-bound scoring vs decode-bound generation). The latency column describes these implementations: the scoring routes issue an item's ~4 option forwards serially at batch 1, and a variant batching an item's options was not measured.

Deployment route	ARC accuracy (198 base-wrong)	Latency p50 / p95	Tokens / item	Relation
Base letter-emission (1 forward, no CoT)	**0.970** (192/198; 2 unparsed as wrong → floor)	**181 / 253 ms**	2.0 generated	**fastest *and* most accurate of the four as implemented: matches SC-CoT, outscores selector and adapter at ~half their latency**
Base self-consistency CoT (k=3)	**0.970** (192/198; k=5 identical → saturates at k=3)	26.7 / 36.6 s	853.9 generated (sum over 3 traces)	ties letter-emission at ~150× the latency
Read-only selector (Section 5.4)	**0.911** (5 probe seeds)	360 / 479 ms	24.7 scored (~4 options)	below letter-emission (2× its measured latency, −6pp); outscores only the *adapter*
Contrastive adapter (cold-MC scoring)	**0.758**	354 / 471 ms	24.7 scored (~4 options)	lowest accuracy of the four; slower than the most accurate

Provenance and one reconciliation. Measured in a single harness that re-derives every route's per-item correctness and validates it against the canonical lm-eval artifacts before timing (`runs/e3_cost/`). Within that harness the adapter scores 0.758 (150/198) and the base cold-MC arm 0.5025, against 0.7626 (151/198) and 0.5050 in lm-eval: a one-item difference in both arms, from the context/continuation tokenisation boundary in the re-implemented log-likelihood. The canonical accuracies elsewhere in this paper are the lm-eval ones; the 0.758 here is the number produced by the same code path that produced the latencies, which is why we report it in this table rather than silently mixing sources.

0.22 rather than 0.5, fold-stability is 0.887 (marginal at $n \ll d = 12288$), and v_{inf} lives strongly on the top principal component (cos 0.55) — but the probe reads v_{inf} 66% *beyond* PC1 (its PC1-routed contribution, $0.134 \times 0.55 = 0.074$, is well under the 0.219 total), so the alignment is a genuine read, not a by-product of dominant variance. The read ≠ write conclusion does not hinge on this leg — it is already carried by the probe (read), the W_{know} null (on-axis write fails), and the off$_\perp$/shuffle pair (off-axis write succeeds) — but the geometry confirms it directionally.

The measured geometry behind that read≠write claim — the matched-control spectrum, the on-axis reference and the direct causal decomposition — is reported with its tables in Sections 4.4 and 4.5; what remains here is what the write costs.

5.5 The deployment pathologies are coupled to injection, not to the lift

Recovering the lift the contrastive way introduces two deployment costs — no transfer to generative CoT (a mechanistic breakage of multi-step reasoning) and retrieval/recall spillover. (A third cost we previously listed — an end-of-turn "collapse" — is retracted at the end of this section as a measurement artifact.) A clean control shows the two surviving costs are coupled to *how* the offset is injected, not to the lift as such. The generic Align-LoRA that recovers most of the lift (Section 5.2) **preserves generation**: gsm8k 0.96→0.95 (n.s.), where the folded contrastive offset destroys it (gsm8k →0.17 baked, and −25 pp adapter-applied; below).

One caveat on that control: the two arms differ in the *form* of the update as well as in its axis, and low-rank adaptation is separately documented to preserve out-of-domain behaviour better than full updates [7] — an account of the dissociation we do not exclude here.

Weight-space edits are an established way to steer a model along a direction, and the paradigm reports *little* change on control tasks [31]; the spillover here runs the other way, and is closer to the unintended downstream consequences documented for knowledge edits [17]. The retrieval spillover, in turn, is consistent with the off-axis push landing on **retrieval-load-bearing heads** — the adapter's heads overlap heads that carry retrieval (in the D.1 SocialIQA adapter: MMLU −19.59 pp, unrecovered by sparsity tuning from K=24 to K=12; correlational, per-head causal test pending). We keep this head-level overlap claim distinct from the Section 4.4 finding that the two functional correctness *faces* are co-located (cos $v_{\text{inf}} \cdot v_{\text{ret}}$ +0.283): face co-location implies that re-selecting heads within the same band cannot dodge the retrieval substrate, but it does **not** assert that the adapter's off-axis direction aligns with v_{ret}. That disjunction — off-axis perturbation of a shared substrate, or the lift partly projecting onto the retrieval face — is settled by measurement in Section 4.4, and it settles toward the first: $\cos_{\text{w}}(\theta_{\text{mc}}, v_{\text{ret}}) = +0.0076$ (BCa95 $[-0.0033, +0.0187]$) sits inside its isotropic null, while the same estimator on the same 48 heads puts the two faces at +0.2586. One caveat runs *against* the result and we state it: the mean $|\cos_{\text{w}}|$ of θ_{mc} against either face (0.0333 and 0.0314) falls *below* the isotropic band for that statistic ($[0.0397, 0.0610]$), so these whitened cosines are smaller in magnitude than random ones and the band is a conservative floor test, not a generous one. The pathologies are the downstream shadow of folding an off-axis-of-degree perturbation into the weights, not an inevitable price of the lift: a different injection (generic SFT) recovers much of the lift without them.

A *third* injection mode confirms the coupling from the other side: blending the offset in at **decode time** at reduced strength — a milder intervention than folding it into the weights — still buys no generative transfer. Sweeping the blend coefficient ε on gsm8k (n=100, $\varepsilon \in \{0, 0.2, 0.5, 1.0\}$), no interior ε beats the base by more than noise (base 0.97; ε=0.2 → 0.98, ε=0.5 → 0.99, both within 1–2 items of base at n=100 and under the 2 pp equivalence tolerance the pre-registration fixes for its reproduction and budget-flatness checks, which we reuse as a noise band on sweeps it does not itself cover), while full-strength decode-time injection (ε=1) reproduces the loss (0.87, −10 pp) — there is no transfer window, so the off-axis lift is not deployable to generation even as a decode-time steering vector. (Its ε=1 −10 pp is milder than the −25 pp of the fully-baked adapter — the blend harness perturbs a subset of heads at generation only — so we read it qualitatively, not as a magnitude.) Even at ε=1 the end-of-turn rate stays at 0.97, independently corroborating the retract below: the generative cost is degraded reasoning, not a failure to terminate.

The contrastive adapter's generative loss on gsm8k is itself a mechanistic loss, not a truncated chain of thought. Run generatively under an apples-to-apples protocol (chat template on, 5-shot, n=400), the adapter scores **0.7275** strict-match against the base's **0.9775** (**−25** pp; R6, the canonical kill-rule run, with an earlier provisional R6probe agreeing at −24 pp). That gap is **flat across a generation-budget sweep**: 0.7275 at caps {512, 2048} (R6probe: 0.7375 flat at {512, 1024, 2048}), with the base **byte-identical** across caps. The base's reasoning runs well under the smallest cap, so the cap never bites it. Turning off the chat template only makes the gap *worse* (base 0.730 / v7mc 0.035, −69 pp): both arms degrade, so the template-ON apples-to-apples figure is the canonical one. The −25 is therefore not a template artifact.

The adapter *does* consume the larger budget — 26/400 of its generations grow with the cap — yet **0/400 documents change strict verdict**. A forensic split of the 105 strict failures at cap 2048 separates the two failure modes: **79** are clean-wrong (they emit a #### answer with the wrong number — reasoning resolved but defective, independent of the cap), and the remaining **26** are genuine non-terminating *reasoning* loops (a real task-specific failure, distinct from the retracted factual-recall artifact below) *already* present at cap 512 (e.g. doc 88: a "... not sold by Harald..." loop filling the budget, no ####), i.e. the failure *precedes* the budget rather than being caused by truncation.

The −25 pp is therefore a mechanistic breakage of the reasoning chain — coupled to the contrastive injection, consistent with Section 5.1's no-transfer — not a measurement artifact of a clipped CoT. These 26 loops are a *task-specific* reasoning failure on gsm8k, **not** evidence of a general termination collapse (which the corrected re-measure below refutes on direct recall).

(We retain the gsm8k caveat: it is OOD and does not measure in-domain transfer, so it bounds the *collateral* claim, not the no-transfer claim of Section 5.1.)

Degeneration, not shorter reasoning — measured on two benchmarks. The gsm8k loops above were found by reading failures. We then measured the same thing systematically, on two multiple-choice benchmarks run generatively, scoring every trace by its distinct-4 ratio and calling a trace degenerate below 0.35. The rates order the injection modes consistently and put the base at the floor in both:

	base	contrastive	Align-LoRA	plain CE (3 seeds)		
				s0	s1	s2
MMLU-Pro (n=1000)	**0**	146	—	298	—	—
ARC (n=1172)	**0**	33	5	**323**	151	278

Three things make this a measurement rather than a threshold artifact. The base's *worst* trace on ARC scores 0.803 — it never approaches the cutoff, so the zero is not a near-miss. Sweeping the cutoff from 0.20 to 0.60 moves the s1 arm's count from 147 to 156: the split is categorical, not a knob. And the degenerate traces are what exhausts the generation budget rather than long reasoning — median lengths of 23–54 tokens with 13–34% of traces at the cap is a bimodal distribution, and 311 of the s0 arm's 397 unparsed traces are loops.

The ordering — plain CE ≫ Align-LoRA > contrastive, base at zero — is the same on both benchmarks, and it complicates the clean story of the preceding paragraphs in a way we state rather than smooth over: Align-LoRA *preserves* gsm8k accuracy while still degenerating five times where the base degenerates none. Preserving a score and preserving generation are not the same property.

On ARC the plain-CE arms also lose chain-of-thought accuracy under *both* brackets of the three-way split, which the MMLU-Pro run did not establish: −13.99 and −15.02 pp for two seeds, disjoint from base on floor and ceiling alike, because the base's floor is 0.97526 and the arms cannot reach it even if every truncated trace is granted a correct answer. Three caveats bind those numbers and none of them is optional. The third seed's −34.04 rests on a two-item margin and is not cited alone. The seeds cannot be *ordered* against each other, because the arm that truncates most has both the lowest floor and the highest ceiling — the ranking inverts with the bracket, so the magnitude is not quotable without naming the seed. And the arms ran at a generation cap of 512 against the base's 640: monotonicity of the bound means a smaller budget cannot manufacture the disjunction (a truncated trace granted correctness can only lower the ceiling when it closes), so the *sign* survives the asymmetry — but the magnitudes are measured against a comparand with more budget and we do not read them as effect sizes. None of these cells is replicated.

The dissociative triangle. Three facts about the contrastive adapter's generative behavior dissociate, and we state them together because the gsm8k loss could otherwise be read as a blanket generative failure. (a) *Direct factual recall is clean:* on single-hop recall the adapter terminates at 100% and answers tersely and correctly — the earlier 48→4% / 60→4% "collapse" is a **retracted** eos-token measurement artifact (below). (b) *Multi-step reasoning is broken:* on gsm8k the adapter loses **−25** pp, a mechanistic breakage (flat across a generation-budget sweep; 26 non-terminating reasoning loops already present at cap 512), not truncation and not a termination failure. (c) *The damage is coupled to injection, not to the lift:* the generic SFT that recovers most of the lift preserves gsm8k (0.96→0.95), and the reasoning loss is reproduced across **four independent injection modes** — folded/baked (−25 pp), decode-time blend (ε=1, −10 pp), activation-scale steering (damage confined to the 0.75→1.0 band), and confirmed from the other side by the Align control that recovers the lift *without* the loss — so it is a property of *how* the offset enters the weights, not of the lift.

A confound we state rather than hide: the (a)-clean-vs-(b)-broken contrast conflates two things — multi-step *reasoning* and *distribution distance* (gsm8k is OOD to the tqa+arc training) — so gsm8k bounds the *collateral reasoning cost*, not the no-transfer claim. The no-transfer per se does **not** rest on gsm8k: it is carried by the per-item arbiter (the fixed ARC items sit at the base's CoT ceiling, 124/124, Section 5.1) and by the out-of-task GPQA cold-MC lift ~0 while base GPQA CoT is 0.71 (Section 4.2). gsm8k carries only the *collateral damage to reasoning*, and that damage is the triangulated four-injection-mode result above.

The collateral reasoning loss is dose-separable from the scoring benefit — in steering, not in the baked adapter. The ε-sweep above varied a decode-time *blend* of one face; a cleaner test places both faces on a *single* magnitude axis by scaling the injected offset itself (activation-scale $s \in \{0, 0.25, 0.5, 0.75, 1.0\}$; s=1 reproduces the adapter's scoring lift — ARC acc_norm 0.505→0.855, +35.0 pp ≈ the baked +35, raw acc 0.5025). On this common axis the two faces cross their thresholds at *different* doses: the scoring benefit **saturates early**, reaching its full value already at s=0.75 (ARC 0.855, identical to s=1.0), while the gsm8k reasoning score holds at base through s=0.75 (0.970→0.950, at the edge of that same 2 pp tolerance) and drops only in the 0.75→1.0 band (→ 0.815). The scoring-benefit threshold thus lies *below* the reasoning-harm threshold, so the 0.75→1.0 magnitude is collateral damage with no accompanying scoring gain (the lift is already at ceiling).

Two qualifications on that band, from a denser sweep between the anchors ($s \in \{0.75, 0.80, \ldots, 1.00\}$, n=200 per cell) that reproduced both endpoints digit for digit. First, the quoted accuracies are *floors*: they count every truncated trace as wrong. The anchors survive that reading — base [0.9700, 0.9750] against s=0.75's [0.9500, 0.9550] and s=1.0's [0.8150, 0.8650] are disjoint either way — but the *size* of the drop across the band is not one number: −13.50 pp on floors, −9.69 among traces that closed, −9.00 on upper bounds. Under the bounds reading, $(13.50 - 9.00)/13.50 = 33\%$ of the floor-to-floor fall is exhausted budget rather than degraded reasoning; the figure is 28% if one instead conditions on the traces that closed, which conditions on a post-treatment variable and is why we quote the bounds. We therefore give the band's magnitude only with the definition attached.

Second, the fall is **progressive, not a cliff**. Under all three definitions five sub-intervals are negative, none carries more than a third of the total, and the largest single step is 0.85→0.90 in each. Against s=0.75 the drop only becomes declarable (disjoint bounds) from s=0.85 on; s=0.80 is not separable from it. Nothing about the dose separation depends on a sharp edge at 0.75 — what the data support is an accumulating cost whose onset sits below the dose at which scoring stops paying, which is the claim the preceding paragraph makes.

Degeneration turns out to sit on the same axis, and to switch off before the lift does. Repeating the dose sweep on MMLU-Pro (n=1000 scored, n=300 generated), the scoring lift is already complete at half dose — +11.40 pp CI [+8.80, +14.10] at s=0.50 against +12.20 at full dose, indistinguishable — while the repetition loops of the preceding paragraphs go 146 → 10 → **0** across $s = 1.00, 0.75, 0.50$. At half dose the worst trace scores 0.406 on the degeneracy metric: not a marginal pass. That the saturation point replicates on a ten-option benchmark with a third of the effect size makes it unlikely to be a property of ARC.

We flag two things about this rather than sell it. It is a steering result and the paper's headline arms are baked, so it bounds what an operating point could look like, not what the adapter is. And a companion measurement of ours did not survive its own control: an apparent +9.02 pp of the half-dose arm *over* the base collapsed to +2.52 CI [−1.86, +7.01] once both arms were given a larger generation budget, because 100% of the base's improvement came from items that had been truncated. The direction that survives is narrower and worth stating plainly — the adapter does not stop early, it degenerates: given a budget of 2,560 tokens it generates *more* than the base (1,488 against 1,398 on the hardest items) and hits the cap six times as often, converting the extra budget into repetition rather than reasoning. Any intervention that lengthens the intervened arm's chain of thought inherits that risk and should report its loop rate beside its accuracy.

This does **not** reopen a transfer window. We are careful with the wording, because two interior points of the ε-sweep above do print *numerically* over base (ε=0.2 → 0.98 and ε=0.5 → 0.99 against base 0.97): at n=100 those are one- and two-item differences, inside that 2 pp tolerance, so we read them as noise and not as a gain — no dose produces a reasoning improvement we would defend, but the honest statement is "no dose rises above base beyond noise," not "no dose rises above base." What is dose-separable is the *collateral damage*, not a deployable gain. The separation is a property of the **injected, scaled offset (steering); the** −25 pp **headline is the fully-baked adapter at** s=1.0, so whether a *trainable* adapter can be held in the low-damage regime is an open question, not a result here (the s-swept gsm8k figures use the steering harness, n=200, and are read as curve *shape*, not against the canonical −25 pp).

We name one prior-art foil directly: an angle–norm account of steering [2] predicts on geometric grounds that a concept-discriminative (angular) readout should saturate at lower magnitude than generative coherence (norm-sensitive) breaks — so a reviewer may read our dose separation as geometrically expected. We distinguish it on three measured grounds that account does not cover: the offset is **off-axis** to the concept direction (Section 4.4, not an on-concept angular push), its mechanism is **non-linear** (the direct-logit reading under-reads it, Section 5.6), and it is **not a norm/temperature rescaling** (cosine with the temperature direction 0.023, Section 5.6) — so the separation here is not the generic angular-saturates / radial-breaks story, and we report it as a dose-refinement of the *collateral cost*, not as a deployable window.

An independent result in a different regime corroborates the *shape* of this dissociation from the write side. A budget-normalized **control-window** law for single-neuron steering [36] finds that coherent behavioral control saturates below a **collapse ceiling**, and — its titular point — that *leverage is not reach*: a single neuron can flip a behavior's readout face (e.g. bypass refusal) in fluent, on-task text while genuine downstream reach appears only later and only for a subset of pivots (three of six audited). That is the single-neuron analogue of our dissociation — control of the *scoring* face at a dose below where generative coherence breaks, with the scoring gain never buying generative reach (Section 5.1) — obtained in an unrelated setting (sparse behavior-gating neurons, not a multi-head correctness offset), so we read it as convergent support for the threshold structure, not as the same experiment.

A previously-claimed end-of-turn collapse is retracted as an eos-token artifact. We earlier reported that the contrastive family collapses end-of-turn termination (V7-mc / D.1 factual 48%→4%, social 56%→4%, an analogous V7-mc 60%→4%). That is a measurement artifact: the verbosity harness (`eval_adaptive_verbosity`) scored termination on the base <eos> token, whereas Gemma 4 closes a conversational turn with <end_of_turn> (token 106). With the corrected turn-stop token, base factual termination is **100%** (reported 48%) and the contrastive adapter's factual termination is also **100%** (reported 4%) — there is *no* collapse; the adapter's factual answers are correct, terse, and terminate ("Paris", "Au", "1945"), and the "degenerate loop" read earlier in this *factual-recall* smoke was junk generated *after* the <end_of_turn> the harness had failed to stop on — a measurement artifact, distinct from the genuine gsm8k *reasoning* loops above.

The honest reframe is that the contrastive adapter **does direct factual recall** (terse, correct, terminating — no degradation vs the base, which is also correct but verbose) and **breaks multi-step reasoning** (gsm8k −25 pp) — which fits the per-item arbiter (Section 5.1: the fixed items are knowledge the base reaches by *reasoning*) and the no-transfer result better than a blanket "cannot terminate" claim did.

The analogous D.1 figures were re-measured: the →4% "collapse" is withdrawn, but a real, modest termination degradation remains on open-ended generation (social 96→76%, creative 68→12%) — adapter-dependent and off the V7-mc headline; this paper makes no →4% termination-collapse claim.

5.6 Mechanism candidates ruled out: not temperature, not token-frequency (a content/symbol signpost)

Two mechanistic accounts of the offset are ruled out, and one is left as a signpost; the exploratory workspace and generation-face analyses that extend this section are collected in Section A and carry no claim here.

Not a temperature / null-space mechanism. A natural hypothesis, given the readout framing, is that the offset acts like a confidence-regulation / entropy neuron — writing into the unembedding null space to rescale the logit temperature without re-ordering the argmax [49]. It does not. A direct-logit decomposition projects the residual write onto the *genuine* temperature direction $d^* = \arg\min \|W_U r - \mathbf{1}\|$ (whose own logit image is 0.9988 uniform, so it is well identified) and finds cosine **0.023** — orthogonal to temperature at chance level (random-direction baseline $\sim 1/\sqrt{d_{\text{model}}} \approx 0.014$). Causal activation-patching at the scoring positions where the lift lives (the option-continuation tokens) confirms it: the **relative** temperature fraction is only **0.215** (CI_{hi} 0.224 ≤ the pre-registered 0.35 kill-rule; in absolute fractions of the effect, temperature 0.177 / ranking 0.803), so the offset re-ranks rather than rescales — it moves logit *differences*, not their mean. *Position matters:* at the next-token generation position the offset flattens the distribution more (temperature fraction 0.38), so the negative is anchored at the scoring positions. (Full decomposition — the d^* identification, the anisotropy caveat on the raw uniform-shift fraction, the causal cosine −0.149, and the linear-attribution-under-reads-the-off-axis-alignment pattern [36] — in Section A.)

Not a token-frequency edit; a content/symbol signpost. Direct-logit attribution over the 48 heads finds **no clean lexical mechanism**; the one robust signature is on the *suppression* side — single uppercase letters, punctuation and structural tokens, exactly the surface tokens of multiple-choice labels and turn formatting — the readable face of the *format-preference* channel that aligned-model calibration distinguishes from answer-decision uncertainty [25]. The most promising unifying candidate is the content/symbol-binding channel MCQA mechanism work isolates [34, 61], but the decisive TEXT-vs-LABEL scoring ablation is **consistent-with, not clean**: on ARC the label-arm lift survives a headroom normalization (Δ_{letter} +0.1575 raw / +0.17 norm) yet the permutation controls that would isolate letter-specificity are ceiling-starved, and on TruthfulQA the letter arm is itself at ceiling (uninformative). We report the symbol channel as a signpost, not a named mechanism. The symmetric cosine test for this third candidate — the same construction standard the temperature and frequency negatives received — has now run: a *mean* structural-token contrast direction (the readout-row difference of means between the ~27k structural tokens and the rest; exact by construction, and its logit image separates the classes cleanly) has cosine **−0.0099** with the offset's aggregate write (**−0.0115** for the uppercase-letter-only variant), at the null floor ($1/\sqrt{d} \approx 0.014$, 3×-null gate 0.041) like the other two; a *full-target* format direction is not linearly identifiable in the unembedding basis at all (identification $r = 0.78 <$ the 0.9 self-gate that d_{freq} passed), consistent with the suppression signature living in the *tail* of the offset's logit image rather than in its mean — which is exactly why the channel stays a signpost: the offset is not the mean format direction either (`runs/oq1_functional_axis/format_control.json`). The *other* Stolfo component — a token-frequency edit — is **closed at the attribution level**: over the full wikitext-103 corpus (121.6M tokens) the recovered frequency direction is identified at r=**0.906** (past its 0.9 self-gate) and the offset's cosine with it is **0.0128** (at the null floor, below the 3×-null gate 0.041), so the offset is not a token-frequency edit. One scope note travels with it: unlike the temperature negative, the frequency control had not received a causal-patching cross-check — and the temperature case showed the causal alignment can exceed the linear read severalfold (0.023 → −0.149) — so we ran that cross-check rather than assume it. It comes back **negative for a sign-flip**: projecting the real base→adapter logit delta onto the frequency signature gives cosine **+0.0059**, CI [−0.0027, +0.0142] — same sign as the direct-logit 0.0128, smaller (×0.46), and crossing zero — where the temperature case flipped sign and grew sixfold

(0.023 → −0.149, CI [−0.158, −0.140]). The sharpest contrast is in sign *consistency* rather than magnitude: the per-item frequency projection has comparable mean absolute size to the temperature one (0.120 vs 0.177) but inconsistent sign (66% positive, against 99% for temperature), and a component whose sign varies item-to-item cannot be the mechanism of a lift that is systematic across items. One limit travels with the negative: the frequency direction *orthogonalized against the base logits* — the part of log-frequency not already shared with the temperature channel, which has no direct-logit counterpart — does carry a small but non-zero causal component (cosine **+0.0298**, CI [+0.0221, +0.0374], ~1/5 of the temperature component and ~1/27 of the ranking one). Frequency is therefore ruled out as the *mechanism*, not as a trace contaminant.

An exploratory Jacobian-lens analysis (single seed) further places θ_{mc} *off* the model's global workspace — an automatic-channel edit, the mechanistic shape a direction that lifts cold scoring but does not transfer to reasoning should have — and an nnsight causal patch on real generation positions finds the same structure-suppression signature (~91% structural, a ×3 enrichment); we report both, and the per-position recall-tail resolution, in Section A and rely on them for no claim above.

5.7 A low-dimensional description of what the arms change: two parameters per cell, none per item

Having ruled temperature and frequency out, we can describe what the arms *do* change to the score distribution. Write the per-item decision margin as $g = \log p(\text{gold}) - \log p(\text{best distractor})$ and compare an arm's margins against the base's on the same items.

What this section does and does not claim. g is a *gold-relative* coordinate: it is defined using the label, and so is every fit below. What has no per-item parameter is the *description* — two numbers estimated per model×arm×task cell, constant across that cell's items. That is a low-dimensional descriptive signature, and we read it as nothing more. It is not a label-free intervention: a map of this form could not be applied to the raw option scores without already knowing which option is gold, and an affine map applied to those raw scores could not change any arg max at all — it is precisely because the intercept acts on the gold-relative margin that accuracy moves. Nor is the absence of fitted per-item parameters evidence that the arms carry no per-item information; we test that separately below, and the test is not conclusive either. The ratio $\text{sd}(g_{\text{arm}})/\text{sd}(g_{\text{base}})$ splits the eighteen arm×task cells we have, across both families, without exception: **the contrastive objective in domain expands the margin scale** (2.10/1.50/1.95 on ARC / HellaSwag / TruthfulQA in Gemma, 1.47/1.63/1.81 in Qwen) and **everything else compresses it** to 0.41–0.72 — ordinary cross-entropy in either domain, and the contrastive objective itself once trained out of domain (0.63–0.70). That last cell is worth stating precisely: out of domain the objective keeps an accuracy advantage over plain cross-entropy at identical parameterization and corpus, but it lands on the *compressing* side of the split, losing the scale signature that in-domain training produces.

Scale is one scalar; the accuracy change needs a second, which is where the rescaled distribution sits relative to zero. Fitting $g_{\text{arm}} \approx a\, g_{\text{base}} + c$ per cell and reading off the effective decision threshold $-c/a$, the shift is large and negative wherever the arm helps (Gemma contrastive: −43.6/−24.7/−47.1 nats) and positive wherever it hurts (both plain-CE arms on TruthfulQA, +4.2 and +11.8). Its *sign* matches the sign of the measured accuracy change in 17 of the 18 cells; the exception is the one cell whose measured effect is itself null (+0.8 pp, a no-op in 2 of 3 seeds). Pushed all the way, a deterministic map built from those two global scalars alone — carrying no information about which item is which — reproduces the direction and rough size of every accuracy change with a non-zero effect to reproduce (17 of 18; the eighteenth measures exactly zero), over-explaining by a median of 29%, cell-wise range 0.70–1.93× the measured effect. We report that as sufficiency of *form*, not as a calibrated fit: the map is deterministic where the real arm is noisy, and noise attenuates.

That fit is in-sample, and a ratio of accuracy changes is a poor way to read it: its denominator is near zero in the small-effect cells, and a harmless-looking 1.43× on ARC is in fact a 15 pp absolute error. We therefore re-ran the whole comparison *cross-fitted* — 40× repeated 5-fold, so every item's prediction comes from parameters estimated without it — scoring the absolute error of the predicted accuracy change in percentage points, and against the trivial alternatives the affine map has to beat. Median absolute error across the eighteen cells: **affine 2.36** pp, intercept-only 4.53, slope-only 9.42, base-margin-ignoring constant 35.62. All fourteen cells with an effect of at least 3 pp are predicted with the correct sign.

Two things follow, and the second qualifies the subsection's own title. First, the map does generalize: it is not an artifact of fitting and scoring the same items, and it beats both one-parameter restrictions. Second, *the scale is not what moves the accuracy*. Slope-only scores identically to the identity map — necessarily, since for $a > 0$ the sign of $a\,g$ is the sign of g, so a pure rescaling cannot cross a single item over the decision boundary. It is the intercept that moves items, and the scale earns its place only by modulating how much intercept is needed (2.36 against 4.53 pp). The margin-scale split reported above is a real and exceptionless signature of the objective; it is not, by itself, the thing that changes the score.

The error is also not uniform: it concentrates in the largest-effect cells (canonical contrastive, 15.1 pp on ARC and 10.3 on TruthfulQA; Qwen contrastive 11.8 and 11.7), where the map over-predicts. Ten of eighteen cells fall within ±3 pp. So the affine description is adequate as a low-dimensional summary and weakest exactly where the phenomenon is strongest — which is the honest limit of reading it as an account of the canonical lift.

The obvious objection is that the arms might rescale *and* aim, so we bound the aiming rather than assume it away. The tempting null — permuting each arm's per-item push across all items — is invalid here, because any rescaling makes that push $(a - 1)\,g_{\text{base}} + c$, anticorrelated with the base margin by construction and without any knowledge of which option is correct; such a null therefore rejects for every arm that rescales, which is all of them. Permuting instead *within strata of the base margin* preserves the conditional law of the push and destroys only item identity, and under it 1 of 18 cells exceeds 2σ where the invalid control gave 9 of 18. The survivor does not hold up: selected post hoc from 18, it fails both Bonferroni and BH-FDR ($p = 0.0066$ against a threshold of 0.00278), and its significance rides on a free parameter — z runs from 1.30 to 3.43 as the stratum width goes from 10 items to 100. Its residual does differ from zero under a paired bootstrap (+1.12 pp, CI [+0.20, +2.09] at a stratum width of 25), so an item-specific component exists in that cell. We decline to quote a single number for its size, because the same free parameter moves it: the residual runs +0.28/+1.31/+1.66/+3.12 pp as the stratum width goes from 10 to 100 items, i.e. between ≈1% and ≈14% of that cell's +22.7 pp effect, with no width privileged a priori. Across the stratifications we tested, the largest observed residual share in the most favorable of the eighteen cells ranged from ≈1% to ≈14% of its effect. Because that range is sensitive to the stratum width and the cell was selected post hoc, we report it as a sensitivity range and *not* as a bound: it does not license a statement of the form "the item-specific component is at most x", and we make no such statement anywhere in this paper.

This is the readout thesis made mechanical, and it accounts for the task pattern without new machinery. ARC sits on the edge of its own decision boundary — median $|g|$ of 9.8 nats in Gemma and 8.5 in Qwen against 17–23 on HellaSwag and TruthfulQA, with 12.5% and 22.2% of its items inside 3 nats — so a global rescaling carries more items across zero there than anywhere else. (Density alone does not order the finer structure: in Gemma, HellaSwag at 7.0% and TruthfulQA at 7.25% are tied while their effects differ, so we report it as a descriptor and not a predictor.) An edit that moves a threshold need not know which questions it is fixing, which is why the items it fixes turn out to be ones the base could already reach by reasoning (Section 5.1), and why the same push has nothing to offer a generation channel that never scores the options against each other at all.

Three limits travel with this. It is measured on raw accuracy throughout, which is the right scale for a mechanism stated in log-probability differences but is *not* the metric behind two of the four headline lifts (Section 6); the two are not interchangeable. The contrastive-canonical cells are a single training run — the published adapter — while the plain-CE cells carry three seeds each. And the stratified null does not show that the arms carry no per-item information; it shows that this statistic cannot distinguish such information from a rescaling, with the audit above bounding what it could be hiding.

5.8 Two failure modes: ARC carries the readout story (suppression), TruthfulQA is answer-prior (promotion)

The readout account is established on ARC and must not be read across all tasks uniformly. On **ARC**, the lift is question-conditional: a PMI / decision-conditional control retains **78%** of it (naive +0.3275 → PMI +0.2550), and the choices-only lift is only +0.0375 (CI $[-0.005, +0.08]$, not clearly positive) — so it is genuine, question-conditional, and the case the Section 5.1 arbiter and Section 4.4 spectrum characterize. On **TruthfulQA**, by contrast, much of the +37.25 is an **answer-prior / surface-form Goodhart** effect: the choices-only lift is +0.2225 (CI [+0.1675, +0.2725]) and the adapter selects the correct option **0.5725 of the time with no question at all** (base 0.35, chance 0.229); PMI retains 68%. This matches known multiple-choice shortcut behavior in which highest-probability scoring favors options by surface form independent of the premise [29], and choices-only accuracy sits above chance [4].

The two tasks also fail the cold readout by **opposite calibration mechanisms**, visible in the base gold-rank distribution on the items the adapter flips. On **ARC** the gold is **suppressed** — buried at median rank 2, with only 46% of flipped items having it at rank 1 — so the adapter must *promote* it past more fluent distractors (gold-mass added 0.35 > wrong-mass removed 0.21). On **TruthfulQA** the gold is instead a **flat near-miss**, already at median rank 1 (51% of flipped items at rank 1), so the adapter mostly nudges an already-near-top answer over the line. TruthfulQA is therefore not merely "excluded from the inducibility control" — it is a **second failure mode** of the same lossy readout (a near-miss the answer-prior resolves), distinct from ARC's active suppression. We report TruthfulQA held-out, as evidence the scoring lift is *real* and exploitable, not as evidence for the readout mechanism, which rests on ARC. The paper-level rule: the lift is not a single phenomenon — TruthfulQA is answer-prior (promotion); ARC is question-conditional, off-axis-of-degree, and non-transferable (suppression).

We scope the TruthfulQA result precisely. It was previously a pre-registered hedge — a selection lift on the mc1 metric (+37.25, paired-significant under exact McNemar, Holm-corrected p_{corr} 6.26×10^{-38}, b=6/c=155) pending a held-out-plus-mc2 control — and that control has now **run** (full set, n=817), removing the hedge in a specific way. The lift **generalizes held-out**: on the ~20% never matched into training (164 UNSEEN of 817), the v7-mc gain is Δ_{UNSEEN} mc1 **+37.8**, indistinguishable from Δ_{SEEN} +36.9, and it holds on the second metric (Δ_{UNSEEN} mc2 **+26.1**) — clearing the pre-registered gate (Δ_{UNSEEN} mc1 $\geq +15$ pp and Δmc2 $> +10$ pp) comfortably. So the +37.25 is *not* memorization of train-on-eval pairs (the matched corpus covers ~79.9% of the eval, yet the held-out gain is undiminished), and it is real beyond the mc1 selection metric. The decisive contrast is with the generic control: the higher-capacity Align-LoRA, which recovers most of the *ARC* lift, **collapses held-out on TruthfulQA** (mc1 Δ_{SEEN} +38.6 → Δ_{UNSEEN} +7.9; mc2 +24.1 → +7.6) — the signature of memorizing / format-familiarity — which is precisely why TruthfulQA is excluded from the inducibility control (Section 5.2), and which marks the contrastive lift here as the genuine, generalizing one. What the held-out control does **not** do is convert the gain into truthfulness: the *character* of the lift is still answer-prior (choices-only +0.2225, above), so the held-out result shows that the answer-prior behavior *generalizes* — a robust selection behavior, not eval-set recall — not that the model has become more truthful. We therefore report TruthfulQA as a real, held-out, generalizing answer-prior lift on mc1/mc2 — a second failure mode of the same lossy readout — never as a truthfulness gain and never as evidence for the off-axis mechanism.

5.9 Status of the evidence

The claims rest on a converging set of controls, all on Gemma 4 31B IT: the per-item arbiter and the GPQA out-of-task control (Section 5.1); the cold-MC calibration signature (Section 5.1, confident burial under fluency; present-but-swamped, residual AUC 0.574 base / 0.667 adapter); the Align-LoRA inducibility control (Section 5.2); the surface-ceiling and same-space-probe channel results (Section 5.3); the read-only selector (Section 5.4); the geometry spectrum, the direct $\text{off}_{\perp}/\text{off}_{\parallel}$/shuffle decomposition, and the on-axis causal controls (Section 4.4); the no-locus sweep (Section 4.5); and the injection-coupling control (Section 5.5).

The load-bearing experiments have run. The **inducibility attribution** is closed (a matched Align-LoRA recovers ~62% of the ARC lift, flat across rank — a lower bound). The **off-axis-of-degree geometry is causal, and identified against random and on-axis alternatives**: a perpendicular push recovers ~100% of the lift, a parallel push 0%, the strongest on-axis reference (W_{know}) 6.4%, and a norm-matched random perpendicular direction ~0% (three seeds) — closing the non-identifiability objection on both the harness side and the write side (Section 4.4). Two quantities that this section previously argued rather than measured are now measured, and both came back the way the off-axis reading predicts. The on-axis references are **not** weak readers — W_{know} separates gold from distractors at 0.9613 (CI95 $[0.9428, 0.9781]$) and v_{inf} at 0.9646, against 0.9545 for a probe *fitted* on the same space and population — so their causal failure is not an artifact of estimating them badly; and the offset's angle to the *retrieval* face, which we previously declined to claim, reads $\cos_{\text{w}}(\theta_{\text{mc}}, v_{\text{ret}}) = +0.0076$ (BCa95 $[-0.0033, +0.0187]$) inside its isotropic null, against +0.2586 for the two faces against each other on those same 48 heads. The **same-space probe** recovers ARC base-wrong gold-vs-distractor at AUC 0.955 (CI $[0.934, 0.973]$, null 0.494) against the 0.502 collapsed-output ceiling, with a surface-form control at 0.442 — a *channel* claim (the output-scalar channel is lossy relative to the activations), not "pure miscalibration" and not "the probe beats the adapter" (different information sets — 12288 activation dims vs collapsed logits, so 0.955 is not a contest with 0.830); its diffuseness (a random-48 head set decodes at 0.949) is a read-test decoupled from the adapter's write-specificity (Section 5.3). The **read-only selector** built on that probe recovers the lift undamaged (selector 0.943; Section 5.4) — a read-only-selector capability prior work establishes [28, 40], used here as the read leg, not claimed as new — and the **read ≠ write** dissociation is four-way: read (selector 0.943), on-axis write fails (W_{know} 6.4%), off-axis write succeeds ($\text{off}_{\perp}$ ~100%, shuffle ~0%), and the probe's read direction is geometrically aligned with v_{inf}, not with θ_{mc} (moderate, Section 5.4). The no-transfer is corroborated **out-of-task** on GPQA (cold-MC base 0.4747 → adapter 0.4697 = −0.5 pp, CI crosses 0, while base CoT is 0.7071). On TruthfulQA the held-out control has run: the lift generalizes (Δ_{UNSEEN} mc1 +37.8) where a generic control collapses (+7.9), so it is real and not memorization, but answer-prior by character (Section 5.8). And the temperature/null-space account of the offset is ruled out **causally** (activation-patching: at the scoring positions the relative temperature fraction is 0.215, with absolute fractions temperature 0.177 and ranking 0.803; the offset moves logit *differences*, not the mean — Section 5.6). The *effect* replicates cross-family (Qwen 14B: same-sign cold-MC lift on all four tasks — three clearly, with Winogrande +3.0 small and not individually significant at n=400, counted as marginal — ARC +18 / HellaSwag +7.8 / TruthfulQA-mc1 +15.8 / Winogrande +3, magnitude below Gemma; null==base gate exact), removing "single-model" from the effect; the off-axis geometry replicates too (whitened cos −0.002, frac⊥ 97.5%), and so does the causal decomposition that reads it — perpendicular component +106%, parallel −6%, on-axis reference again inside its own noise floor (Section 4.6).

A note on multiplicity and sample-sharing scopes these statistics. The paper's tests fall in two families, and only the first carries formal correction. The *confirmatory* family — the four headline lifts and the pre-registered kill-rules (R2, R6) — is Holm-corrected (m=4, Section 4.1). The *mechanism* family (arbiter, calibration, PMI-residual, probe, selector, geometry arms) we read as converging description rather than as a set of independent significance claims, and no conclusion in it rests on an uncorrected marginal p-value:

every load-bearing p is either far below any correction over the full campaign (selector 3.7×10^{-47}; negative control 1.4×10^{-4} and its premise-file robustness 3.6×10^{-5} — the weakest load-bearing figure, the all-ARC contrast $p = 0.0015$, still survives a Bonferroni correction across all ~25 tests run) or is *interpreted as a null* (W_{know} $p = 0.136$), where multiplicity cuts the other way. These diagnostics were also developed on one sample: the arbiter, calibration, probe, selector, and geometry arms were all computed on the vinf_causal n=400 draw and its nested subsets, so their agreement there is consistency across facets of one sample, not independent replication. Three of them have since been re-run outside it (below); calibration and geometry have not, and for those the caveat stands as written. What is independently replicated is the headline magnitude (the A1 run; the R4 full-split run, every delta inside the n=400 CI; the Qwen family) — and, as the corresponding strengthening on the mechanism side, **the arbiter has now been re-run on the full ARC-Challenge test split** (n=1172, outside the shared n=400 draw). It replicates: the base scores 0.4932 and the adapter 0.8558 (+**36.3** pp, against +35.0 on the n=400 slice), the adapter fixes 444 items, and the base already answers **442/444** = **0.9955** of them with chain-of-thought (CI [0.984, 0.999], against 124/124 on the slice, CI [0.970, 1.000]). The negative control replicates with far more power: items neither arm fixes are answered at **134/150** = **0.8933** (CI [0.834, 0.933]), against the fixed items' 0.9955 (Fisher OR 26.4, $\mathbf{p = 1.4 \times 10^{-8}}$, against $p = 1.4 \times 10^{-4}$ on the slice) — so the arbiter discriminates, and its near-ceiling reading on the fixed items is not an artifact of the sample it shares with the other diagnostics. The power caveat is unchanged: ARC remains near its chain-of-thought ceiling, so this confirms the benign reading without distinguishing "recovers knowable" from "fabricates" as a harder benchmark would.

The probe and the selector re-run outside the slice. The arbiter's replication leaves the read side untested, so we froze the pipeline and re-ran it on the complement: activations extracted for the full split, probe and selector *fit* on items 0–399 — the development slice — and *scored* on the 772 items outside it. No cross-validation and no refitting: a single held-out split, which is out-of-sample in the sense the objection asks about, where cross-validation over the pooled split would only be out-of-fold. Both replicate. The probe reads base-wrong gold-vs-distractor at **0.9553** (n=395 base-wrong items) against 0.9545 on the slice, a difference of eight ten-thousandths; the read-only selector scores **0.9453** against 0.9433, inside the slice's binomial interval [0.917, 0.962]. Both permutation nulls sit at chance (probe 0.501, max 0.535; selector 0.244 against 0.25), so the instrument measures on the new items rather than fitting them. The base is slightly *worse* on the complement (0.4883 against 0.505), so the selector's margin over it widens rather than narrows. Calibration and geometry have no train/test mode and were re-run on the pooled split instead (n=1172, the slice inside it): the lift reads 0.3626 against 0.3500, calibration's pre-registered verdict is unchanged, and the geometric contrast that carries the off-axis reading strengthens — the probe's read direction aligns with v_{inf} at 0.283 and with θ_{mc} at 0.072, a factor of 3.9 against 3.3 on the slice. That is more power, not independence, and we count it as such.

Two of those numbers move against us and we report them as they came. The burial statistic — the fraction of S_{flip} items whose gold sits at rank ≥ 2 — falls from 0.543 to **0.502**: still a majority, by two tenths of a point, and the reading survives only because its other half (gold probability 0.239 against the control's 0.464) is unchanged and wide. And the geometry's own verdict does not close on the larger sample either, for the same reason it did not close on the slice: $\cos(\text{PC1}, v_{\text{inf}}) = 0.548$ exceeds $\cos(\text{probe}, v_{\text{inf}}) = 0.283$, so we cannot exclude that v_{inf} lies trivially in the leading principal component. The replication confirms that this ambiguity was not a property of the sample; it does not resolve it. Finally, the split is out-of-sample in *items* and not in *features*: the 48 (layer, head) cells are still the ones the adapter's own training selected by correctness on ARC, which is the Tier-1 caveat of Section 5.3 and is untouched by this run.

Where each claim stands. Settled by executed controls: the canonical gsm8k loss magnitude is −**25** pp (R6, base 0.9775 vs v7mc 0.7275, apples-to-apples), flat across a generation-budget sweep → a *mechanistic* loss (multi-step reasoning breakage) and not truncation (Section 5.5), with the loss replicating in sign across three independently trained adapters at −24.2 ± 7.3 pp (Table 12); the temperature/null-space negative is **causal** (patching, temperature fraction 0.215 at scoring positions — Section 5.6); and the effect replicates cross-family (Qwen 14B, 4/4-sign lift). Claims we hold with stated limits rather than as closed: the generic recovery fraction is *capacity-dependent* — ~37% at the adapter's own parameterization against ~62% at 1,800×–58,000× capacity, flat within each grid but plainly moving across them — so we report the contrastive-specific residual against a named control rather than as a bound on generic fine-tuning (Section 5.2); the present-but-swamped claim's base-arm residual CI lower bound (0.520) is *intermediate* and rests on the adapter arm (0.584 > 0.55) — the power increase ran and does not rescue it: no defensible pool, including the adapter-independent $S_{\text{cot\_coldwrong}}$ (CI lower bound 0.512) and the pro-gate union (0.535), clears 0.55, so the base channel is frozen intermediate; the earlier end-of-turn "collapse" figures are **retracted** as an eos-token artifact — the corrected re-measure shows no collapse, with only a modest, adapter-dependent open-generation degradation in the D.1 re-measure (full account and corrected numbers in Section 5.5); the logit-lens frequency control (Section 5.6) — flagged pending in earlier passes — is now **closed at the attribution level** (over the full wikitext-103 corpus the frequency direction is identified at r=0.906, past its 0.9 self-gate, and the offset's cosine with it is 0.0128 at the null floor, so the offset is not a token-frequency edit; its causal-patching cross-check has now run and confirms it — no sign-flip, causal cosine +0.0059 with a CI crossing zero, Section 5.6); and the read-only selector's feature set is item-held-out but not feature-held-out, pending the cross-benchmark check (ARC-Easy / OpenBookQA). The cross-family universality of the *effect* now has one replication (Qwen 14B, 4/4 sign), and the off-axis-of-degree *mechanism* now replicates too — geometrically (Qwen whitened $\cos(\theta_{\text{mc}}, v_{\text{inf}})$ −0.002 within null, perpendicular fraction 97.5%) and causally (the perpendicular component carries the lift, the parallel one and the on-axis reference do not; Section 4.6). The collateral reasoning loss is now shown **dose-separable from the scoring benefit in steering** (single activation-scale axis: the scoring lift saturates at s=0.75 where gsm8k is still at base, and the reasoning damage falls only in the 0.75→1.0 band — Section 5.5), a dose-refinement of the *collateral* cost (not a deployable window: no dose lifts reasoning above base beyond noise, Section 5.5) whose dissociation-of-thresholds claim survived a prior-art gate (no-scoop; the nearest foil, an angle–norm account [2], is distinguished in Section 5.5), with whether a *trainable* adapter inherits the low-damage regime left as open (steering ≠ baked).

6 Limitations

This paper characterizes a single observable: contrastive-correctness PEFT (the LoFiT/ITI family) recovers a large cold-MC lift that is **mostly inducible by generic fine-tuning** (~62% of the ARC lift), along a learned **off-axis-of-degree** direction that does not transfer to generation and, recovered this way, introduces deployment pathologies. The recovery exploits a fact established in prior work and confirmed here — that cold multiple-choice scoring under a chat template is a **lossy** readout (present-but-swamped, not empty) of knowledge the model already has. The scope is deliberately narrow: a mechanistic study of that effect, not a routing architecture. Several limitations bound even this restricted claim.

The headline numbers are pre-registered with kill-rules; the anti-floor and gen-face rules have now executed. The scoring lifts and the gen-face losses are pre-registered with explicit kill-rules and matched-control designs in our evaluation protocol. Throughout this paper, "pre-registered" means pre-specified in a time-stamped internal protocol before the confirmatory run, following exploratory discovery — a dated document versioned in the project repository (git-timestamped, released with the code), not an entry in an external registry filed before we knew there was an effect to test: the headline kill-rules and matched-control designs (protocol of 2026-06-22, runs from 2026-06-27), the arbiter negative-control thresholds (2026-07-15, run 2026-07-21), and the on-axis-penalty lift-matched gate (2026-07-08, run 2026-07-09) — internal and verifiable from the release, not a third-party registry. The anti-floor 5-shot kill-rule (R2, n=400) ran 2026-06-27 and the lift survived, but the rule was pre-registered without fixing *which* residual instantiates it, so we state the formula explicitly: the anti-floor quantity is **a@0 − b@5** — the zero-shot adapter against a fully-prompted base — giving ARC **+22.0** pp, HellaSwag **+10.0** pp, Winogrande **+8.25** pp. It is distinct from the matched-shot quantity a@5 − b@5 (ARC +22.25 pp, McNemar b=99/c=10, $p{\approx}10^{-19}$; HellaSwag +17.5 pp, Winogrande +13.75 pp), which answers whether the adapter still wins once few-shot is granted to both arms. The +35 is not a zero-shot-floor artifact under either reading, but the margin is strongly task-dependent — few-shot closes 37% of the ARC headline, 41% of HellaSwag's and 57% of Winogrande's — so Winogrande is the weakest anchor under the anti-floor reading (+8.25 pp), and we do not cite its +13.75 as an anti-floor figure (TruthfulQA carries no anti-floor control at all: lm-eval fixes num_fewshot=0). All four MC anchors are paired-significant beyond the anti-floor control — an exact McNemar test on byte-identical base/adapter items gives ARC $p = 1.32\times10^{-32}$, TruthfulQA $p = 1.57\times10^{-38}$, Winogrande $p = 1.99\times10^{-12}$, HellaSwag $p = 1.63 \times 10^{-8}$, and all four survive Holm–Bonferroni correction ($p_{\text{corr}} \leq 1.63 \times 10^{-8}$); none is dropped. This certifies the effect exists and is paired-significant, not that it is on-axis — the on-axis recovery ceiling is 6.4% (W_{know}, Section 4.4), a *bounded* null rather than a zero: the design detects 12.6% of the ARC lift at power 0.80 and 36.2% at power 1.0000, so it is blind only below ~9–13% (Section 4.4). The full-set Δ is now confirmed (the v7mc full-set run R4 closed 2026-06-27: ARC acc_norm +36.2, HellaSwag acc_norm +19.2, Winogrande acc +20.1, TruthfulQA acc +37.1, every delta within the n=400 CI — so the n=400 headline values are conservative, not inflated). All four deltas are differences within a *matched pair*: base and adapter arms share the same commit, the same context window, the same batch size and the same n per task, so no full-split delta here is assembled from runs of different campaigns. What the pair does *not* carry is a pinned model snapshot — neither arm stamps a model SHA, the reproducibility limitation recorded in Section 3.1. The gen-face canonical gsm8k kill-rule (R6) has now run — −25 pp, flat across the generation budget, so the loss is mechanistic and not truncation — and the EOS re-measure has run too, with a consequential result: the earlier end-of-turn "collapse" figures are **retracted as an eos-token measurement artifact** (the verbosity harness scored the wrong turn-stop token; corrected, the adapter terminates factual recall at 100%), so we make no termination-collapse claim; the D.1 re-measure shows only a modest, adapter-dependent open-generation degradation — the full account and every corrected number are in Section 5.5. The gsm8k loss magnitude, previously run-dependent (−5/−17/−25/−78 pp), is now canonically **−25** pp (R6). The *direction* of each effect is robust; the remaining un-certified magnitude is the contrastive-specific fraction (below).

Reproducibility and provenance: the lift is not a leakage or input artifact, and the headline mixes two scoring metrics. Four provenance audits each independently *strengthen* the effect rather than threaten it. (i) *Test↔train overlap is negligible.* On ARC, near-duplicate contamination between the eval set and the adapter's training data is 1.0% (4/400 at Jaccard ≥ 0.8, robust across three matching methods); the stricter genuine-leak rate (matching stem + options + answer) is ≤ 0.5% (2/400), and exact eval-ID overlap is zero. This sits at the clean side of our pre-registered gate (< 1% clean / 3–5% retract), far from the retract line — we report it honestly rather than claiming zero. (ii) *The transfer benchmarks are genuinely held out.* The offset was trained on only two datasets — TruthfulQA and ARC (verified across three independent artifacts: the offset config, the head-probe record, and the eval config; top_k = 48, 800 steps, val_acc 0.94 — a training monitor computed, due to dataset ordering in the validation slice, on TruthfulQA items only, so checkpoint selection never saw ARC signal; Section 3.1) — so HellaSwag, Winogrande, gsm8k, and MMLU never entered training (Winogrande has no training builder at all). Their lift is therefore genuine cross-task transfer, not in-distribution recall. (iii) *Base and adapter see byte-identical inputs.* For every task the base and adapter prompts are byte-for-byte identical across all 400 items (1600/1600 per task), with zero truncation (max ≈ 228 tokens ≪ the 4096 context) — the arms differ only in the inference-time hooks, eliminating an "input distribution differs" confound. (iv) *The setup is deterministic and pinned.* The offset resolves to a single artifact across 8/8 runs (sha256 `fcf74ffc...`; the per-shard safetensors hash is pod-only and not reproducible locally), under transformers 5.8.0 and lm-eval 0.4.12. One honest reporting caveat carries through to the headline metric itself: Table 1 mixes two cold-MC metrics — acc_norm for ARC (+35.0) and HellaSwag (+17.0), and plain acc for TruthfulQA-mc1 (+37.25) and Winogrande (+19.0), because mc1 and Winogrande do not define a length-normalized variant. The choice is not cosmetic: on HellaSwag acc_norm (+17.0) is *smaller* than acc (+22.25), so the headline is the more conservative figure and the lift is not a length artifact. We state the per-task metric explicitly rather than imply a single unified score.

The same choice is *anti*-conservative in the other direction, for the inducibility fractions rather than the lift, and we flag it rather than let the two uses of one metric pass as if they behaved alike. Recomputed under raw accuracy, HellaSwag's inducibility collapses — 69.5% → 8.9% for the same-parameterization control, 92.6% → 19.1% for the higher-capacity arm, and in Qwen it becomes unreadable (the acc lift is −0.5 pp, two items). ARC's survives in the generic arm (0.6214 → 0.6099) but not everywhere: the same-parameterization control's 36.2% becomes 26.9% in Gemma and its 27.2% becomes 59.7% in Qwen, so that figure carries the metric with it too. The four lifts themselves are robust. What does not survive is the *cross-task ordering* of inducibility — nor, as the recomputations just given show, the recovered *fractions* themselves. Both are acc_norm statements and are cited as such throughout, not as properties of the models.

The contrastive-specific residual is grid-relative, not a bound on generic fine-tuning. We report the recovered fractions as the best recovery observed under the generic-SFT family we tested, and two facts fix the limits of that claim. Within each grid the fraction is flat: the higher-capacity control's r8–r256 slope-CI includes zero and a best-effort r512 flank does not raise it (frac 0.22 at lr1e-4; a below-base broken adapter at lr2e-4), so the residual is not a gap more rank closes there. *Across* capacity, however, the fraction plainly moves — ~37% at the adapter's own 12,336 parameters against ~62% with three-to-five orders of magnitude more — so no single number is "the" generic fraction, and we assert no ceiling for generic fine-tuning: a different architecture, intervention site, optimizer or search budget could recover more, and we have not tested those. The contrastive-specific residual should therefore be read against a stated control: ~60–64% of the ARC lift at equal capacity, ~38% against the higher-capacity arm. We also note the higher-capacity control is clean only on ARC: TruthfulQA was excluded there as train-on-eval leakage (held-out, that control collapses 92%→30%), so its generic-fine-tuning claim is an ARC result. The same-parameterization control does measure TruthfulQA — and recovers 0.9% ± 3.3 of its lift, i.e. nothing (Section 5.2).

"Generic" versus "domain exposure": one substitution, and it favors "any domain" — within multiple-choice format. The inducibility controls train on the same TruthfulQA+ARC budget the effect is measured on (Section 3.6), which left the ~62% consistent with two readings: that non-contrastive SFT recovers most of the lift regardless of *what* it is trained on, or that training *on this domain* is what does it. We ran the missing arm. Holding parameterization, budget and schedule fixed and swapping only the corpus for MedMCQA [42] — same pair format, medical domain, no overlap with the four anchors — the non-contrastive control recovers 33.8% ± 5.5 of the ARC lift out of domain against 36.2% ± 5.0 in domain, a ratio of **0.93** at the best checkpoint and **1.00** at the final one. **In the single out-of-domain corpus we tested, the domain is not what matters:** plain cross-entropy trained on medicine buys back as much of an ARC lift as plain cross-entropy trained on ARC itself. One substitution cannot establish that *no* domain matters — MedMCQA shares the multiple-choice pair format with the anchors, and a corpus that did not would be a separate test — so we state the claim at the strength the experiment supports: swapping the training domain did not reduce recovery here.

Two things follow that the earlier reading could not say. First, the contrastive objective does work *outside* its domain: cell C recovers +17.5 pp (best) and +21.3 pp (final) more than plain CE at identical parameterization and corpus. Second, the off-axis ratio moves with the domain as well as with the objective (Figure 1 gains two points, 2.75 ± 0.26 and 2.98 ± 0.10): the domain shift is a real axis at +0.90 ± 0.17 (5.3σ), but it differs from the objective shift by only +0.07 ± 0.25 (0.28σ), so *which* axis weighs more is not decided by these data and we do not order them. Both figures are the larger of the two estimates the 2×2 affords, and we say so: measured under the other level of the opposite factor, the domain shift is +0.30 ± 0.20 (1.5σ) and the objective shift +0.23 ± 0.16 (1.4σ). The interaction is therefore real (+0.60 either way you slice it) and neither main effect is a single number — which is a further reason not to order the axes.

The out-of-domain arms also carry a cost we did not anticipate and report as a finding rather than a footnote: both damage TruthfulQA (at the best checkpoint, contrastive −4.2 pp and plain CE −10.1 pp relative to base; −1.0 and −9.8 at the final one, harmful in 3/3 seeds throughout), where the in-domain non-contrastive control was merely inert.

The headline anchors are knowledge benchmarks with low chain-of-thought headroom — by design, and stated as scope. All four headline tasks sit at or near the base's CoT ceiling (ARC 0.98, HellaSwag 0.83, Winogrande 0.9425). That is not an oversight but the condition that makes the per-item arbiter informative — there must be a latent, reachable answer for cold scoring to misrank — and it is what the mechanism predicts the exploit needs: on GPQA, the one hard anchor, with genuine reasoning load and ~23 pp of cold-scoring headroom, the same adapter recovers approximately nothing (Section 4.2).

Saturation and difficulty are nonetheless separable, and we since separated them. Evaluated on MMLU-Pro — ten options, chance 0.100, base cold 0.2610 — the adapter still lifts scoring by +12.20 pp and still fails to transfer (Section 4.7). So the effect is *not* confined to saturated four-option benchmarks, and the GPQA null is not a difficulty result: both benchmarks are hard, both leave the base large cold-to-CoT headroom, and the exploit fires on one and not the other. What separates them is distance from the adapter's training distribution, not how hard they are — a protocol-bound inference, since we vary benchmark identity rather than manipulating distance directly, and benchmark identity carries format, domain and option count with it (Section 4.7).

What remains unrun is the narrower converse arm: an adapter trained *in-distribution* on a hard, non-saturated benchmark, rather than one trained elsewhere and evaluated on it. That arm would test whether contrastive-correctness PEFT *produces* a large off-axis lift where the base has few buried-but-reachable answers to amplify — a question our MMLU-Pro evaluation cannot answer, because there the base has many (0.2610 cold against 0.7060 under CoT). Claims about the training-time behavior of the objective are accordingly scoped to the corpora measured.

Pretraining contamination is not audited — and the claims are robust to it by construction. The leakage audit above covers the adapter's fine-tuning corpus, not the base model's pretraining corpus, which is closed to us; ARC, TruthfulQA, HellaSwag, and Winogrande are public benchmarks and plausibly present in any frontier pretraining mix. Two features of the design make the conclusions insensitive to this. First, every comparison is paired: both arms share the base model, hence whatever pretraining exposure exists, and the deltas subtract it out. Second, the mechanism claims are agnostic about the *origin* of the knowledge: the arbiter establishes that the base reaches the fixed items' answers by reasoning while cold scoring misranks them — equally true whether that knowledge came from generalization or from memorized benchmark text — so the attribution of what the adapter's lift *is* does not change. What pretraining contamination could inflate is the base's absolute CoT ceiling, which we use descriptively, not as a claim.

Adapter replicates: the structure is stable across re-trains, one magnitude is not. The canonical analyses here characterize one V7-mc adapter, so we re-trained it twice more, varying both the data split and

Table 12: Adapter replicates. Three independently trained contrastive adapters, varying the data split *and* the batch-visit order; identical recipe otherwise. Lifts are pp over the same base arm (n=400, chat-template, lm-eval); gsm8k is the generation loss at full activation scale (n=200, k-shot 4); `ratio` is the net-effect on-axis alignment with v_{inf} normalized to a random direction (on-axis reference W_{know}: 3.93). Termination is the natural end-of-turn rate (factual / creative / social).

Adapter	ARC	TQA-mc1	HellaSwag	Winogrande	gsm8k Δ	ratio	Termination
seed 0 (canonical)	+35.25	+36.00	+17.50	+18.25	−25.0	1.74	preserved
seed 1	+36.50	+35.00	+20.25	+20.00	−31.0	1.81	100/92/100%
seed 2	+35.75	+33.75	+19.00	+19.00	−16.5	2.00	100/92/100%
mean ± sd	**+35.83** ±0.63	+34.92 ±1.13	+18.92 ±1.38	+19.08 ±0.88	**−24.2** ±7.3	**1.85** ±0.14	—

Reading. The scoring lift and the off-axis geometry are stable (ARC sd 0.63 pp; ratio sd 0.14, computed on the unrounded per-seed ratios rather than on the rounded values printed in the column); the generation loss is not (range −16.5 to −31, sd 7.3), though its sign is invariant — 3/3 seeds damage multi-step reasoning. Termination is preserved in every seed, including the creative prompts where the *base* terminates 68% of the time and the adapters reach 92% by generating shorter — an independent corroboration of the end-of-turn retraction (Section 5.5).

The gsm8k column is a floor, and the arms truncate asymmetrically. Each accuracy counts every truncated trace as wrong, and truncation is far from balanced between arms: at scale 0 it runs 0.5%, at full scale 9.0% in seed 1 and 5.5% in seed 2 — an asymmetry of up to 18× in the very variable the comparison conditions on. Scoring the truncated traces both ways, the drop is bounded by [−31.5, −22.0] in seed 1 and [−17.0, −11.0] in seed 2, so the tabulated −31.0 and −16.5 are floor-to-floor differences sitting near the pessimistic end of their bounds, not point estimates. What the bounds do not touch is the *sign*: the treated and reference intervals are disjoint in both seeds, and the no-transfer claim rests on the sign and on the mechanism being flat along the generation budget, not on the magnitude.

Artifacts: `runs/v7_lofit_seed{1,2}/` (Hugging Face, Table 14); truncation audit `outputs/remediation_20260729/gsm8k_truncation_e2seeds.json`.

the optimization order, and re-ran the core spine on each (Table 12). The headline effect is stable: across three independently trained adapters the ARC lift is **+35.83 ± 0.63** pp (TruthfulQA +34.92 ± 1.13, HellaSwag +18.92 ± 1.38, Winogrande +19.08 ± 0.88), the off-axis net-effect ratio is **1.85 ± 0.14** against the on-axis reference's 3.93, and natural termination is preserved in every seed (Section 5.5). The replicates also happen to bracket the released artifact: they were trained with a corrected validation-split construction and recover lifts equal to or slightly larger than the canonical run, so that run is not an outlier favored by the earlier construction.

One magnitude does *not* replicate tightly, and we cite it accordingly: the gsm8k generation loss is **−24.2 ± 7.3** pp (range −16.5 to −31), so its *sign* is invariant — all three seeds damage multi-step reasoning — while its size varies by about a factor of two. The canonical −25 pp is near the middle of that range, not a constant, and the no-transfer claim rests on the sign and on the mechanism (flat across the generation budget), not on the magnitude.

Two restrictions remain. Every geometric and causal number is a property of the $K = 48$ head slice this recipe trains — a hyperparameter inherited from LoFiT (Section 3.6), with no K-ablation — so the claims characterize this family's offset, not Gemma's correctness circuitry at large. And the finer mechanistic analyses (the J-lens and dose studies, Sections A.2 and A.3) remain single-seed and single-model, as flagged there.

The "signal lives upstream" claim is a *channel* claim, now sharpened by the activation-probe — and it remains a channel claim, not "pure miscalibration." Our surface-ceiling result (Section 5.3) — a supervised, held-out re-scoring of the output scalars caps at AUC 0.574 against the adapter's 0.830 — is information-set-confounded: the adapter mutates activations while the ceiling re-weights collapsed scalars. The in-space activation-probe that closes this confound has now **run** (no fine-tuning, per-head `o_proj`-input, held out by item, same cold-MC option-continuation distribution): it recovers ARC base-wrong gold-vs-distractor at **AUC 0.955** (CI [0.934, 0.973], null 0.494) against the **0.502** collapsed-output ceiling, with a surface-form control at **0.442** (≈ chance). This widens and cleans the channel gap (0.502→0.955) and licenses a **sharpened channel claim** — the cold-MC output-scalar channel is lossy relative to the activation channel — and nothing stronger. It does **not** become "pure miscalibration / signal intact upstream": a linear probe decoding correctness ~0.95 from activations is near-universal in capable LMs, so it shows the information *exists*, not that the deployed readout merely under-uses it. Nor is it "the probe beats the adapter" or a representational edit — the probe reads 12288 activation dimensions while the adapter is constrained to nudge collapsed output logits, so 0.955 is not on the same axis as 0.830. The diffuseness the probe reveals (a random-48 head set decodes at **0.949** ≈ the adapter's heads) is a **read-test** that the channel is lossy model-wide, decoupled from the **write-tests** that establish the adapter's specificity (a random norm-matched offset recovers ~0% of the lift, Section 4.4); it does not undercut the adapter story. We do not generalize the channel claim past ARC: TruthfulQA's probe reaches 0.974 but its surface-form control is non-trivial (0.639, with a gold-first position artifact at 1.000), so it is anchored out of the channel claim and read as answer-prior (Section 5.8).

The no-transfer is corroborated on GPQA but the verifier-guided arm is untested by design. We establish no transfer on ARC (the fixed items are at the base's CoT ceiling) and confirm it on GPQA, where the base reasons successfully (CoT 0.70) yet a scoring shortcut still has no generative headroom. Because the base already reasons GPQA out, a verifier-guided deployment (base generates, adapter scores) has nothing to harvest and we did not build it; that is a design decision following from the result, not an independent test, and a setting with genuine selection headroom (which we did not find here) would be required to revisit it.

The off-axis verdict is of degree and method-dependent, established geometrically and causally on Gemma — and now replicated, geometrically and causally, on a second family. The recovery is off the functional correctness direction by a *method-dependent* amount: the random control recovers ≈0% (the direction is learned, not arbitrary), the strongest on-axis reference recovers 6.4%, the contrastive net effect sits at cosine 0.088 and generic SFT at 0.199 (2.3× more on-axis). We deliberately do not claim "purely off-axis" or "off-axis universally" — a weight-space proxy suggested that and the clean activation experiment refuted it. Both halves of the verdict now hold on Qwen 14B: geometrically, the cold-MC effect is 4/4-sign and the offset's whitened $\cos(\theta_{\mathrm{mc}}, v_{\mathrm{inf}})$ is -0.002 (vs Gemma's -0.003, both within the null band) with 97.5% of the offset energy perpendicular to v_{inf}; causally, re-injecting the two components separately puts the lift on the perpendicular one and nothing on the parallel one, and the on-axis reference again fails to beat same-norm noise (Section 4.6). One honest heterogeneity of degree remains: Qwen's *raw* on-axis energy budget (2.47%, z+8.9 vs null) is ~4× Gemma's (0.59%), though the covariance-controlled whitened cosine is identical (Section 5.9). What is still single-model is the *per-item* evidence — the arbiter — and the capacity arm of the inducibility spectrum (Align-LoRA and its rank sweep).

The lift is task-heterogeneous, and TruthfulQA carries a different mechanism than ARC. The ARC +35 is question-conditional (PMI retains 78%; choices-only +0.0375, CI crossing zero) and carries the readout story. TruthfulQA's +37.25 is largely answer-prior surface-form Goodhart (choices-only +0.2225; correct option 0.5725 with no question, base 0.35). Any blanket statement that "the +35 is off-axis" or "the +35 is surface-form" is wrong; the mechanism is task-conditional and our readout/off-axis claim is anchored on ARC.

The direct-logit reading has no clean lexical mechanism; the causal patching test has now run, and the symbol-channel test came back mixed. A logit-lens analysis finds no clean lexical story; the one robust signature is suppression of structural single-uppercase tokens, the readable face of the format-preference channel (not of a termination collapse — that claim is retracted, Section 5.5) rather than an explanation of the lift. The frequency control is now closed at the attribution level: over the full wikitext-103 corpus the recovered frequency direction is identified at r=0.906 (past its 0.9 self-gate) and the offset's cosine with it is 0.0128 (at the null floor, below the 3×-null gate 0.041), so the offset is not a token-frequency edit — and the causal-patching cross-check (which the temperature negative did receive) has now run and does not overturn it: no sign-flip, causal cosine +0.0059 with a CI crossing zero, against the temperature's 0.023 → −0.149 (Section 5.6). The activation-patching test of the offset has now run: it confirms the temperature negative causally (temperature fraction 0.215 at scoring positions), but the decisive TEXT-vs-LABEL symbol-channel test is only *consistent-with*, not clean (mixed across ARC and TruthfulQA, Section 5.6), so the unifying mechanism is still a signpost, not a named channel.

The within-format difficulty control and independent base-CoT for HSwag/Wino have now run — "out-of-task, not difficulty-bound" is supported. The activation-probe in-space — the decisive fairness control for the channel claim — has **run** (Sections 5.3 and 5.9): it lands at AUC 0.955 against the 0.502 collapsed-output ceiling, resolving Section 5.3 to the sharpened channel claim (the output-scalar channel is lossy relative to the activation channel), not to "pure miscalibration" or a representational edit. The two controls that previously left the difficulty-vs-distribution question open have now run as well (Section 5.1). (a) The *within-format* difficulty split — ARC-Easy vs ARC-Challenge under one cold-MC pipeline — moves the base-wrong fix-rate only **−0.10** (Easy 0.861 → Challenge 0.763) as difficulty rises within ARC, whereas going out-of-task to GPQA collapses it **−0.56** (to 0.202): distribution, not within-format difficulty, drives the collapse. (b) Independent base-CoT for HSwag (0.83) and Winogrande (0.9425) confirms both are high — like ARC's 0.97, well above GPQA's 0.70, though the arbiter is run in a lettered-MCQ format distinct from the cold-MC channel on every anchor (Section 3.2), so it is a knowledge-accessibility probe rather than a channel-faithful readout — and the recovery is recoverability-gated (the adapter fixes CoT-right items more than CoT-wrong ones: HSwag 0.64 vs 0.31, Winogrande 0.76 vs 0.42), so the +17/+19 lifts also rescue CoT-accessible knowledge rather than exploiting a format shortcut. We therefore now report "out-of-task, not difficulty-bound" as **supported on all three anchors** rather than suggestive — with a standing power caveat: the hard cells are small (HSwag n=32, Winogrande n=12), the Easy→Challenge range stays below GPQA's, and the burial gradient is shallow throughout (corr(fix, z-gap) −0.17). Two further caveats on the GPQA arm remain non-findings and we do not state them as contrasts: recoverability-gating could not be detected there (CoT-right 0.186 = CoT-wrong 0.186 is power-starved at n=43, with 31 unparsed items contaminating the hard cell), and the item-level churn (~21 fixed / ~22 broken) is churn within noise (no McNemar), not compensatory damage. What genuinely remains open is the cross-family universality of this difficulty-vs-distribution reading (below) and the verifier-guided arm (untested by design, Limitations).

A training-time on-axis constraint — the constructive follow-up has now run, and it does not rescue transfer. The experiment that killed the process-reward-model reframe killed an *injected, eval-time* on-axis offset; the constructive hypothesis is instead to train the adapter under a penalty forcing the learned offset toward the functional correctness direction v_{inf}, and ask whether a lift constrained to be on-axis *then* transfers to generative CoT. That penalty has now been wired into training and swept (target v_{inf}, $\lambda \in \{0, 5, 15, 50\}$), against a pre-registered **lift-matched** gate — an on-axis adapter counts only if, at equal MC lift, it retains *more* gsm8k than simply scaling the off-axis offset down, otherwise a merely weaker adapter would trivially do less damage. No λ passes. At λ=5 the on-axis adapter recovers only ARC 0.5725 (against the off-axis 0.8575) while its gsm8k sits 9.9pp (3.8σ) below the off-axis dose curve at matched lift, and the TQA lift is ~0 at every

λ (the TQA lift is *entirely* off-axis mass): the on-axis penalty buys no transfer, and at the doses that actually constrain the offset on-axis it kills the lift itself (collapsing toward the W_{know} 6.4% floor). This is **fork (b)** — it *strengthens* the off-axis reading, adding a second causal leg (**necessity**, complementing the **sufficiency** of the off$_\perp$ ~100% / random ~0% spectrum of Section 4.4): the +35 *is* the off-axis mass, not an artifact of injection. It is not under-fitting — λ=0 converges to val_acc 0.92 while λ>0 plateaus flat-and-worse (0.72/0.64), on the trainer's own held-out slice of the training corpus, scored as the correct continuation outscoring the incorrect one rather than on any benchmark; and evaluation used the best-val-acc snapshot, so the on-axis arm was shown its most favorable face and still failed. One door remains untried: a penalty-warmup schedule (W_{know} predicts it will not rescue). A follow-up, not a headline result.

The strongest possible on-axis arm is built and not run: it needs a GPU we do not have. The on-axis references we inject are constructed, and Section 4.4 now shows they are also excellent *readers* — which removes the estimation-quality account of their causal failure but not the last version of the objection, that some better on-axis direction exists. There is a principled candidate: the decision direction of the probe *fitted* on these activations, the empirically optimal linear correctness reader in this space. Injecting it norm-matched through the identical harness would put the on-axis arm at its ceiling by construction. We built that arm and did not run it, and we say which half is missing. Built and verifiable: the offset blob is norm-matched to the trained offset per head (maximum deviation below 10^{-6}), and out of sample — on the 395 base-wrong items of the complement, disjoint from the items the probe was fitted on — it still reads gold-vs-distractor at 0.9486 (CI95 $[0.9318, 0.9637]$), so the direction that would be injected remains a strong reader where it would be judged. Not done: what it *does*, which requires generative and scoring forward passes through a 31B model in BF16 and therefore a GPU class we do not have locally. One design constraint is load-bearing and is recorded in the blob's own configuration rather than left to the person who runs it: unlike v_{inf} and W_{know}, which are estimated on a separate corpus, the probe is fitted *on ARC*, so it is fitted on items 0–399 and any evaluation of it must score the complement, 400–1171. We report no estimate of the outcome. Until it runs, the on-axis null rests on two constructed references that read as well as a fitted probe, and not on the fitted probe itself.

A cheap test-time-compute baseline, narrowed (PMI is already settled). The arbiter implies the adapter approximates knowledge the base already reaches by CoT, which invites the question of whether a cheap test-time-compute baseline matches or beats it on the scoring task. One leg is already settled and does **not**: PMI / decision-conditional re-scoring caps at base-wrong AUC 0.574 against the adapter's 0.830 and actively *hurts* the adapter when applied (Section 5.3), and a same-scaffold DC-PMI head-to-head leaves the adapter recovering items PMI does not (TQA, 28 items, CI lower bound +0.622). Two more legs have now **run** and are reported as the **deployment triangle** in Section 5.4. The *reasoning* TTC family — self-consistency over sampled CoT — recovers 0.970 on the 198 base-wrong ARC items, outscoring both the read-only selector (0.911) and the adapter's cold-MC scoring (0.758) — at ~150× their measured latency. And the cheapest of the four routes we timed — the base emitting the answer letter in a single greedy forward, no CoT — also recovers **0.970** (192/198), and on measured per-item latency it ties the reasoning baseline at ~150× less cost while outscoring the selector and the adapter at roughly half their measured latency (181 ms against 360 and 354 ms). That latency gap is a fact about the implementations we timed, not about the routes in principle: our scoring harness runs each of the ~4 option continuations as its own batch-1 forward, and an implementation that batched an item's options into one forward is a comparand we did not measure. So the adapter loses to a reasoning baseline *and*, in this comparison, to the most trivial alternative on accuracy as well as on measured latency, and the selector's value is evidentiary, not deployment (letter emission outscores it too). The one scope caveat is that letter emission is a lettered-MCQ readout distinct from the cold-MC continuation channel (Section 3.2), so this contrasts two readouts of the same base — the lossy-readout thesis in its sharpest form — rather than a like-for-like scoring test. And on the mc_only set — the items the adapter fixes in cold-MC that v_{inf} does not — self-consistency reaches **124/124**: the Section 5.1 arbiter taken to its extreme, the base reaching by reasoning every item the adapter "fixes" by cold scoring.

Cross-family universality: what replicates, and what is still one model. The behavioral *effect*, the off-axis geometry and the causal decomposition all hold on a second frontier family — Qwen 2.5 14B IT, run through the identical pipeline, detailed in Section 4.6: a same-sign cold-MC lift on all four tasks (ARC +18.0 / HellaSwag +7.8 / TruthfulQA-mc1 +15.8 / Winogrande +3.0 in the gated reference run — the last small and not individually significant at n=400, counted as marginal), whitened $\cos(\theta_{\text{mc}}, v_{\text{inf}}) = -0.002$ with 97.5% perpendicular energy, the lift carried by the perpendicular component (+106%) and not by the parallel one (−6%), an on-axis reference that again fails against same-norm noise, a matched non-contrastive control that recovers 27% of the ARC lift, and the same non-transfer to generation (−49.0 pp on gsm8k, turn termination preserved). Two qualifications hold this at N=2. The per-task pattern differs — Winogrande +3 against Gemma's +19, magnitudes below Gemma's throughout — a difference we do not explain. And what stays single-model is the *per-item* evidence and the capacity dimension: the arbiter, the higher-capacity inducibility arm with its rank sweep, the activation-patching decomposition, and the no-locus result. Two families are suggestive of a general property, not evidence for one.

7 Open questions and future work

A dose-window efficiency candidate — gated, exploratory. The single-magnitude activation-scale axis of Section 5.5 surfaces an unexpected *positive*: at intermediate doses the offset **shortens** generative CoT without breaking the computation. But it is measured on the injected, scaled *steering* offset (steering ≠ baked), on a single seed, and whether a trainable adapter inherits the efficient regime is exactly the open question fork (b) leaves standing — so we report it as a candidate, not a claim, in Section A.3.

Open questions inherited from the system-level negatives. Two further questions bound the deployment reading rather than the mechanism. The false-positive routing gate is unbuilt (the only damage-screen is weak, AUC 0.609), so we make no hazard-free deployment claim; and the per-item fine-dispatch nulls were estimated against a permutation floor at n=120 and an underpowered cross-task screen at n=62, so they are reported as no-signal-at-this-power, not established absence. Neither affects the central result, but both must be settled before the routed system is read as more than a mitigation vehicle.

8 Conclusion

We asked what a large cold multiple-choice lift from contrastive-correctness PEFT is, on a frontier 31B instruction model, and our answer is that it should not be read as a capability gain. Non-contrastive controls attribute much of the lift to supervised fine-tuning without a contrastive term — though still on correct-option continuations in the same multiple-choice format, which is the class we tested and not fine-tuning at large — and locate what the objective does contribute: at the adapter's own capacity — the identical 12,336 parameters, only the training objective changed — a generic objective recovers 36.2% ± 5.0 of the ARC gain (best-checkpoint; ~36–40% across the selection rule), rising to ~62% when the control is given three-to-five orders of magnitude more parameters, and ranging from 79.5% on Winogrande to nothing at all on TruthfulQA (headline metric per task; the fractions and their cross-task ordering are metric-dependent — under raw accuracy the 36.2% reads 26.9% — the lifts are not). The contrastive adapter's write lives in a learned direction that is off-axis of degree to the correctness geometry measured at the 48 heads this recipe intervenes on — and with the parameterization held fixed, it is the contrastive objective, not the apparatus, that goes furthest off-axis — a geometry that is causal, and identified against random and on-axis alternatives on both the harness and the write sides. That mechanistic chain — arbiter, calibration, PMI-residual, probe, selector and geometry — is a convergent analysis developed post-discovery on one fixed 400-item slice in native order, not an independent confirmatory replication — though it is no longer confined to that slice: the full split re-runs the lift, the arbiter, and, fit on the slice and scored on the 772 items outside it, the probe and the selector; calibration and geometry still share the sample (Section 5.9). And the lift does not become generation: the items it fixes already sit at the base's chain-of-thought ceiling, a read-only selector on the same activations recovers it without the deployment costs, and — most plainly — the base asked to emit the answer letter in a single forward recovers more of the base-wrong items (0.970) than either the adapter (0.758) or that selector (0.911), and in our implementation it does so at about half the measured latency — 0.980 against 0.855 across all 400 items — while tying a reasoning baseline some 150× more expensive. That last contrast reads the base in a lettered-MCQ format rather than in cold-MC option scoring, so it compares two readouts of the same base rather than two models like for like, and the cost comparison is between the two implementations we timed, not a property of the readouts as such. The effect and its off-axis geometry replicate on a second model family — though only partly: the cross-family lift is same-sign on all four tasks but marginal on Winogrande at n=400, and what remains single-model is the *per-item* evidence (the arbiter), the capacity dimension of the spectrum (the higher-capacity inducibility arm and its rank sweep), the activation-patching decomposition and the no-locus result; the matched-capacity inducibility control and the perpendicular/parallel re-injection decomposition *do* replicate on the second family.

One finding points the other way, and we state it as a candidate rather than a claim. The scoring lift and the generative damage are separable by *dose*: at three quarters of the activation scale ARC is already at 0.855, identical to full scale, and on MMLU-Pro the lift is complete at half dose while the repetition loops that mark the degeneration fall 146 → 10 → 0. There is, on *each of these two benchmarks separately*, a dose at which the scoring benefit is complete before the collateral cost arrives — s=0.75 on ARC (where gsm8k is nonetheless already 2 pp below base) and s=0.50 on MMLU-Pro (where the loops reach zero). No single dose is shown to be full-benefit and zero-pathology on both. It is a steering result and the arms this paper leads on are baked, so it bounds what an operating point could look like rather than telling us what the adapter is — but it is the first thing a practitioner would want to know, and the lift being cheaply separable from its collateral is itself evidence about what kind of thing the lift is.

The practical consequence is a caution for the contrastive-PEFT-for-multiple-choice program, drawn from one adapter of that family rather than from its published recipes. A cold multiple-choice gain of this size can be an off-axis, non-transferring edit to a lossy readout — obtainable read-only, without touching weights or generation, and beaten by a reasoning baseline — so a benchmark improvement of this kind is not, on its own, evidence of a better model.

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Code and data availability

Code to reproduce every analysis, the evaluation protocols and their pre-registered kill-rules, and the result artifacts behind each table are released with this paper; the adapter offsets and the heavy activation/evaluation artifacts are hosted as Hugging Face datasets referenced per-number in Section B (run → artifact map).

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A Extended mechanism detail and exploratory analyses

This appendix collects (Section A.1) the full mechanism-decomposition detail behind Section 5.6's verified negatives, moved out of the body for length, and (Sections A.2 and A.3) the exploratory single-seed / single-model analyses that extend the mechanism story but that the paper relies on for **no** claim. Sections A.2 and A.3 are exploratory (single seed / single model / not load-bearing).

A.1 Full decomposition for Section 5.6 (temperature negative, symbol channel, frequency control)

Not a temperature / null-space mechanism. A natural mechanistic hypothesis, given the readout framing, is that the offset acts like a confidence-regulation neuron: writing into the unembedding null space to rescale the residual-stream norm and thereby the logit temperature, scaling all logits without re-ordering the argmax [49]. A direct-logit decomposition refutes this for θ_{mc}. We split the offset's effect on the vocabulary into a uniform component (a temperature shift, parallel to the all-ones vector) and a ranking component (a re-ordering, perpendicular to it), and project the residual write onto the *genuine* temperature direction $d^* = \arg\min \|W_U r - \mathbf{1}\|$ (whose own logit image is 0.9988 uniform, confirming it is well identified). The offset's cosine with d^* is **0.023** — orthogonal to the temperature direction at chance level (the random-direction orthogonality baseline is $\sim 1/\sqrt{d_{\text{model}}} \approx 0.014$), and no single head is temperature-pure (per-head maximum uniform-shift fraction 0.58). (The raw logit-space uniform-shift fraction, 0.301, does *not* discriminate, because Gemma's unembedding is anisotropic and a random direction already produces ~0.24 ± 0.18 of apparent uniform shift; we therefore rest the negative on the d^* projection, not on the raw fraction.) The offset moves *differences* of logits, not their mean — it is a ranking edit, not a temperature edit — which closes a "the lift is just a confidence/temperature rescaling" alternative a reviewer could raise. **This is now confirmed causally.** A real-forward activation-patching test — the causal upgrade of the direct-logit reading — measures the offset at the scoring positions where the lift lives (the option-continuation tokens) and finds a **relative** temperature fraction of only **0.215** ($\text{temp}/\sqrt{\text{temp}^2 + \text{rank}^2}$, CI_{hi} 0.224 ≤ the pre-registered 0.35 kill-rule; in absolute fractions of the effect: temperature **0.177**, ranking **0.803**): the offset re-ranks, temperature is minority (the softmax-invariant uniform component, **0.4592** at the option positions, is excluded from the verdict). The causal cosine with the temperature direction, −0.149, is ~6× the direct-logit 0.023 and ~10× the random-orthogonality baseline (~0.014) — and of *opposite sign*: the linear attribution mis-stated the alignment in sign as well as magnitude, which is the expected signature of a first-order read of an off-axis causal component (whose first-order gradient is near zero, so its linear projection is dominated by noise terms that carry no sign information) rather than an anomaly of the measurement — yet the causal temperature fraction remains a clear minority (0.215 relative), so the negative rests on the causal measurement, not on the direct-logit cosine. The same logic is why we ran the frequency control's own causal cross-check rather than rest it on the attribution level (below): it comes back without a sign-flip, so the two Stolfo components dissociate under causal measurement. This under-reading of a causally effective, off-axis component by a *linear* (direct-logit) attribution is itself a documented pattern in the steering literature: for single-neuron control, first-order gradient attribution *anti-predicts* controllability precisely because effective controllers write **off the readout axis** and carry a near-zero first-order gradient, and a forward-only contrastive screen recovers the controllers that attribution misses [36] — so a linear read of our offset under-stating its causal alignment is the expected signature of an off-axis mechanism, not an anomaly. *Position matters:* measured instead at the next-token (generation) position the offset flattens the distribution more (temperature fraction 0.38, Δentropy +0.99, γ −0.59) — the generation-face flattening we connect to the format face below — so the temperature negative is anchored at the scoring positions, not the generation one.

The signpost: a content/symbol channel, not a lexical one. Direct-logit attribution over the 48 adapter heads finds **no clean lexical mechanism**: the tokens it *promotes* are diffuse and multilingual, with no coherent semantic theme. The one robust signature is on the *suppression* side — single uppercase letters and structural tokens (`B, S, P, T, C, L, Y, H, I, G, D, O, V, W` and punctuation), exactly the surface tokens of multiple-choice labels and turn formatting — the offset's logit-level correlate of the *format-preference* uncertainty that aligned-model calibration distinguishes from answer-decision uncertainty [25]. This is the readable face of the **format-preference channel** — the same generation-face flattening the causal test finds at the next-token position (above) — not of a termination collapse (that claim is retracted, Section 5.5) and not an explanation of the scoring lift, and it is an aggregate-of-heads property (a coherent sum across heads, not a per-head one). With temperature ruled out, the most promising unifying-mechanism candidate is the **content/symbol-binding channel** that MCQA mechanism work isolates: models first select a winning answer in *content* space and then bind or route it to the *symbol* to emit [61], a "correct-letter" routing studied at larger scale [34]. The decisive behavioral test — a TEXT-vs-LABEL scoring ablation (whether the lift survives permuting the option labels) — has now **run, and the result is consistent-with but not clean**. On ARC the label-arm (letter) lift exceeds the text-arm lift (Δ_{letter} +0.1575, CI [+0.087, +0.228]) and survives a headroom normalization (norm Δ_{letter} +0.17, CI [+0.088, +0.249]), consistent with a symbol-channel contribution — but the two permutation controls that would separate a *letter-specific* effect from a *generic symbol slot* are themselves ceiling-starved (caps_random base 0.78, numeric 0.97), so letter-specificity is **untestable** here. On TruthfulQA the letter arm is itself at ceiling (base 0.81), so its raw Δ_{letter} −0.29 is a **headroom artifact** — the task is *uninformative* rather than contradictory (a point we surface only after normalizing; the raw sign looked opposite). The symbol channel is therefore **consistent-with, not a clean mechanism**: the lift already lives in the text-face arm (which reproduces +0.3275 ARC / +0.3575 TQA on its own), so permuting the labels does not collapse it. We report this as a signpost, not a named mechanism. This suppression signature is now confirmed *causally on real generation positions* — an nnsight patch of the offset over 50 gsm8k traces (26 non-terminating loops + 24 clean-wrong), measuring the base→adapter logit delta per generated token: of the tokens the offset most suppresses, ~**91**% **are structural** (punct/symbol), a ×**3 enrichment** over the vocabulary base-rate — the same structure-suppression face found at the scoring positions, now on the generation face — and the structure share **grows mildly along the generation** (slope +0.04, slightly steeper in the loops). But the signature is strong only in the tail (diffuse in raw mass) and its growth is sub-threshold, so it does **not** license a "commitment-machine" reading of the reasoning loss; we leave that frame as an undecided conjecture, not a mechanism. A control that the promotion is not a token-frequency artifact — the *other* Stolfo component (token-frequency neurons) — has now been **re-derived, persisted, and closed** (`derive_freq_control`, reusing the same aggregate-write construction, ‖agg‖ 100.28). Over the full wikitext-103 training corpus (121.6M tokens, 112.8k of the 262k vocabulary observed) the recovered frequency direction d_{freq} is well identified — its logit image reproduces the target log-frequency at $r = \mathbf{0.906}$, past the pre-registered 0.9 self-gate — and the offset's cosine with it is **0.0128**, below the 3×-null gate (0.041) and at the null floor ($1/\sqrt{d} \approx 0.0136$). The identification rose monotonically with corpus size ($r = 0.855 \rightarrow 0.887 \rightarrow 0.897 \rightarrow \mathbf{0.906}$ at 5M/20M/40M/121.6M tokens), crossing the gate only at the full corpus. So the offset is **not** the token-frequency component of a Stolfo mechanism either — a closed control, not merely consistent-with. **The causal-patching cross-check has now run** (the same protocol as the temperature one: real forward, ARC, n=400, scoring positions; the six previously published decomposition scalars reproduce to $\Delta = 0.00$, so the frequency arm is strictly additive). Projecting the real base→adapter logit delta onto the mean-centered frequency signature gives cosine **+0.0059**, CI [−0.0027, +0.0142] — **same sign** as the direct-logit 0.0128, ×0.46 its size, and crossing zero — against the temperature's flip to −0.149 (CI [−0.158, −0.140], ×6.5). The per-item view is sharper than the means: the frequency projection's mean *absolute* size is comparable to the temperature's (0.120 vs 0.177), but its **sign is inconsistent across items (66% positive, vs 99% negative for temperature, with γ<0 in 99.3%)** — the two components dissociate not in how much they load but in whether they load *coherently*, and only a coherent component can drive a lift that is systematic across items. **One declared limit:** the frequency direction orthogonalized against the base logits (the part of log-frequency not already shared with the temperature channel — a quantity with no direct-logit counterpart, so it neither confirms nor contradicts the 0.0128) does carry a small but non-zero causal load, cosine **+0.0298**, CI [+0.0221, +0.0374]: ~15× the vocabulary-space null ($1/\sqrt{V} \approx 0.002$) but ~1/5 of the temperature component and ~1/27 of the ranking one. We therefore state the negative at its true strength: token frequency is ruled out as *the mechanism*, not as a trace contaminant of the offset's logit image.

A.2 Off-workspace, not workspace-broadcast (J-lens + recall-tail) — exploratory, single seed

Off-workspace, not workspace-broadcast: the off-axis push is an automatic-channel edit, not a global-workspace one. If the off-axis push is a shortcut that the cold-scoring readout reads but generation does not (Section 5.1), a natural question is whether it enters the model's *global workspace* — the broadcast subspace whose contents a Jacobian-lens ablation degrades — or acts off-workspace as an automatic channel. The workspace account [23] predicts the second: it reports that multiple-choice scoring is *not* degraded by J-space ablation (automatic) while GSM8K-CoT is robust because the model externalizes what it would otherwise hold in the workspace — both of which map one-to-one onto θ_{mc}-lifts-MC-but-not-CoT. We fit a Gemma Jacobian-lens (`anthropics/jacobian-lens`, 114 prompts, above its ~100-prompt saturation, source layers → L58) and test whether θ_{mc} behaves like a workspace-broadcast direction. Four independent reads converge on **off-workspace**. (i) An MLP-gain kill-switch — testing whether the MLPs amplify θ_{mc} the way they amplify broadcast directions — returns gain ≈ 1.0–1.4×, indistinguishable from a random vector (~1.0×) and far below a J-lens broadcast reference (`jlens_dir_highref` 2.3–4.0× at L37–40); θ_{mc} is not amplified like a workspace direction (the kill-switch did not fire). (ii) The J-space fraction of θ_{mc} equals a norm-matched random vector's (excess ≈ 0). (iii) At option positions the offset inserts *no* answer-letters (insertion rate 0.0 at every layer — the +35 is option-text, not letter, consistent with Section 5.8), yet it *does* rewrite the late readout (recall@25 falls from 0.98 at L30 to 0.34 at L49, strongest in the offset's own layers). (iv) The tokens it rewrites the readout toward are **structure, not knowledge** — bare capitals, punctuation and markup, multilingual fragments, zero semantic/answer tokens — the same structure-suppression signature the causal patch found. This read is resolved *per-position* on the real base-vs-adapter activations (the recall-tail): across the tail layers where recall@25 falls (L38→L55), the adapter-only inserted ids are **60–88% structure/format** (bare capitals, quotes/punctuation, non-latin script) with **zero answer-tokens at every one of the twelve layers**, and the non-structural remainder is generic multilingual function-words (`and`, `de`, `own`, `ability`, `same`), *not* answer-knowledge — so even the 60% floor under-states the no-answer-knowledge share. So the offset perturbs the late readout toward output *format*, not via workspace broadcast and not toward the answer: an automatic-channel edit, which is the mechanistic shape a direction that lifts cold scoring but does not transfer to reasoning should have. We separate what is measured from what is inferred here: the load-bearing E2 signals are occupancy and recall@25 (the readout *is* rewritten late, recall@25 0.98→0.34); the "zero answer-*letters* inserted" figure is a designed null-control (`run_p10_pod.sh` expects ~0 for A/B/C/D insertion, consistent with Section 5.8's option-text-not-letter finding), not a discrimination of off-workspace from an answer-directed push. The `structure_fraction` (60–88%) is biased toward structure by the top-50 cut (item-specific content tokens have low counts and fall outside). A gold-option-text occupancy metric (top-25 vs the gold continuation tokens) would firm up this read; it is flagged as the Section 5.6 follow-up, not run.

The J-lens sanity gate is marginal, and we scope the claim to what it licenses. The fitted lens clears its mandatory sanity gate — recovering the model's own next-token from intermediate residuals — only *marginally*: at just 1 of 18 fitted layers (L55 @ 0.50, the layer nearest the L58 target), 0.0 elsewhere. We diagnose this as the known Jacobian-lens limit, not a broken fit: recovery is governed perfectly by distance-to-target (Spearman(recovery, distance) −1.000; every scale/shape metric of the averaged Jacobian grows monotonically toward the target — $\|J\|_F$ ρ +0.996, mean-diagonal ρ +1.000), and far from the target the averaged Jacobian collapses toward rank-1 with near-zero trace (one linear map cannot bridge many nonlinear layers). The load-bearing read (iv, the structure-not-knowledge readout) **reproduces at L55, the one sanity-valid layer** (90% structure, zero knowledge/answer tokens — the same class as the marginal layers L30–49). We therefore scope the workspace claim precisely: *off-workspace is the convergent read across the four probes above and an independent causal channel (the nnsight patch, Section 5.6), and the J-lens is quantitatively valid near its target, where the direction read reproduces* — **not** that the J-lens is globally valid. A per-source-target re-fit would harden the early-layer reads; more prompts would not (the marginality is structural, not sampling noise). Exploratory, single seed.

A.3 Dose-window efficiency candidate — exploratory, single seed; steering ≠ baked

A dose-window efficiency candidate — a positive worth chasing, but gated. The single-magnitude activation-scale axis that dissociates the collateral cost (Section 5.5) surfaces an unexpected *positive*: at intermediate doses the offset **shortens** generative CoT without breaking it. Conditioning on the intersection where both the base and the scaled offset answer correctly (n=200 gsm8k), scale 0.75 cuts generation length **−34.7**% while the real arithmetic stays intact — verifiable calculation steps fall only **−7.2**% against a −19% drop in «`...`» annotation markers and a −52% drop in prose, with 98% of remaining steps still valid, and a qualitative arbiter confirms complete reasoning even where the «» markers are gone. The genuine shortcut/breakdown — calc −18%, step-omission, creative-degeneration loops — appears only at full strength s=1.0, where accuracy collapses to 0.815. So *inside* gsm8k there is a threshold dissociation: an intermediate dose prunes prose while preserving the computation — the same phenomenon as the dose-separable collateral cost (Section 5.5), read on a different metric. We flag this as a **candidate, not a claim**: it is measured on the *injected, scaled* steering offset (steering ≠ baked), on a single seed, and whether a trainable adapter inherits the efficient regime is exactly the open question that fork (b) above leaves standing.

B Run provenance

Run → population. The paper's numbers come from a small set of runs with deliberately different item populations; we list them in Table 13 so no two counts are conflated.

Two ARC base-wrong counts appear in this paper — **198** and **204** — and the difference is a **scaffold** difference *inside the same run*, not a difference between runs. Both are the vinf_causal 400. The **lm-eval scaffold** (`d_null.json`, acc_norm, the scoring that produces the headline lift) leaves **198** items wrong; the **premise-file scaffold** (`answer_only_arc_challenge.json`, a length-normalized argmax over option continuations scored under one shared neutral premise) leaves **204**. Every derived subset therefore belongs to one scaffold or the other, and we label them rather than let a reader subtract across the boundary. On the **lm-eval** scaffold: 124 `mc_only`, 151 S_{flip}, 192 $S_{\text{cot\_coldwrong}}$, 202 $S_{\text{base\_right}}$, and **47** adapter-residual (151 + 47 = 198). On the **premise-file** scaffold: 204 base-wrong and **73** adapter-residual, which is the population of the PMI residual-discrimination analysis in Section 5.1. The arbiter and its negative control are reported on the lm-eval scaffold so they are apples-to-apples with the 124, with the premise-file partition given as robustness; both pass an alignment guard on gold index and option count (0 mismatches over 400). The A1 run that supplies the Table 1 headline is a separate run, and its base also leaves 198 ARC items wrong under the same lm-eval scoring (base acc_norm 0.505 × 400 = 202 correct).

Table 13: Run → population map. Every headline number traces to one of these runs; base-wrong subsets are labeled by scaffold so no two counts are conflated.

Run (artifact)	Date	Scoring n	Base-wrong subset(s)	Feeds
A1 canonical trio (`msap-a1-trio-b2-planner-2026-06-10`; private, available from the authors on request — the same four lifts are reproduced at full split, with more power, in the public `msap-eval-bulletproof-lean-20260627`, Table 14)	2026-06-10	400	ARC 198 base-wrong@0-shot	T1 headline (+35.0 / +37.25 / +17.0 / +19.0); T2 gsm8k −25
vinf_causal (`msap-vinf-causal-20260623-public`)	2026-06-23	400 (own within-run baselines; ARC lift +35.0)	*lm-eval scaffold:* 198 base-wrong; 124 mc_only; 151 S_{flip}; 192 $S_{\text{cot\_coldwrong}}$; 47 adapter-residual. *premise-file scaffold:* 204 base-wrong; 73 adapter-residual	T3 arbiter (124/124, parser-fixed) + its negative control (residual 41/47, `arbiter_negative_control_b2.json`); T3b calibration / present-but-swamped; T6 v_{inf} 11%; T8 PMI-DC / choices-only
wknow_causal (`msap-wknow-onaxis-20260623`)	2026-06-23	400 (null-adapter baseline 0.500)	—	T5/T6 W_{know} 6.4%; significance (McNemar $p = 0.136$)
R2 anti-floor 5-shot (pre-registered eval)	2026-06-27	400	ARC 198 base-wrong@0 (adapter fixes 151; few-shot 69)	T1 caption: anti-floor +22.0 pp (a@0−b@5) and matched-shot +22.25 pp (a@5−b@5; McNemar b=99/c=10)
R4 full-set (`msap-eval-bulletproof-lean-20260627`)	2026-06-27	full test sets	—	full-Δ confirmation (ARC +36.2 / HSwag +19.2 / Wino +20.1 / TQA +37.1)
Same-cost letter baseline (`B5_base_letter_greedy_arc400.json`, same repo)	2026-07-21	400 (198 base-wrong)	ARC 198 base-wrong	base greedy, no-CoT, 1-forward letter emission → 0.980 all / 0.970 base-wrong; deployment triangle 4th point (Section 5.4)
mechanism_v8 (`msap-mechanism-v8-20260701`)	2026-07-01/02	400	—	T2 R6 gsm8k −25 (cap sweep); Section 5.5 EOS retract + Stage-E blend; Section 5.6 patching; Section 5.9 Qwen 4/4-sign
Three later analyses, all re-derivations over artifacts already listed above; no new generation and no GPU. Local files, versioned in the repo	2026-08-09	—	48 heads (U-01); MMLU-Pro 1,000 paired `doc_id` (U-16); ARC 198 base-wrong, lm-eval scaffold (U-08a)	`runs/oq1_functional_axis/u01_thetamc_vret.json` → $\cos_{\text{w}}(\theta_{\text{mc}}, v_{\text{ret}})$ and its positive control (Section 4.4); `runs/paired_gen_dl/arbiter_mmlu_pro_cot.json` → the MMLU-Pro rows of Table 3; `runs/vinf_causal/u08_onaxis_reader_auc.json` → the on-axis reader AUCs. The blob of the un-run arm and its build report are `runs/vinf_`

Run → HF artifact (reproducibility). Several load-bearing numbers come from runs whose result files (small JSON) are synced into `runs/` from a Hugging Face dataset backup, while the heavy binaries they were derived from (per-head cold-MC activations, J-lens jacobians, LoRA adapter weights — each 2–6 GB) remain on HF as pointers rather than versioned in the repo. The pointer for each is given in Table 14. The sync is partial, and we mark which is which: rows tagged *HF-only* have no local copy in the released checkout and must be fetched from the dataset backup before the corresponding analysis will run. Every dataset named in Table 14 is public, so each such fetch is available to a reader. Two families of heavy binary are the exception and are *not* in the public backup: the per-head cold-MC activations (`coldmc_perhead_*.pt`, 2.8 GB) and the Align-LoRA adapter weights (`adapter_model.safetensors`, ∼50 GB), which stay in a private archive. Nothing in Table 14 points at them, and, as noted below, none of the paper's numbers require re-deriving them; they are available from the authors on request.

The post-parser-fix `arbiter_arc_cot_v2.json` (124/124) is versioned in the repo, so `scripts/analyze_arbiter_arc_cot.py` reproduces the headline arbiter count from a clean checkout; the pre-fix `arbiter_arc_cot.json` (123/124, one item lost to a letter-parser artifact) stays HF-only, and the script prefers the v2 when both are present. **Every quantity derived from the arbiter is read from the v2**, and we say so because the correction propagates further than the headline count: two downstream analyses (the cold-MC calibration and the base-arm power check) resolved the arbiter path to the pre-fix v1 by default, which silently excluded doc 128 — base-wrong, and unparsed only because of the very letter-parser artifact the fix removed — from the adapter-independent $S_{\text{cot\_coldwrong}}$ pool. Re-derived on the v2, that pool is **192** rather than 191, its CI lower bound 0.512 rather than 0.513, and the calibration signature is unchanged at the reported precision (z-gap 1.45σ, gold-rank median 2, frac rank≥2 0.59); the base-right control likewise reads 200/202 rather than 196/202. The verdicts are unchanged; we report the re-derivation rather than leave two arbiter versions in circulation. `scripts/analyze_p9_gate.py` recomputes the on-axis-penalty gate from the per-λ files above and self-checks against the reference anchor. The heavy binaries (`vinf_causal/coldmc_perhead_*.pt`, `jspace/{jacobians,gemma_jlens}.pt`, `align_lora_control/*/adapter_model.safetensors`) are HF-only; none of the paper's numbers require re-deriving them.

Table 14: Number/analysis → local result file → Hugging Face dataset backup.

Number / analysis	Local file (`runs/...`)	HF dataset (`sebacascardo87/...`)
Arbiter 124/124 (post-parser-fix v2)	`vinf_causal/arbiter_arc_cot_v2.json`	`msap-vinf-causal-20260623-public`
P9 on-axis-penalty lift-matched gate (λ=5 −9.9 pp / 3.8σ)	`p9_onaxis/{mc,gsm8k}_lam{5,15,50}.json`, `mc_dose_lam0_s*.json`, `gsm8k_dose_lam0.json` (HF-only)	`msap-p9-onaxis-20260709`
Recall-tail (Section 5.6 / Section A.2)	`jspace/stage_e2_recall_tail.json` (insertion rate, inserted-token categories) and `jspace/stage_e2.json` (the `recall_at_25` decay) (HF-only)	`msap-jspace-gemma-20260709`
Align-fairness rank sweep (~62%)	`align_fairness/align_fairness/R1_{v7mc,null}.json`, `align_direction_r*.json` (HF-only; the same-named files under `runs/derisk/` are the geometry sweep, not these accuracies)	`msap-align-fairness-20260702`, `msap-phase1-evals-20260624`
Table 6 random norm-matched	`phase1/derisk/random_s{0,1,2}.json`; `offsets_random_s*.pt` (HF-only)	`msap-phase1-evals-20260624`
Section 4.4 $\text{off}_{\parallel}$ / $\text{off}_{\perp}$ / W_{know} spectrum arms	`eval_bulletproof/RD_{offpar,offperp}.json` (HF-only); the W_{know} arm is `wknow_causal/onaxis_significance.json`	`msap-r5-align-p0-20260628`
Section 4.4 perpendicular-shuffle write-side control (3 seeds)	`RD_offperp_shuffle_seed{0,1,2}.json` (HF-only)	same
Table 12 adapter replicates (ARC/TQA/HSwag/Wino, gsm8k, net-effect ratio)	`v7_lofit_seed{1,2}/eval/` (HF-only)	`msap-review-remediation-20260724`
Section 5.4 deployment triangle (latencies 181/354/360/26,704 ms, VRAM, accuracies 0.970/0.911/0.758)	`e3_cost/e3_cost_benchmark.json`	same
Same-parameterization checkpoint-selection provenance (best step per seed)	`e1_plain_ce/offsets_seed*.log.json`	same
Same-parameterization objective provenance (`loss_type`, head set, 12,336)	`v7_lofit_gemma4_31b_chat/offsets_mc.pt` and `e1_plain_ce/offsets_seed*.pt` (embedded `config`), `e1_plain_ce/train.log`	same
Full-split (n=1172/10042/1267/817) matched pair, adapter and base arms	`eval_bulletproof/R4_v7mc_FULLSPLIT.json` (HF-restored, unversioned by size) and `eval_bulletproof/R4_base_full/`; pairing checks and digest in `eval_bulletproof/R4_fullsplit_provenance.json`	`msap-eval-bulletproof-lean-20260627`
Section 4.5 layer-band patching sweep	`phase1/derisk/patch_layers/` (HF-only)	`msap-phase1-evals-20260624`